

Online Appendix

The Effect of Minimum Wages on Low-Wage Jobs

Doruk Cengiz

Arindrajit Dube

Attila Lindner

Ben Zipperer

University of Massachusetts

University of Massachusetts

University College London,

Economic

Amherst

Amherst, NBER, IZA

CEP, IFS, IZA, MTA-KTI

Policy Institute

Appendix A	
Additional Figures and Tables	3
Appendix B	
Upper Tail Employment Effects in the Neoclassical Model	33
Appendix C	
Washington State Case Study	35
Appendix D	
Event-by-event analysis	43
Appendix E	
Data Appendix	50
Appendix F	
Comparison of Administrative Data to CPS	52
Appendix G	
Reconciling the Results from the Two-Way Fixed Effects Panel Regression with Log Minimum Wage and Event Based Regressions	67

Appendix A

Additional Figures and Tables

Figure A.1 shows all minimum wage increases between 1979 and 2016. We use the time series of state-level minimum wage changes from Vaghul and Zipperer (2016). Blue circles show the minimum wage events that are used in the event study analysis. The light orange triangles represent small minimum wage changes that we do not analyze (but control for). For these changes, the minimum wage increased either by less than \$0.25 (the size of our wage bins) or less than 2 percent of the workforce earned between the new and the old minimum wage. Finally, the green circles indicate federal changes, which we also exclude from our primary sample of treatments because the change in missing number of jobs, Δb , is identified only from time-series variation for these events as there are no “control states” with wage floors lower than the new minimum wage. The figure highlights that around 70% (99/138) of the minimum wage changes in our sample occurred after 2000.

Some wages in the CPS are imputed. In most of our analysis we use only non-imputed wages. This might be of concern if the imputation rate changes in response to the minimum wage, or is correlated with minimum wage changes for some other reason. Figure A.2 shows event study estimates where the outcome is the state-level imputation rate. The figure shows that minimum wage events studied here have no apparent effect on the imputation rate.

Our definition of "overall employment" does not include self-employed workers, who are not covered by the minimum wage. (Note that the QCEW does not include self-employed workers either). The exclusion of self-employed can be problematic if minimum wages shift employees to self-employment. Figure A.3 (“Impact of Minimum Wages on the Self- Employment”) shows that the self-employment rate (i.e., self employed workers divided by wage and salary plus self-employed workers) is not affected by the minimum wage. This confirms that there is not any shift to self-employment induced by the minimum wage.

Figure A.4 shows the average change in the minimum wage after the 138 events we use in our baseline specification. The figure depicts a sizable, statistically significant and persistent increase in the minimum wage starting from the first event year ($\tau = 0$) as compared to the controls. On average, states with minimum wage events experience an increase of 8.4% (0.7%) in the 5 post-treatment years, validating our definition of minimum wage event.

Figure A.5 plots the evolution of wage and total employment changes for affected workers over annualized event time using our baseline specification with wage-bin-period and wage-bin-state fixed effects. The upper graph in Figure A.5 illustrates the clear, statistically significant rise in the average wage of affected workers at date zero, which persists over the five year post-intervention period. In contrast, the lower panel in Figure A.5 shows that there is no corresponding change in employment over the five years following treatment. Moreover, employment changes are similarly small during the three years prior to treatment.

Figure A.6 shows the effect of the minimum wage on the wage distribution when we take into account that sometimes minimum wage increases are phased in over multiple events. In 65% of the

cases we study, a primary minimum wage increase is followed by a secondary one within 5 years, on average at \$0.56 above the minimum for the primary event. In contrast to the main results of the paper where we show the partial effect of each event, here we show the cumulative effect of both primary and secondary events by taking into account the incidence and size of secondary increases averaged across our sample of events. The cumulative effect of primary and secondary events on missing jobs is 2.5%, which is larger than the partial effect of the primary events, which is 1.8% (see Figure II). Therefore, the presence of multiple events can explain some of the difference between the jobs below the new minimum wage—which is around 8.6%—and the missing jobs below the new minimum wage—which is around 1.8%— in the main analysis.

Figure A.7 compares our main estimates of own wage elasticity of employment to the estimates in the previous literature. The estimates from the previous literature are obtained from Harasztosi and Lindner (forthcoming), using studies that reported both employment and wage estimates. We report the benchmark estimates from Column 1 in Table I and the Card and Krueger high probability groups from Column 6 in Table II. The dashed line shows the lower bound estimates of our benchmark specification. The Figure A.7 points out that our benchmark estimates can rule out 7 out of 11 negative estimates in the literature. When we additionally use demographic information and focus on the Card and Krueger high-probability group, the precision of our estimates increases, and we can rule out 8 of those 11 negative estimates.

Panel (a) in Figure A.8 plots the relationship between missing jobs below (multiplied by -1) and the excess jobs above the new minimum wage for the various subgroups in Table II. While there is large variation in the missing jobs across various demographic groups, they are matched closely by excess jobs above the new minimum wage. The dashed line is the 45-degree line and depicts the locus of points where the missing and excess jobs are equal in magnitude ($\Delta a = -\Delta b$). In all cases, except for the black or Hispanic group, the excess jobs are larger than the missing jobs indicating a positive albeit statistically insignificant employment effect. For black or Hispanic individuals, the difference between excess and missing jobs is negligible.

Panel (b) in Figure A.8 plots the relationship between missing jobs below (multiplied by -1) and the excess jobs above the new minimum wage for fully partitioned education-age groups. We use 4 education categories and 6 age categories, yielding a total of 23 education-by-age groups.³⁸ For each of these 23 groups, we separately estimate a regression using our baseline specification, and calculate changes in missing (Δb_g) and excess jobs (Δa_g) for each of them. Each grey circle represents one age-education group, while the blue squares show the binned scatterplot. We also report the linear fit (red line) and the 45-degree (dashed) line that depicts the locus of points where the missing and excess jobs are equal in magnitude ($\Delta a = -\Delta b$). The figure can be used to assess labor-labor substitution across various demographic groups. If there is no employment effect in any of the groups, the slope coefficient μ_1 from regressing $\Delta a_g = \mu_0 + \mu_1 \times (-\Delta b_g)$ should be close to one; under this scenario, differences across groups in the number of excess jobs at or

³⁸Education categories are less than high school, high school graduate, some college and college graduate. Age categories are teens, [20, 30), [30, 40), [40, 50), [50, 60), and 60 and above. We exclude teens with college degrees from the sample.

above the minimum wage exactly mirrors the difference in the number of missing jobs below. In contrast, if employment declines are more severe for lower skilled groups—for whom the bite ($-\Delta b$) is expected to be bigger—then we should expect the slope to be less than one, especially for larger values of $-\Delta b$. As shown in in Figure A.8, the slope of the fitted line is very close to one, with $\hat{\mu}_1 = 1.070$ (s.e. 0.075). The binned scatter plot shows that there is little indication of a more negative slope at higher values of $-\Delta b$. While some specific groups (e.g., individuals with less than high school education between 30 and 40 years of age) are above the 45 degree line, others (e.g., individuals with less than high school education between 40 and 50 years of age) are below the line. Overall, these findings provide little evidence of heterogeneity in the employment effect by skill level.

Figure A.9 shows the event-by-event relationship between missing jobs, excess jobs, employment change and the minimum to median wage (Kaitz index). We plot the bin-scattered non-parametric relationship without controlling for other characteristics of the event. The figure is very similar to our benchmark estimates in Figure V where we do control for observable characteristics including urban share, decade dummies and whether the state leans Republican.

Figure A.10 shows the event-by-event relationship between the change in employment and the minimum to median wage ratio (the Kaitz index). Here we show the raw (and not binned) scatter plots, where each dot represents one of the 138 events studied in the event study. The red circles show the 8 minimum wage changes in Washington DC, while the green circles show the remaining 130 events. The figure highlights that events from Washington DC are often outliers, which is not surprising given that the Washington DC sample sizes are very small in the CPS. To alleviate the influence of outliers when comparing across events, we decided to drop Washington DC from our event-by-event analysis in Figure V and in Figure A.9. However we keep those events in the rest of the paper where we report the pooled event study estimates.

Figure A.11 shows the impact of minimum wages on the wage distribution in *weighted* and *unweighted* TWFE-logMW specifications. Panel (a) reports Figure VI from the main text estimated using (level) fixed effects. Panel (b) reports the *unweighted* version of Figure VI. The use of weights has a modest impact on the results.

Obtaining a meaningful “first stage” effect of the minimum wage on average wages is essential for interpreting the estimated employment effects of the minimum wage. Table A.1 compares the t-statistics obtained from estimates of wage elasticities using our preferred (bunching) estimator focusing locally around the minimum wage using equation 1, and the estimator that runs equation 1 at the state-level and uses log of average state level wage as the outcome variable. Both sets of estimates use the paper’s same underlying 138 events for the minimum wage increases. In nearly every demographic group, the local estimator’s wage effects are much more precisely estimated and the aggregated estimator’s wage effects are often not distinguishable from zero at conventional levels of statistical significance. For all workers, the t-statistic for the local bunching estimator is 12 times as large as the t-statistic from the aggregate estimator. Only in the smaller subgroup of teens does the aggregate estimator’s precision modestly outperform that of the bunching estimator. In almost all cases, the bunching estimator is able to estimate a wage effect statistically different from zero

at the 1 percent level of significance. The only exception is for the low probability CK group, for which our estimator obtains a positive wage effect statistically distinguishable from zero at the 5 percent level, and where the aggregate estimator obtains a negative and highly imprecise wage effect estimate.

In this paper we infer job losses from employment changes around the minimum wage. This has a potential advantage even in the absence of large upper tail employment changes: filtering out random shocks to jobs in the upper part of the wage distribution can improve precision of the estimates. Table A.2 compares the point estimates and standard errors of the localized (bunching) estimator and an estimator that uses equation 1 at the state-level, and specified group’s aggregate employment as the outcome variable for calculating the elasticity of employment with respect to the minimum wage. For almost all the groups, the bunching estimator is at least as precise as the aggregate estimator, sometimes substantially more so in the case of smaller demographic groups. Row 1 shows that for all workers, the point estimates of both approaches are rather similar when estimating the policy’s employment elasticity, with the standard error of the bunching approach being modestly smaller at 88% of the aggregate estimator. For workers with lower education, the standard errors of the employment elasticity using the bunching approach are between 65% and 76% of those using the aggregate approach. The last three rows of the table examine the high probability, middle, and low probability groups described in section II.B.. Only for the middle group does the aggregate estimator largely outperform the bunching estimator’s precision. (As we discuss in the paragraph above, however, for this middle group there is no significant wage effect detectable using the aggregate approach, which makes the precision meaningless.)

As a further check on the correlation between minimum wages and the imputation rate of wages, Table A.3 shows the effect of the minimum wage on the imputation rate using various alternative specifications. All specifications confirm that minimum wages have no impact on the imputation rate.

Table A.4 explores the robustness of the benchmark analysis shown in Column 1 of Table I. In column (1) of Table A.4, we focus on the effect for events that take place in the 7 states without a tip credit, where the same minimum wage is applied to tipped and non-tipped employees.³⁹ Even though the share of the workforce earning below the new minimum wage (9.9%) in these states are similar to those in the primary sample, the bite of the policy is larger in the no-tip-credit states: missing jobs are 2.7% of pre-treatment employment in the no-tip-credit sample as compared to 1.8% in the full sample. However, the larger number of missing jobs is almost exactly compensated by an excess number of jobs above the minimum wage, which amount to 2.6% of pre-treatment employment. The resulting employment elasticity with respect to own wage is -0.139 (s.e. 0.530).

In the second column of Table A.4, we expand the event definition to include (nontrivial) federal minimum wage increases, which produces a total of 369 events. Here we find the missing jobs (Δb) to be slightly larger in magnitude at 2.0% of pre-treatment employment. The wage effect for affected workers is 6.7% and statistically significant. The employment elasticities with respect to

³⁹These states are Alaska, California, Minnesota, Montana, Nevada, Oregon and Washington.

the minimum wage and own wage are both close to zero at -0.009 (s.e. 0.019) and -0.157 (s.e. 0.32), respectively. For federal increases, the change in the number of missing jobs below, Δb , is identified only using time series variation, since there are no covered workers earning below the new minimum in control states. However, $\Delta a + \Delta b$ is identified using cross-state variation, since at least for the 1996-1997 increase and especially for the 2007-2009 increase there are many control states with covered employment \$4 above the new federal minimum wage. Overall, we find it reassuring that the key finding of a small employment elasticity remains even when we consider federal increases.

In column (3) of Table A.4, we consider the number of hours employed and estimate the effect of the minimum wage on full-time equivalent (FTE) workers. These estimates are not very different from Table I. The actual number of FTE jobs below the minimum wage (relative to the pre-treatment employment) is lower ($\bar{b}_{-1}$ = 6.7% as opposed to 8.6% in Table I), indicating that low-wage workers work fewer hours. Consistent with this, missing jobs estimate is also smaller in magnitude when we use an FTE measure (-1.3% instead of -1.8%). The average wage change for affected workers accounting for hours is 7.3% (s.e. 1.2%), while the employment change is 4.4% (s.e. 3.3%). After accounting for hours, the employment elasticity with respect to the minimum wage and the own wage are 0.029 (s.e. 0.022) and 0.601 (s.e. 0.442), respectively. The analogous estimates for headcount employment in Table I were 0.024 (s.e. 0.025) and 0.411 (s.e. 0.430).

In column (4) of Table A.4, we restrict the sample to hourly workers; we expect these workers to report their hourly wage information more accurately than our calculation of hourly earnings (as weekly earnings divided by usual hours) for salaried workers. Although the actual number of workers below the new minimum wage is close to our benchmark sample (10.4% vs. 8.6% in Table I) the missing jobs estimate almost doubles (3.3% vs. 1.8% in Table I). As a result, the wage effects are more pronounced for this subset of workers than the overall sample (9.4% versus 6.8% in Table I), which is consistent with measurement error in wages being smaller for those who directly report their hourly wages. Nevertheless, the employment elasticities with respect to the minimum wage (0.029, s.e. 0.035) and with respect to the own wage (0.306, s.e. 0.392) are very similar to our benchmark estimates.

In column (5), we exclude workers in tipped occupations, as defined by Autor, Manning and Smith (2016). Tipped workers can legally work for sub-minimum wages in most states, and hence may report hourly wages below the minimum wage (as tips are not captured in the reported hourly wage). As we explained in Section II.C., such imperfect coverage creates a discrepancy between the actual level ($\bar{b}_{-1}$) and the change (Δb) in the number of workers below the new minimum wage; however, it does not create a bias in our estimate for the change in employment ($\Delta a + \Delta b$). Excluding tipped workers reduces the average bite, $\bar{b}_{-1}$ = 6.1%, while the estimate of missing jobs of -1.6% is close to our benchmark estimate of -1.8% in Table I. Consequently, estimated wage effects are larger by around 20% (8.2% versus 6.8% in Table 1). However, excluding tipping workers has a negligible impact on the employment estimates: the own-wage employment elasticity is 0.337 as opposed to 0.41 in Table I.

In column (6), we present estimates using the raw CPS data instead of the QCEW benchmarked

CPS. The missing jobs estimate of -1.8% is essentially the same as the baseline estimate. The wage (7.7%) and employment (4.6%) estimates as well as the employment elasticities with respect to the minimum wage (0.039) and own wage (0.590) are slightly more positive. The benefit of using the QCEW benchmarked CPS is the increased precision of the estimates. Without benchmarking, the standard errors for the minimum wage and the own-wage elasticities are 44% and 25% larger than those in column (1) of Table I.

In column (7) we provide estimates without using population weights. These results are virtually identical to our benchmark estimates (Column (1) of Table I). For instance, the employment elasticity with respect to the minimum wage is 0.401 (s.e. 0.418), which is virtually identical to the weighted estimate of 0.411 (s.e. 0.430). The similarity of the weighted and unweighted estimates is reassuring, since a substantial difference between the two could reflect potential misspecification (Solon, Haider and Wooldridge 2015).

In column (8), we limit the sample to 1993-2016. The similarity of the employment elasticity with respect to the minimum wage estimates obtained from post-1992 sample and from the baseline sample (0.006 (0.026) instead of 0.024 (0.025)) is used below in Appendix G to explain differences between the findings of the event-based approach and the TWFE-logMW specification.

Our data is in 25-cent bins and the baseline specification treatment indicators are in 1-dollar increments. To allay any concerns, in column (9), we also check the robustness of our results where the treatment indicators are also in 25 cent increments. In other words, there are 4 times as many regression coefficients for this specification as in our benchmark specification. Obviously, the specific \$0.25 wage bins estimates are noisier than the \$1 bin estimates. However, once we sum up these more noisily estimated coefficients, we obtain estimates that are highly similar to our baseline results (0.023 (s.e. 0.026) and 0.401 (s.e. 0.447) instead of 0.024 (s.e. 0.025) and 0.411 (s.e. 0.430), respectively).

Table A.5 explores the sensitivity of the results using alternative thresholds, $\overline{W}$, for calculating the excess jobs at or above the minimum wage. In our baseline specification, we calculate the excess jobs by adding up the impact in the interval between MW and $\overline{W} = MW + \$4$. In the table we report results using values for $\overline{W} - MW$ between \$2 and \$6. The table shows that the excess jobs estimate increases when the threshold is increased from \$2 (column 2) to \$3 (column 3), but beyond that the estimates remain stable. Therefore, our results are not sensitive to the particular value of $\overline{W}$ once we take into account the presence of spillovers up to \$3 above the minimum wage.

In Table A.6, we consider the robustness of our results to using alternative event windows. Column 1 repeats our baseline results using a window between event dates -3 and 4 (i.e., the 3rd year before the minimum wage increase and 4th year after). Columns 2 and 4 show that reducing the post-treatment window end-date to 2, or extending it to 6 has little impact on the wage or employment estimates. Similarly, columns 4 and 5 show that extending the pre-treatment start date to -5 or reducing it to -1 also has very limited impact on the estimates. For example, across all 5 columns, the employment elasticity with respect to the minimum wage varies between 0.008 and 0.025; the associated standard errors vary between 0.021 and 0.027. Overall, these estimates show

that our findings are not driven by our specific choice of the event window.

Table A.7 reports estimated wage and employment effects using the aggregate event-based (panel A), and local bunching-based (panel B) estimators for the Card and Krueger probability groups. While the aggregate event-based approach considers wage and employment of the full group, the bunching approach looks locally at wage and employment changes of affected workers near the minimum wage. Note that the percentage change in overall average wage will be considerably smaller than the percentage change in wage at the bottom of the distribution. Take the case where both employment fell by 5% and wages rose by 5% for affected workers, but affected workers were only half of total employment. Then aggregate employment would fall by 2.5%, but average wage will rise by even less, since unaffected workers have higher wages than affected workers. As a result the common way of calculating employment elasticity—that takes the ratio of the employment effects and wage effect—will be biased using the aggregate approach; and the smaller the share of affected workers in the group (so that the average wage of the group is much larger than the wage of affected workers), the bigger is the bias.

Column 1 of Table A.7 shows the estimates for the high probability group. Both approaches estimate sizable and statistically significant wage effects with no indication of disemployment. The wage and employment elasticities with respect to the minimum wage are 0.187 (s.e. 0.062) and 0.081 (s.e. 0.084) in panel A, respectively, using the aggregate approach; these are consistent with the findings in panel B using the local estimator. However, the former approach fails to detect a statistically significant wage effect of the policy for the middle and the low probability groups in columns 2 and 3. The wage elasticity estimates in columns 2 and 3 are 0.065 (s.e. 0.057) and -0.005 (s.e. 0.038). This limits the ability of using the CK probability group approach by itself to examine the employment effects of the minimum wage. Since the “first stage” wage effect is missing for the latter two groups, it is difficult to assess the size of the estimated employment effects (0.057 (s.e. 0.047) and 0.001 (s.e. 0.023) for the middle and low probability groups, respectively). On the other hand, the bunching estimator captures a sizable and statistically significant wage effect for all of the groups (0.051 (s.e. 0.013) and 0.060 (s.e. 0.032) for the middle, and low probability groups). By examining changes in the frequency distribution for wages locally around the minimum wage, the bunching approach enables us to establish a causal relationship between the policy and the employment effects for each of the groups.

Table A.8 shows the impact of the minimum wage for incumbents and for new entrants to the labor force. Since CPS interviews individuals twice (one year apart), we can only assess a short term impact of the minimum wage for these two subgroups. However, columns (1) and (2) highlight that the short term and the long term impacts of the minimum wage are very similar in the overall sample. By matching the CPS over time, we lose observations either because matching is not possible, or because there are “bad” matches (see Appendix E for details). Finally, we can only observe past employment status in the second period, so we can only use half of the observations in the matched sample. This shrinks our primary sample size from 4,694,104 to 1,505,192. The results from this matched sample are shown in column (3). The missing jobs are exactly the same as in

the baseline (column 1), however, the excess jobs are slightly lower (1.8% in column 3 vs. 2.1% in baseline). As a result, the change in affected jobs is slightly smaller than in the baseline estimate, but it is still statistically insignificant and positive in sign. Columns (4) and (5) decompose these changes by incumbents and new entrants. Two thirds of the missing jobs come from incumbents, while one third from new entrants. However, the change in missing jobs matches the change in excess jobs in both groups, so the employment effects are very similar (0.9% for incumbents and 0.8% for new entrants). At the same time, the wage effects are different, since new entrants do not experience any spillover effects (see Figure IV).

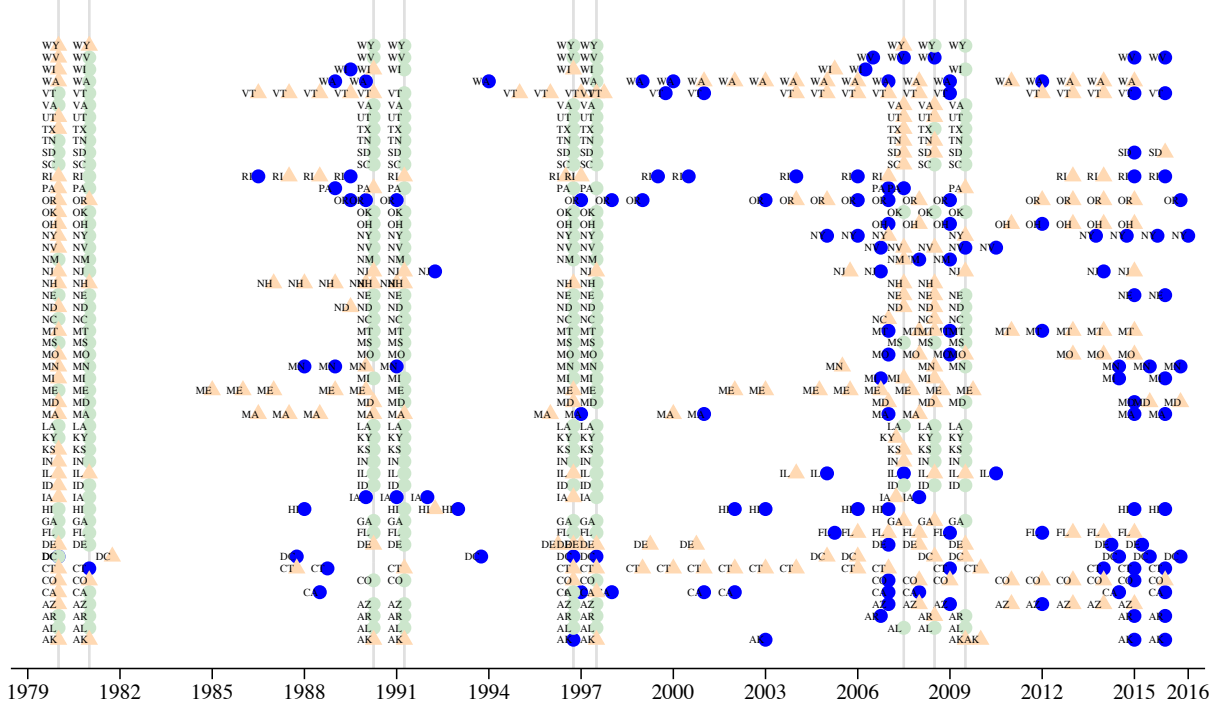
Table A.9 shows estimates for the event-by-event analysis presented in Figure V using alternative specifications. The estimated relationship between the Kaitz index on the jobs below, on the missing jobs, on the excess jobs, and on the employment change are similar across various specifications, which underlines the robustness of the results presented in Figure V.

Table A.10 shows the estimated employment elasticities using our event-based approach, as well as distributed lag specifications in log minimum wage (with 4 years of lags, contemporaneous, and 3 years of leads) estimated in both TWFE-logMW and in first differences (FD) specifications (see the details in Appendix G). We report employment estimates on aggregate employment in (columns 1, 2 and 5) and employment under \$15 (columns 3, 4 and 6) in Panel A. There is a wide range of estimates for aggregate employment, as we pointed out in Figure VI. When we exclude employment variation in the upper tail and focus on employment in jobs under \$15, the range of estimates narrows considerably. For example, for the weighted estimates, the employment elasticity with respect to the minimum wage is -0.020 (s.e. 0.028) in the fixed effect specification, -0.005 (s.e. 0.019) in first difference specification, and 0.027 (s.e. 0.022) in the event-based specification. These estimates cannot be distinguished statistically from each other, or from zero. This highlights that variability in the estimates is mainly driven by variation in employment above \$15, which is unlikely to reflect the causal effect of the minimum wage. Column 6 estimates event-based regressions of the minimum wage on jobs below \$15. We refer to this specification as the “simpler method” in Section II.C. and we report the estimates in Column 7 of Table I. (The slight difference between Column 6 in Table A.10 and Column 7 in Table I is that the former is based on annual data while the latter is based on quarterly data.) Column 7 shows our baseline estimates where we estimate the effect of the minimum wage on job counts in each wage bin, calculate the missing and excess jobs and then add them up. Both the point estimates and the standard errors are very close to each other in the “simpler method” and in our baseline regressions.

Panel B of Table A.10 shows the TWFE-logMW, first difference (FD), and event-based (EB) regressions for teens (see the details in Appendix G). The variability in the estimates for teens is not driven by changes in employment in the upper tail. This is not surprising, since most teens earn below \$15, and so variation in the upper tail can only have limited impact on the estimates. Column 6 estimates event based regression of the minimum wage on jobs below \$15. Column 7 shows our baseline estimates where we estimate the effect of the minimum wage on job counts in each wage bin, calculate the missing and excess jobs and then add them up. The estimates with the “simpler

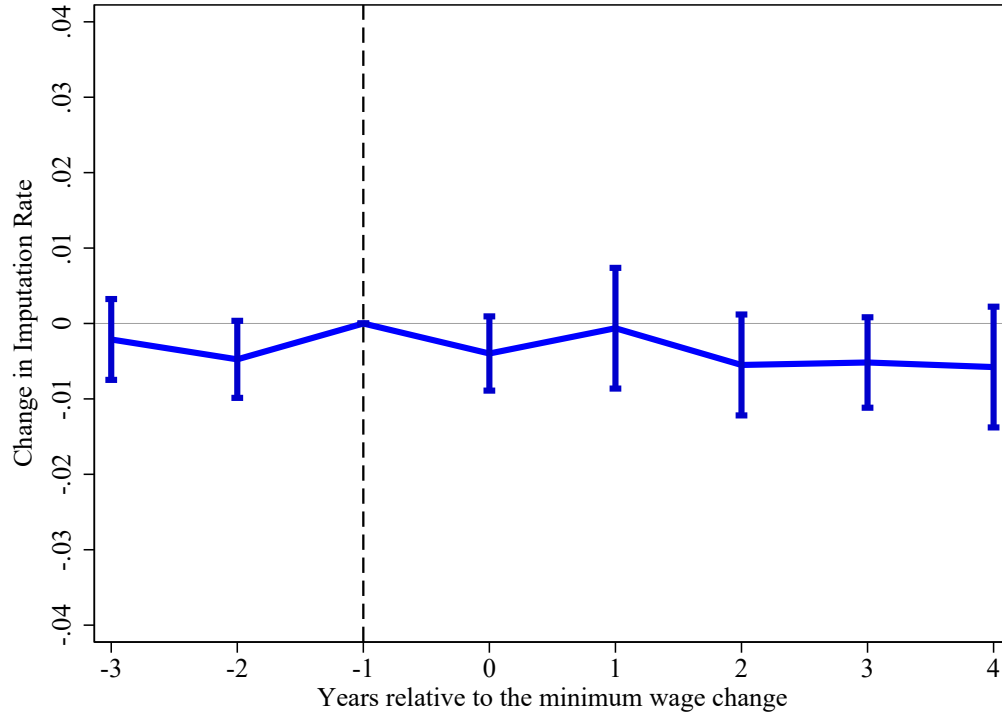
method” (column 6) and with our baseline method (column 7) are very similar. In general, we find that the teen estimates from fixed effects models tend to be more negative than the first difference ones—similar to Allegretto et al. (2017), and to the estimates for overall employment. Moreover, event-based estimates are much closer to those using first differencing, again mirroring the findings for overall employment. In [Appendix G](#), we show that the variability across specifications for teen (and aggregate) employment disappears in the post 1992 sample, and are driven by idiosyncratic shocks in the late 1980s and early 1990s—a period with few state level minimum wage changes.

Figure A.1: Minimum Wage Increases between 1979 and 2016



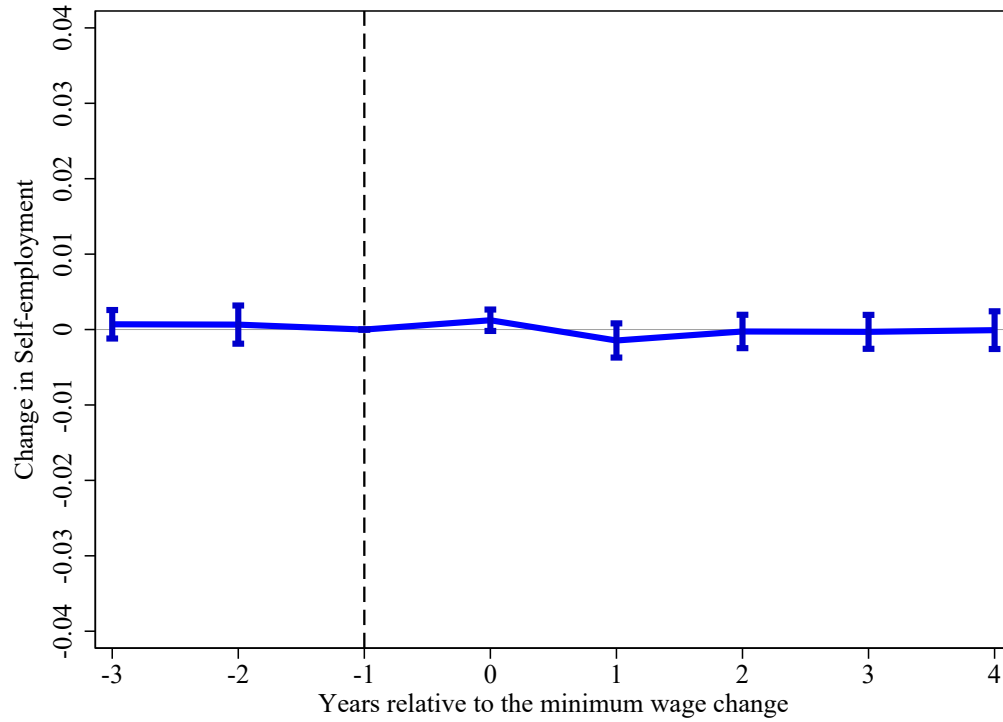
Notes: The figure shows all minimum wage increases between 1979 and 2016. There are a total of 623 minimum wage increases. The blue circles show the primary minimum wage events used in estimating equation 1; the light orange triangles highlight small minimum wage changes where minimum wage increased less than \$0.25 (the size of our wage bins) or where less than 2 percent of the workforce earned between the new and the old minimum wage. The green circles indicate federal changes, which we exclude from our primary sample of treatments because the change in missing number of jobs, Δb , is identified only from time-series variation for these events as there are no “control states” with wage floors lower than the new minimum wage (see the text for details).

Figure A.2: Impact of Minimum Wages on the Imputation Rate



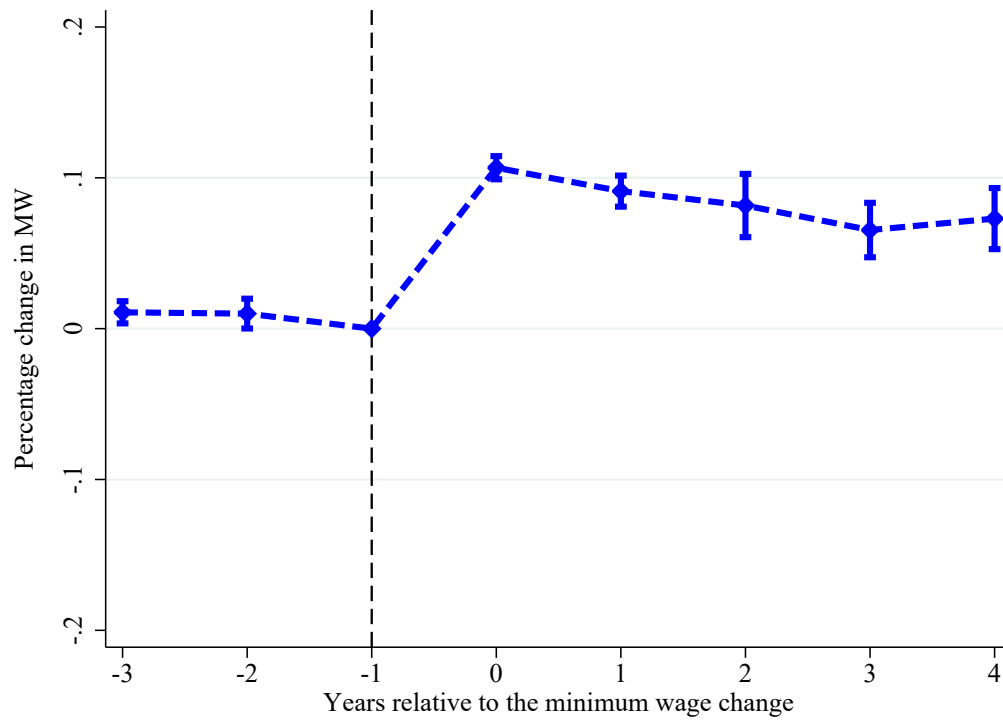
Notes: The figure shows the effect of the minimum wage on the imputation rate. In our event study analysis we only use non-imputed hourly wages. To alleviate the concern that imputation has an effect on our estimates, we implement an event study regression where the outcome variable is state-level imputation rate. Events are the same 138 state-level minimum wage changes between 1979-2016 that we use in our benchmark specification. Similarly to our benchmark specification we include state and time fixed effects in the regression. In the Appendix Table A.3 we report results with other specifications. The blue line shows the evolution of the imputation rate (relative to the year before the treatment). We also show the 95% confidence interval based on standard errors that are clustered at the state level.

Figure A.3: Impact of Minimum Wages on the Self-Employment Rate



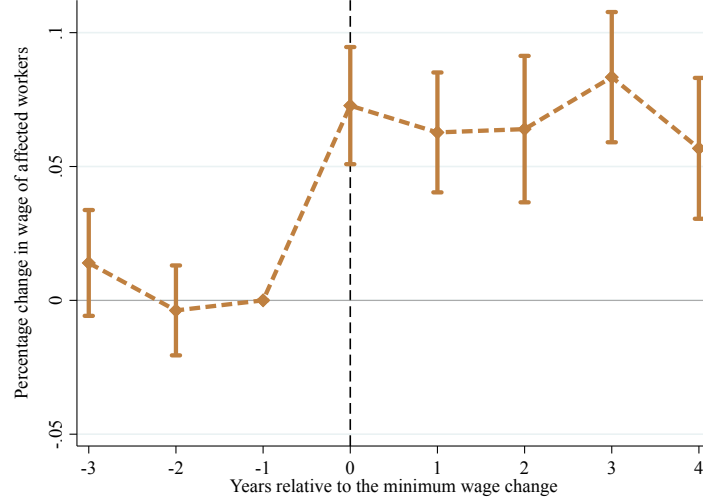
Notes: The figure shows the effect of the minimum wage on the self-employment rate. In our event study analysis we only use wage workers. To alleviate the concern that changes in self-employment rate have effects on our estimates, we implement an event study regression where the outcome variable is state-level self-employment rate. Events are the same 138 state-level minimum wage changes between 1979-2016 that we use in our benchmark specification. Similarly to our benchmark specification we include state and time fixed effects in the regression. The blue line shows the evolution of the self-employment rate (relative to the year before the treatment). We also show the 95% confidence interval based on standard errors that are clustered at the state level.

Figure A.4: Average Progression of Minimum Wages Around 138 Events

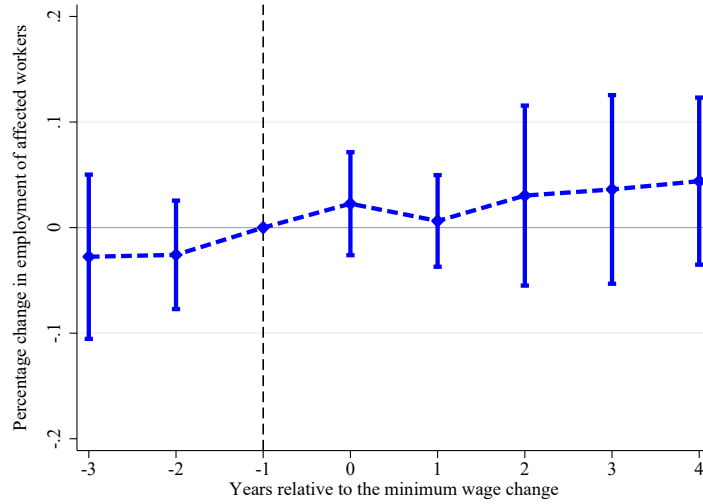


Notes: The figure shows the average increase of the minimum wage in the event window. Events are the 138 state-level minimum wage changes between 1979-2016 that we use in our benchmark specification. Similarly to our benchmark specification we include state and time fixed effects in the regression. The blue line shows the evolution of the minimum wage (relative to the year before the treatment) compared to the counterfactual. We also show the 95% confidence interval based on standard errors that are clustered at the state level.

Figure A.5: Impact of Minimum Wages on Average Wage and on Employment Over Time



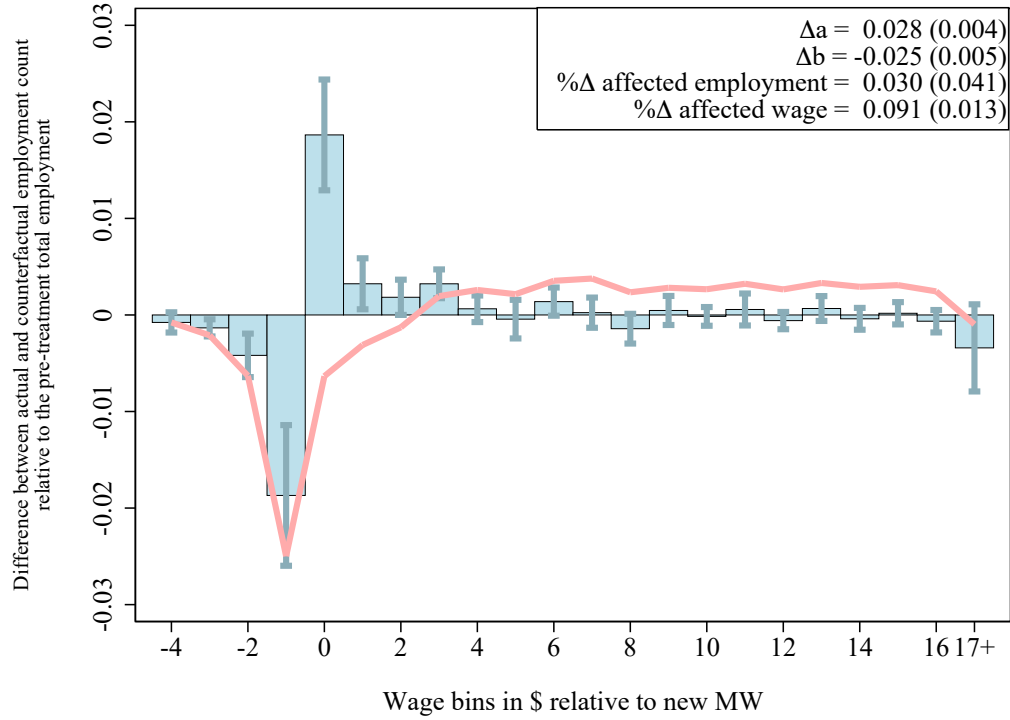
(a) Evolution of the average wage of the affected workers



(b) Evolution of the employment of the affected workers

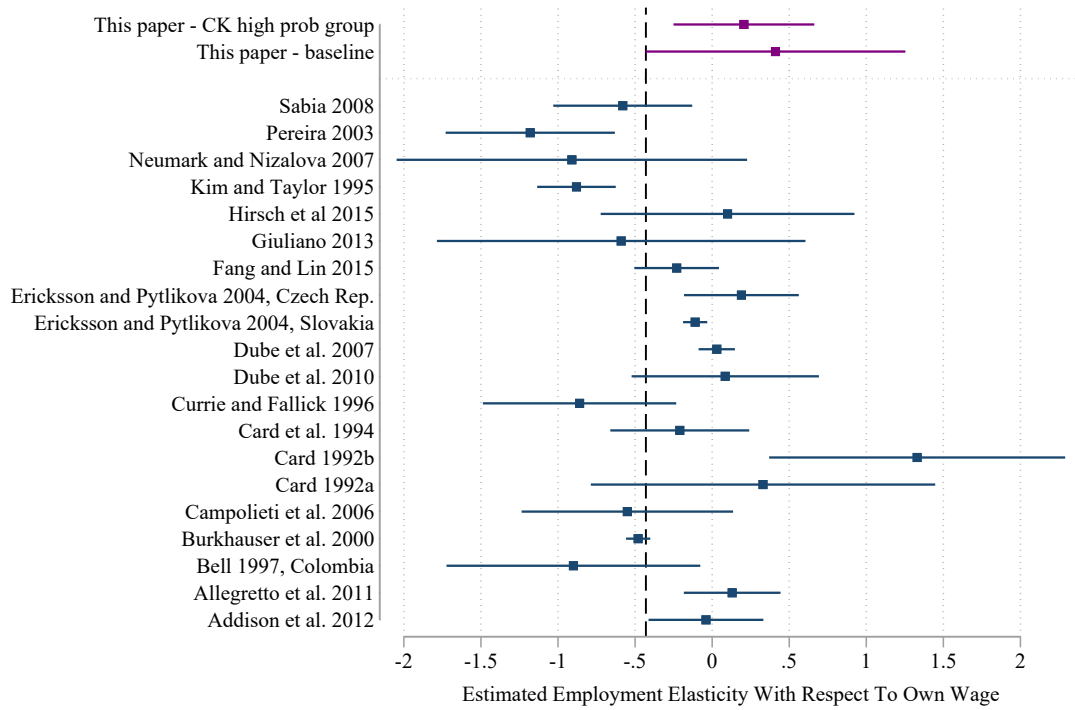
Notes: The figure shows the main results from our event study analysis (see equation 1) exploiting 138 state-level minimum wage changes between 1979-2016. Panel (a) shows the effect on the average wage over time, which is calculated using equation 2. Panel (b) shows the evolution of employment between \$4 below the new minimum wage and \$5 above it (relative to the total employment 1 year before the treatment), which is equal to the sum of missing jobs below and excess jobs at and slightly above the minimum wage, $\Delta b + \Delta a$. The figure highlights that minimum wage had a positive and significant effect on the average wage of the affected population, but there is no sign of significant disemployment effects.

Figure A.6: Change in Employment by Wage Bins after Aggregating Multiple Treatment Events



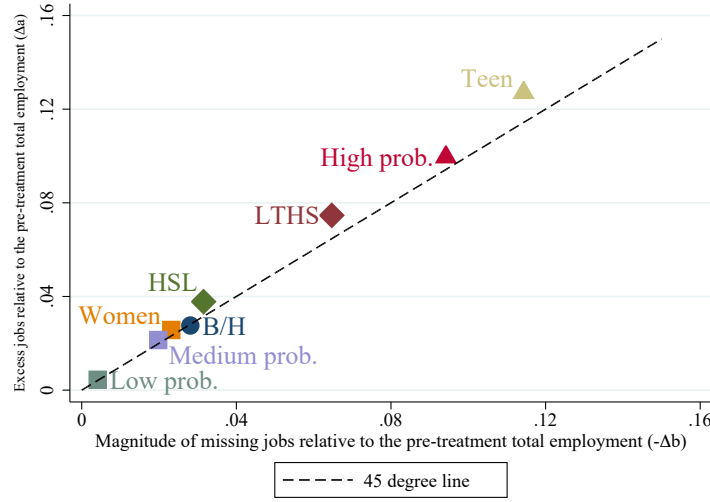
Notes: This figure replicates Figure II in the main text, but calculates a cumulative effect when there are multiple events in the 5-year post-treatment window. Overall, 65% of the time, a primary minimum wage increase is followed by a secondary one within 5 years, on average at \$0.56 above the minimum for the primary event. Figure II shows the partial effect of each event. Here we show the cumulative effect of all events within a 5-year post-treatment window by taking into account the incidence and size of secondary increases averaged across our sample of events. The blue bars show for each dollar bin (relative to the minimum wage) the estimated average employment changes in that bin during the 5-year post-treatment relative to the total employment in the state one year before the treatment. The red line is the running sum of the bin-specific impacts. Adjusting for multiple events increases the estimate for missing jobs below the new minimum from 1.8% to 2.5%. Therefore, some of the difference between jobs below the new minimum wage, which is around 8.6%, and the missing jobs below the new minimum wage can be explained by multiple events following each other.

Figure A.7: Employment Elasticity with Respect to Own Wage in the Literature and in this Paper

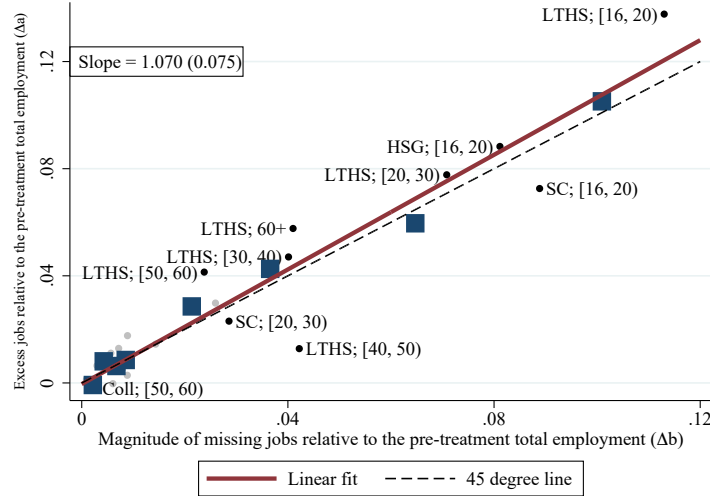


Notes: This figure summarizes the estimated employment elasticity with respect to wage and compares it to the previous estimates in the literature. The estimates in the literature are collected by Harasztosi and Lindner (forthcoming). The two estimates from our paper is the benchmark estimate on overall employment (Column 1 in Table 1) and the estimates for the Card and Krueger high probability group Column 6 in Table II. The dashed vertical line shows the lower bound of our benchmark estimates. The benchmark estimates can rule out 7 out of the 11 negative estimates provided in the previous literature.

Figure A.8: Impact of the Minimum Wage by Demographic Groups



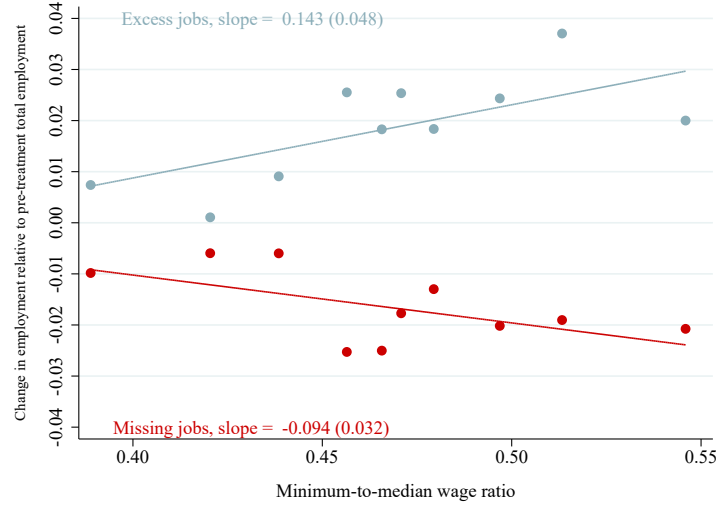
(a) Effect of the minimum wage by demographic groups



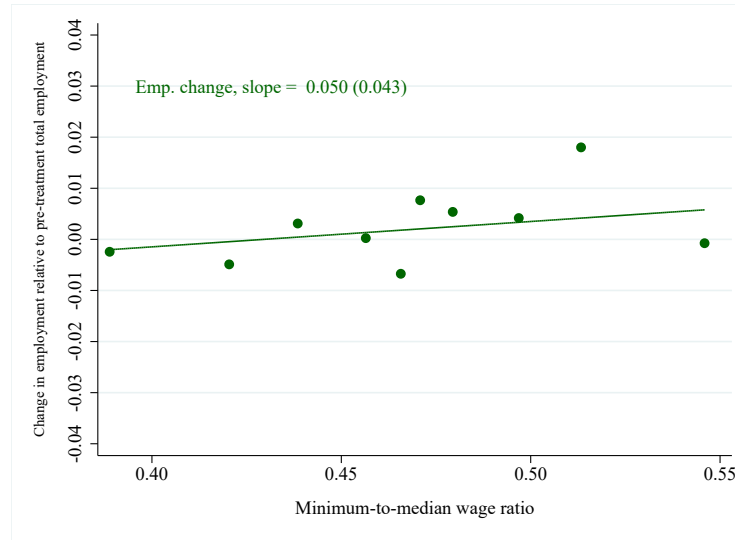
(b) Effect of the minimum wage by age-education groups

Notes: Both figures show the excess jobs (relative to the pre-treatment total employment in that group) above the new minimum wage (Δa) and magnitude of missing jobs below it ($-\Delta b$) for various demographic groups. The black dash line in both of the graphs are the 45 degree line indicating the locus of points where the excess number of jobs above and the missing jobs below the new minimum wage are exactly the same, and so the employment effect is zero. Estimates above that line indicate positive employment effects, and estimates below the line indicate negative ones. Panel (a) shows the estimates for demographic groups in Table II: those with less than high school (LTHS) education, high school or less (HSL) education, women, teen, black or Hispanic workers (B/H), and groups with low, medium and high probability of being exposed to the minimum wage increase. Panel (b) shows the estimates for education-by-age groups generated from 6 age and 4 education categories. The small light gray and black points correspond to each of the groups, while the large blue squares show the non-parametric bin scattered relationship between the excess jobs (Δa) and the magnitude of missing jobs ($-\Delta b$). The red line shows the linear fit. A slope of that line below one would indicate the presence of labor-labor substitution across age and education groups.

Figure A.9: Relationship between Excess Jobs, Missing jobs, Employment Change and the Minimum-to-Median Wage Ratio Across Events (Replicating Figure V in the Main Text without using Controls)



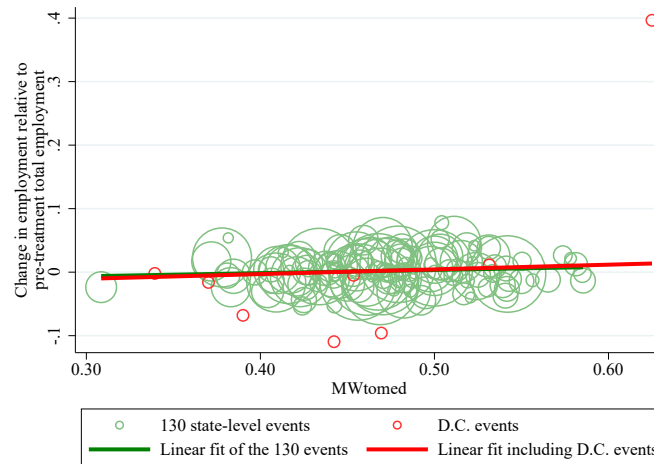
(a) Missing and excess jobs



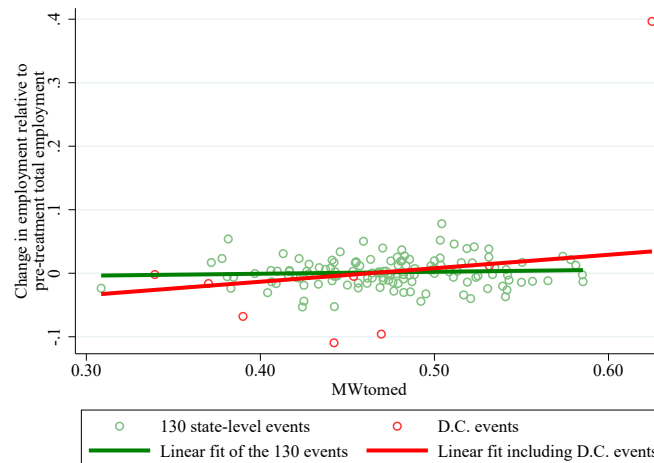
(b) Employment change

Notes: This figure replicates Figure V in the main text without using controls in the regression. The figure shows the binned scatter plots for missing jobs, excess jobs, and total employment changes by value of the minimum-to-median wage ratio (Kaitz index) for the 130 event-specific estimates. The minimum-to-median wage ratio is the new minimum wage MW divided by the median wage at the time of the minimum wage increase (Kaitz index). The 130 events exclude 8 minimum wage raising events in the District of Columbia, since those events are very noisily estimated in the CPS. The bin scatters and linear fits plot the relationship without any control variables. Estimates are weighted by the state populations. The slope (and robust standard error in parentheses) is from the weighted linear fit of the outcome on the minimum-to-median wage ratio.

Figure A.10: Relationship between Employment Change and the Minimum-to-Median Wage Ratio Across Events, Scatterplot



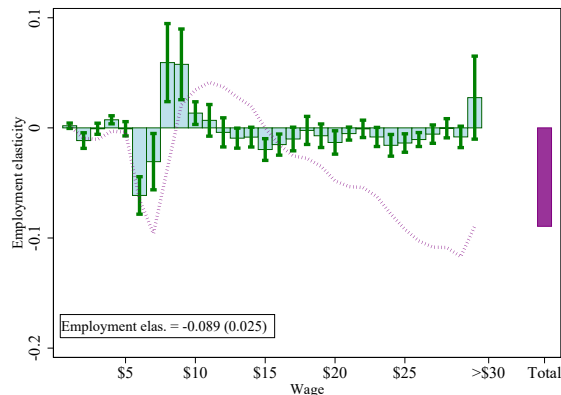
(a) Population weighted



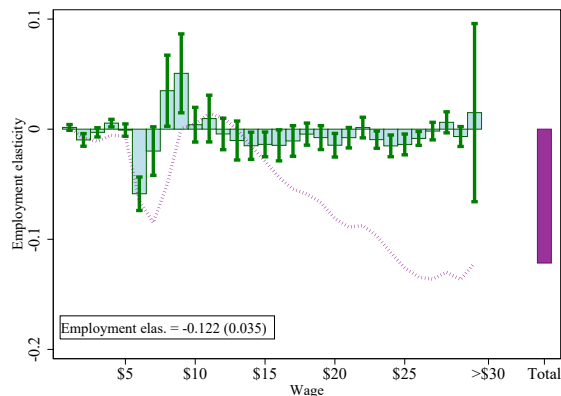
(a) Unweighted

Notes: The figure shows the population weighted and unweighted scatter plots of the estimated percentage change in employment in $[MW - \$4, MW + \$5)$ bins of each of the 138 events during the 5-year post-treatment relative to the 1-year pre-treatment period against the minimum-to-median wage ratio. The estimated employment change of each event is created from 138 regressions corresponding to each event, as explained in Section III.C.. The red circles indicate D.C. events, and the green circles the remaining 130 events. The lines are linear fits. The green line employs the 130 events; while the red one all events.

Figure A.11: Impact on Employment throughout the Wage Distribution in the Two-Way Fixed Effects Model on log Minimum Wages - Weighted and Unweighted Estimates



(a) Weighted



(b) Unweighted

Notes: The figure shows the effect of the minimum wage on the wage distribution using fixed effects specifications (TWFE-logMW), with and without population weights. Both panels estimate two-way (state-bin and year) fixed effects regressions on contemporaneous as well as 2 annual leads, and 4 annual lags of log minimum wage (panel (a) is the same as Figure VI in the main text). For each wage bin we run a separate regression, where the outcome is the number of jobs per capita in that state-wage bin. The cumulative response for each event date 0, 1, ..., 4 is formed by successively adding the coefficients for the contemporaneous and lagged log minimum wages. The green bars show the mean of these cumulative responses for event dates 0, 1, ..., 4, divided by the sample average employment-to-population rate—and represents the average elasticity of employment in each wage bin with respect to the minimum wage in the post-treatment period. The 95% confidence intervals around the point estimates are calculated using clustered standard errors at the state level. The dashed purple line plots the running sum of the employment effects of the minimum wage up until the particular wage bin. The rightmost purple bar in each of the graphs is the elasticity of the overall state employment-to-population rate with respect to minimum wage, obtained from regressions where outcome variables are the state level employment-to-population rate. In the bottom left corner we also report the point estimate on this elasticity with standard errors that are clustered at the state level. Regressions in panels (a) are weighted by state population; whereas the ones in panels (b) are not weighted.

Table A.1: T-statistics for the Wage Effects of the Minimum Wage - Bunching and Aggregate Approaches

	EB-bunching (1)	EB-aggregate (2)
All workers	6.942	0.577
Less than high school	5.526	1.359
High school or less	5.487	0.549
Teens	4.603	4.965
Women	6.261	0.796
Black or Hispanic	3.585	0.584
High prob. group	6.822	3.003
Middle group	3.973	1.140
Low prob. group	1.866	−0.136

Notes. Each cell reports the t-statistic from the estimated wage effect with respect to the minimum wage for various demographic groups. The (local) bunching approach is the preferred specification in this paper, estimating the wage effect from bin-specific employment changes near the relevant minimum wage. The aggregate approach uses as the outcome overall aggregate employment. For the local case, the wage effect is the estimated percentage change of affected workers. For the aggregate estimator, the wage effect is the elasticity of the wage with respect to the minimum wage. Regressions are weighted by state averaged population of the demographic groups. T-statistics are obtained by dividing the estimated wage effects by robust standard errors clustered by state.

Table A.2: Precision of the Employment Elasticities with Respect to the Minimum Wage - Bunching and Aggregate Approaches

	EB-bunching (1)	EB-aggregate (2)	Ratio of bunching to aggregated standard errors (3)
All workers	0.024 (0.025)	0.016 (0.029)	0.878
Less than high school	0.097 (0.061)	0.178* (0.094)	0.654
High school or less	0.061 (0.042)	0.041 (0.055)	0.756
Teens	0.125 (0.134)	0.128 (0.132)	1.011
Women	0.025 (0.027)	−0.006 (0.033)	0.825
Black or Hispanic	−0.005 (0.058)	−0.004 (0.082)	0.716
High prob. group	0.052 (0.062)	0.081 (0.071)	0.876
Middle group	0.016 (0.049)	0.057* (0.034)	1.443
Low prob. group	0.003 (0.014)	0.001 (0.026)	0.558

Notes. Columns 1-2 report the separately estimated employment elasticity with respect to the minimum wage for the bunching and aggregate approaches, for various demographic groups. Column 3 reports the ratio of the local to aggregate approach standard errors. The bunching approach is the preferred specification in this paper, using wage-bin-specific employment per capita changes as the outcome. The aggregate approach uses overall employment per-capita as the outcome. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table A.3: Impact of Minimum Wages on the Imputation Rate in Various Regression Specifications

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ imputation rate	-0.000 (0.004)	0.001 (0.004)	0.001 (0.004)	0.002 (0.003)	-0.004 (0.003)	-0.002 (0.003)	-0.002 (0.003)	-0.001 (0.003)
# observations	7,242	7,242	7,242	7,242	7,242	7,242	7,242	7,242
Mean of the dep. var	0.249	0.249	0.249	0.249	0.280	0.280	0.280	0.280
<i>Controls</i>								
State trends		Y		Y		Y		Y
Division-by-year FE			Y	Y			Y	Y
Weighted					Y	Y	Y	Y

Notes. The table reports 5-year averaged change in the imputation rate of the CPS from 1979 to 2016 after the primary 138 events. The dependent variable is the imputation rate, defined as the number of imputed observations divided by the number of employed observations. The estimates are calculated by employing an event based approach, where we regress state imputation rates on quarterly leads and lags on treatment spanning 12 quarters before and 19 quarters after the policy change. All specifications include state, and quarter fixed effects. Columns 2, 4, 6, and 8 controls for state linear trends; whereas columns 3, 4, 7, and 8 allow census divisions to be affected differently by macroeconomic shocks. The regressions are not weighted in columns 1-4; and they are population weighted in columns 5-8. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table A.4: Robustness of the Impact of Minimum Wages to Alternative Workforce, Treatment and Sample Definitions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Missing jobs below new MW (Δb)	-0.027*** (0.003)	-0.020*** (0.003)	-0.013*** (0.003)	-0.033*** (0.008)	-0.016*** (0.004)	-0.018*** (0.004)	-0.017*** (0.003)	-0.019*** (0.004)	-0.016*** (0.004)
Excess jobs above new MW (Δa)	0.026*** (0.002)	0.019*** (0.003)	0.016*** (0.003)	0.036*** (0.007)	0.017*** (0.003)	0.022*** (0.003)	0.019*** (0.002)	0.020*** (0.003)	0.019*** (0.002)
% Δ affected wages	0.065*** (0.007)	0.067*** (0.012)	0.073*** (0.012)	0.094*** (0.020)	0.082*** (0.014)	0.077*** (0.011)	0.070*** (0.010)	0.058*** (0.011)	0.069*** (0.010)
% Δ affected employment	-0.009 (0.034)	-0.010 (0.021)	0.044 (0.033)	0.029 (0.035)	0.028 (0.039)	0.046 (0.042)	0.028 (0.030)	0.007 (0.029)	0.028 (0.030)
Employment elasticity w.r.t. MW	-0.010 (0.036)	-0.009 (0.019)	0.029 (0.022)	0.029 (0.035)	0.017 (0.024)	0.039 (0.036)	0.022 (0.024)	0.006 (0.026)	0.023 (0.026)
Emp. elasticity w.r.t. affected wage	-0.139 (0.530)	-0.157 (0.326)	0.601 (0.442)	0.306 (0.392)	0.337 (0.496)	0.590 (0.536)	0.401 (0.418)	0.122 (0.495)	0.401 (0.447)
Jobs below new MW ($\bar{b}_{-1}$)	0.099	0.083	0.067	0.104	0.061	0.087	0.079	0.087	0.086
% Δ MW	0.093	0.096	0.101	0.101	0.101	0.101	0.100	0.097	0.101
Number of events	44	369	138	138	138	138	138	116	138
Number of observations	847,314	847,314	847,314	847,314	847,314	847,314	847,314	531,063	847,314
Number of workers in the sample	4,694,104	4,694,104	4,561,684	2,824,287	4,402,488	4,694,104	4,694,104	2,503,803	4,694,104
Set of events	No tip credit states	State & Federal	Primary	Primary	Primary	Primary	Primary	Primary	Primary, treatment in 25-cent increments
Sample	All workers	All workers	FTE	Hourly workers	Non-tipped occupations	CPS-Raw	Unweighted	All workers, post-1992	All workers

Notes. The table reports robustness checks for the effects of a minimum wage increase based on the event study analysis (see equation 1) exploiting minimum wage changes between 1979 and 2016. All columns except column (2) are based on state-level minimum wage changes. The table reports five year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment and wages. Column (1) reports estimates for the 44 events which occurred in states that do not allow tip credit. Column (2) reports estimates using 369 state or federal minimum wage increases. Column (3) uses full time equivalent job counts and so takes changes in hours worked into account. Column (4) uses workers who directly reported being hourly workers in the survey. Column (5) uses workers in non-tipped occupations only. Column (6) does not use the QCEW benchmarking, and instead reports the estimates calculated using the raw CPS counts (see Section 4.2 for details). All regressions are weighted by state-quarter aggregated population except Column (7), where we report unweighted estimates. Column (8) only considers minimum wage events that happened on or before 2012q1 to ensure a full five year post-treatment period. Column (9) shows the results for the post-1992 sample. Column (10) defines treatment indicators in 25 cent increments. All specifications include wage bin-by-state and wage bin-by period fixed effects. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Line-by-line description. The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pre-treatment total employment. The third row, the percentage change in average wages in the affected bins, (% ΔW), is calculated using equation 2 in Section 2.2. The fourth row, percentage change in employment in the affected bins is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{b_{-1}}$). The fifth row, employment elasticity with respect to the minimum wage is calculated as $\frac{\Delta a + \Delta b}{\% \Delta MW}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{b_{-1}}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

Table A.5: Impact of Minimum Wage Increase on the Average Wage and Employment of Affected workers - Robustness to Alternative Wage Windows

	Alternative wage window				
	(1)	(2)	(3)	(4)	(5)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.018*** (0.004)	-0.018*** (0.004)	-0.018*** (0.004)	-0.018*** (0.004)
Excess jobs above new MW (Δa)	0.018*** (0.003)	0.021*** (0.002)	0.021*** (0.003)	0.020*** (0.003)	0.021*** (0.002)
% Δ affected wages	0.046*** (0.009)	0.064*** (0.008)	0.068*** (0.010)	0.068*** (0.013)	0.081*** (0.012)
% Δ affected employment	-0.002 (0.025)	0.029 (0.031)	0.028 (0.029)	0.024 (0.031)	0.033 (0.034)
Employment elasticity w.r.t. MW	-0.001 (0.021)	0.025 (0.027)	0.024 (0.025)	0.020 (0.026)	0.028 (0.029)
Emp. elasticity w.r.t. affected wage	-0.038 (0.539)	0.452 (0.479)	0.411 (0.430)	0.349 (0.443)	0.410 (0.390)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.086	0.086	0.086	0.086
% Δ MW	0.101	0.101	0.101	0.101	0.101
Number of event	138	138	138	138	138
Number of observations	847,314	847,314	847,314	847,314	847,314
Number of workers in the sample	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104
Upper endpoint of wage window ($\bar{W}$):	MW+\$2	MW+\$3	MW+\$4	MW+\$5	MW+\$6

Notes. The table reports the effects of a minimum wage increase based on the event study analysis (see equation 1) exploiting 138 state-level minimum wage changes between 1979-2016. The table reports five year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs, employment and wages. The different columns explore the robustness of the results to alternative upper end points, $\bar{W}$, for calculating excess jobs. The first column limits the range of the wage window by setting the upper limit for calculating the excess jobs to $\bar{W} = \$2$, and the last column expands it until $\bar{W} = \$6$. All specifications include wage bin-by-state and wage bin-by period fixed effects. Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Line-by-line description. The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pre-treatment total employment. The third row, the percentage change in average wages in the affected bins, (% ΔW), is calculated using equation 2 in Section 2.2. The fourth row, percentage change in employment in the affected bins is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{\bar{b}_{-1}}$). The fifth row, employment elasticity with respect to the minimum wage is calculated as $\frac{\Delta a + \Delta b}{\% \Delta MW}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{\bar{b}_{-1}}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

Table A.6: Impact of Minimum Wage Increase on the Average Wage and Employment of Affected workers - Robustness to Alternative Time Windows

	(1)	(2)	(3)	(4)	(5)	(6)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.021*** (0.004)	-0.018*** (0.002)	-0.018*** (0.002)	-0.018*** (0.003)	-0.021*** (0.004)
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.021*** (0.003)	0.019*** (0.002)	0.021*** (0.002)	0.019*** (0.003)	0.020*** (0.003)
% Δ affected wages	0.068*** (0.010)	0.065*** (0.010)	0.064*** (0.009)	0.068*** (0.009)	0.067*** (0.009)	0.066*** (0.009)
% Δ affected employment	0.028 (0.029)	0.010 (0.025)	0.022 (0.031)	0.029 (0.032)	0.013 (0.029)	-0.011 (0.022)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.008 (0.021)	0.018 (0.027)	0.025 (0.027)	0.011 (0.025)	-0.009 (0.019)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.148 (0.380)	0.335 (0.461)	0.427 (0.445)	0.197 (0.436)	-0.163 (0.335)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.086	0.086	0.086	0.086	0.086
% Δ MW	0.101	0.101	0.101	0.101	0.101	0.101
Number of events	138	138	138	138	138	138
Number of observations	847,314	847,314	847,314	847,314	847,314	847,314
Number of workers in the sample	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104
Time window	[-3, 4]	[-3, 2]	[-3, 6]	[-5, 4]	[-1, 4]	[-1, 1]

Notes. The table reports the effects of a minimum wage increase based on the event study analysis (see equation 1) exploiting 138 state-level minimum wage changes between 1979-2016. The table reports averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment and wages. The different columns explore the robustness of the results to alternative time windows. The first column reproduces our baseline estimate in Table 1 column 1. Compared to the baseline specification, columns 2 and 3 change the post-treatment period to 2 and 6 years, respectively. Similarly, in columns 4 and 5, we start the pre-treatment window from 5 years and one year prior to the event. All specifications include wage-bin-by-state and wage-bin-by-period fixed effects. Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Line-by-line description. The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pre-treatment total employment. The third row, the percentage change in average wages in the affected bins, (% ΔW), is calculated using equation 2 in Section 2.2. The fourth row, percentage change in employment in the affected bins is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{b_{-1}}$). The fifth row, employment elasticity with respect to the minimum wage is calculated as $\frac{\Delta a + \Delta b}{\% \Delta MW}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{b_{-1}}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

Table A.7: Impact of Minimum Wages on Employment and Wages for Card and Krueger Probability Groups - Bunching and Aggregate approaches

	(1)	(2)	(3)
Panel A: Aggregate			
%Δ average wage	0.020*** (0.007)	0.007 (0.006)	-0.001 (0.004)
%Δ employment	0.008 (0.009)	0.006 (0.005)	0.000 (0.002)
Employment elasticity wrt wage	0.435 (0.371)	N/A	N/A
Panel B: Bunching			
%Δ affected wages	0.073*** (0.011)	0.051*** (0.013)	0.060* (0.032)
%Δ affected employment	0.015 (0.018)	0.015 (0.048)	0.011 (0.055)
Emp. elasticity w.r.t. affected wage	0.206 (0.233)	0.304 (0.904)	0.184 (0.841)
Jobs below new MW ($\bar{b}_{-1}$)	0.358	0.104	0.027
%Δ MW	0.102	0.102	0.101
Number of events	138	138	138
Number of observations	847,314	847,314	847,314
Group:	High prob.	Middle prob.	Low prob.

Notes. The table reports the wage and employment elasticities with respect to the minimum wage for the high, middle, and the low probability groups using the Card and Krueger predictive model of exposure to minimum wage changes. Both panels A and B are based on the 138 state level events and an event-based approach with five year post-treatment period. Panel A reports the estimates for aggregate employment and wages for the three groups. Panel B reports the estimated employment and wage effect for affected workers using the bunching approach. Regressions are weighted by state averaged population of the relevant demographic group. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table A.8: Impact of Minimum Wage Increase by Pre-Treatment Employment Status: New Entrants and Incumbents

	Matched CPS				
	(1)	(2)	(3)	(4)	(5)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.023*** (0.004)	-0.018*** (0.003)	-0.012*** (0.002)	-0.005*** (0.001)
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.025*** (0.004)	0.018*** (0.002)	0.013*** (0.002)	0.006*** (0.001)
% Δ affected wages	0.068*** (0.010)	0.073*** (0.011)	0.059*** (0.013)	0.095*** (0.020)	0.019 (0.013)
% Δ affected employment	0.028 (0.029)	0.023 (0.024)	0.009 (0.046)	0.009 (0.068)	0.008 (0.034)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.019 (0.021)	0.006 (0.032)	0.003 (0.026)	0.003 (0.011)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.311 (0.320)	0.145 (0.747)	0.094 (0.704)	0.431 (1.682)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.086	0.072	0.042	0.384
% Δ MW	0.101	0.101	0.103	0.103	0.103
Number of events	138	138	137	137	137
Number of observations	847,314	847,314	733,941	733,941	733,941
Number of workers in the sample	4,694,104	4,694,104	1,505,192	1,373,696	131,496
Sample:	All workers	All workers	All matched workers	Incumbents	New entrants
Time window:	5 years	1 year	1 year	1 year	1 year

Notes. The table reports 1 year post-treatment estimates of employment and wages of the affected bins for all workers (incumbents and new entrants) using state-quarter-wage bin aggregated CPS data from 1979-2016, and matched CPS data from 1980-2016. Incumbent workers are employed in the 4th interview month of CPS, and new entrants are not employed in the 4th interview month. The first column replicates column 1 in Table I for comparability. The second column includes all workers in the primary CPS sample and employs the baseline specification, but reports only the first year effects. The third and fourth columns use matched CPS and consider only the first year effects on incumbent, and new-entrant workers. Specifications include wage bin-by-state, wage bin-by period, and state-by-period fixed effects. Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Line-by-line description. The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pre-treatment total employment. The third row, the percentage change in average wages in the affected bins, (% ΔW), is calculated using equation 2 in Section 2.2. The fourth row, percentage change in employment in the affected bins is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{\bar{b}_{-1}}$). The fifth row, employment elasticity with respect to the minimum wage is calculated as $\frac{\Delta a + \Delta b}{\% \Delta MW}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{\bar{b}_{-1}}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

Table A.9: Robustness of the Relationship Between Employment Changes and the Minimum-to-Median Wage Ratio (Kaitz Index) Across Events

	Jobs below new MW ($\bar{b}_1$)		Missing jobs (Δb)		Excess jobs (Δa)		Employment change ($\Delta a + \Delta b$)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Main estimates								
Minimum-to-median ratio	0.314*** (0.063)	0.361*** (0.056)	-0.094*** (0.032)	-0.133*** (0.034)	0.143*** (0.048)	0.139** (0.057)	0.050 (0.043)	0.006 (0.048)
Panel B: With D.C.								
Minimum-to-median ratio	0.312*** (0.061)	0.358*** (0.055)	-0.075** (0.035)	-0.111*** (0.037)	0.149*** (0.048)	0.148** (0.057)	0.074 (0.049)	0.037 (0.055)
Panel C: Unweighted								
Minimum-to-median ratio	0.275*** (0.035)	0.286*** (0.035)	-0.112*** (0.024)	-0.128*** (0.026)	0.142*** (0.037)	0.134*** (0.041)	0.031 (0.038)	0.006 (0.042)
<i>Number of observations</i>								
Panels A, C	130	130	130	130	130	130	130	130
Panel B	138	138	138	138	138	138	138	138
Controls	Y		Y		Y		Y	

Notes. The table reports the effect of the minimum-to-median wage ratio (Kaitz index) on four outcomes: jobs below the new minimum wage, missing jobs, excess jobs, and the total employment change. The minimum-to-median wage ratio is the new minimum wage divided by the state-level median wage. Odd columns reports simple linear regression estimates. Even columns include the controls in Table A.9. Regressions are weighted by state-populations. Robust standard errors are in parentheses; significance levels are * 0.10, ** 0.05, *** 0.01.

Table A.10: Employment Elasticities of Minimum Wage from Alternative Approaches

	Continuous treatment - ln(MW)				Event based		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Fixed Effects	First Difference	Fixed Effects	First Difference			
Panel A: Overall							
Employment elasticity wrt MW	-0.089*** (0.025)	0.027 (0.031)	-0.020 (0.028)	-0.005 (0.019)	0.016 (0.029)	0.027 (0.022)	0.024 (0.025)
Panel B: Teen							
Employment elasticity wrt MW	-0.238*** (0.088)	0.094 (0.122)	-0.210** (0.091)	0.080 (0.120)	0.163 (0.115)	0.152 (0.107)	0.125 (0.134)
Aggregate Under \$15 [MW-\$4, MW + \$5)	Y	Y	Y	Y	Y	Y	Y
Data aggregation	State- year	State- year	State- year	State- year	State- year	State- year	Wage-bin- state- quarter

Notes. The table reports estimated overall (panel A) and teen (panel B) employment elasticities of minimum wage from alternative approaches. All columns show average post-treatment elasticities calculated from regressions of state-level employment to population rate on contemporaneous and 4 annual lags and 2 annual leads of log minimum wages. We use state-by-year aggregated CPS data from 1979-2016. Columns (1) and (3) estimate two-way (state and year) fixed effect regressions, while in columns (2) and (4) we employ first differences. Column (3) and (4) exclude workers with hourly wages greater than \$15. Columns (5)-(7) report estimates employment elasticities using an event study framework where we exploit the same 138 events as in our benchmark specifications. In column (5), we use state by quarter aggregated CPS data. In column (6) we directly estimate the effect of the minimum wage on jobs below \$15. We refer to this specification as simpler method in Section 4.2., since it directly estimate the sum of missing and excess jobs. Finally, column (7) shows estimates from the local approach (same as in column 1 of Table I, and column 3 of Table 2). In all cases we show estimates with and without population weighting. Standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Appendix B

Upper Tail Employment Effects in the Neoclassical Model

In this Appendix, we quantitatively assess the plausible magnitudes of upper tail employment effects of a minimum wage increase in a competitive neoclassical model of the labor market.

Consider a three-factor economy where output is a function of low-skilled, minimum wage labor (L); higher skilled, non-minimum wage labor (H); and capital (K). The elasticity of substitution between high and low-skilled labor is σ_{HL} .

In this setup, the effect of a change in low-skilled wage, w^L , on higher skilled labor demand is given by the well-known formula Hicks-Marshall rule of derived demand:

$$\frac{\partial \ln L^H}{\partial \ln w^L} = s_L(\sigma_{HL} - \eta)$$

where s_L is the share of minimum-wage labor in total production, and η is the output demand elasticity.

In the United States, averaged over our sample (1979-2016), minimum wage workers' share of the wage bill was around 3%. During this same time, labor's share of output was roughly 2/3, which implies a low-skilled share of production of $s_L \approx 2\%$.

In terms of the the elasticity of substitution between high and low skilled workers, Katz and Murphy (1992) estimates $\sigma_{HL} \approx 1.4$. For output elasticity of demand in low-wage intensive sectors, η is often assumed to be 1 (Aaronson and French, 2007).

Overall, these parameter estimates imply a cross-wage elasticity of $\frac{\partial \ln L^H}{\partial \ln w^L} \approx 0.008$. To get a minimum wage elasticity, we note that we find that for a 10% increase in the minimum wage, hourly wages of affected workers increase by 7%, or $\frac{\partial \ln w^L}{\partial \ln MW} \approx 0.7$. Putting all of these estimates together implies a very small minimum wage elasticity for higher-skilled employment of $\frac{\partial \ln L^H}{\partial \ln MW} = \frac{\partial \ln L^H}{\partial \ln w^L} \times \frac{\partial \ln w^L}{\partial \ln MW} \approx 0.008 \times 0.7 = 0.0056$.

How sensitive are these estimates to reasonable variations in the key parameters? Here we vary σ_{HL} between 0.5 and 2, and η between 0.5 and 2. The table below shows that the relevant minimum wage elasticity for upper tail employment, $\frac{\partial \ln L^H}{\partial \ln MW}$, falls between -0.024 and 0.024. When the output elasticity exceeds the elasticity of substitution in magnitude, the upper tail effect is negative, as the scale effect dominates. When the elasticity of substitution is larger in magnitude, the effect on upper tail is positive as the substitution effect dominates. Either way, however, given the small output share of minimum wage workers, a plausible estimate of minimum wage impact on upper-tail employment should be quite small in the neoclassical model. Indeed, these bounds for the upper tail are smaller in magnitude than the standard error for the elasticity of minimum wages for aggregate employment and employment above \$15/hour, as shown in Table G.1.

We also empirically show the absence of an effect on the upper tail of the distribution using our event-based design in section III.

Table B.1: Sensitivity Analysis of the Neoclassical Model for the Upper Tail Employment

	$\eta = 0.5$	$\eta = 1$	$\eta = 1.5$	$\eta = 2$
$\sigma_{HL} = 0.5$	0	-0.008	-0.016	-0.024
$\sigma_{HL} = 1$	0.008	0	-0.008	-0.016
$\sigma_{HL} = 1.5$	0.016	0.008	0	-0.008
$\sigma_{HL} = 2$	0.024	0.016	0.008	0

Notes. The table shows predicted minimum wage elasticities for upper tail employment for alternative output elasticity of demand (η), and elasticity of substitution between high and low skilled workers (σ_{HL}) values.

Appendix C

Washington State Case Study

In this Appendix, we report estimates using administrative data on hourly wages for a case study of a large state-level minimum wage increase. The state of Washington increased its real hourly minimum wage by around 22% from \$7.51 to \$9.18 (in 2016 dollars) in two steps between 1999 and 2000. Moreover, this increase in the real minimum wage was persistent, since subsequent increases were automatically indexed to the rate of inflation. In addition to the size and permanence of this intervention, Washington is an attractive case study because it is one of the few states with high quality administrative data on hourly wages.⁴⁰ Using hourly wage data, we can easily calculate the actual post-reform wage distribution (blue line in Figure I). However, the key challenge is that we do not directly observe the wage distribution in the absence of the minimum wage increase (red line in Figure I). To overcome this challenge, the previous literature constructed the counterfactual by imposing strong parametric assumptions (Meyer and Wise 1983) or simply used the pre-reform wage distribution as a counterfactual (Harasztosi and Lindner (forthcoming)).⁴¹ Here we improve upon these research designs by implementing a difference-in-differences style estimator.

In particular, we discretize the wage distribution, and count per-capita employment for each dollar wage bin k . For example, the \$10 wage bin includes jobs paying between \$10 and \$10.99 in 2016\$. We normalize these counts by the pre-treatment employment-to-population rate in Washington,

$$e_{WA,k,Post} = \frac{1}{\frac{E_{WA,Pre}}{N_{WA,Pre}}} \frac{E_{WA,k,Post}}{N_{WA,Post}}$$

where $\frac{E_{WA,k,t}}{N_{WA,t}}$ is per-capita employment for each dollar wage bin k in state Washington at time t , and $N_{WA,t}$ is the size of the population. We use administrative data on hourly wages from Washington State to calculate $e_{WA,k,Post}$.

We calculate the post treatment counterfactual wage distribution for each wage bin, $e_{WA,k,Post}^{CF}$, by adding the (population-weighted) average per capita employment change in the 39 states that did not experience a minimum wage increase during the 1998-2004 time period to the Washington state's pre-treatment per-capita wage distribution. After the appropriate normalization, this leads to the following expression:

⁴⁰The state of Washington requires all employers, as part of the state's Unemployment Insurance (UI) payroll tax requirements, to report both the quarterly earnings and quarterly hours worked for all employees. The administrative data covers a near census of employee records from the state. One key advantage of the method proposed here is that there is no need for confidential or sensitive individual-level data for implementation. Instead, we rely here on micro-aggregated data on employment counts for 5-cent hourly wage bins. Workers with hourly wages greater than \$50 are censored for confidentiality purposes. We deflate wages to 2016 dollars using the CPI-U-RS.

⁴¹As shown in Dickens, Machin and Manning (1998), estimates using the Meyer and Wise (1983) approach is highly sensitive to the parameterization of the wage distribution.

$$e_{WA,k,Post}^{CF} = \underbrace{\frac{1}{\frac{E_{WA,Pre}}{N_{WA,Pre}}}}_{\text{normalization}} \times \underbrace{\left[\frac{E_{WA,k,Pre}}{N_{WA,Pre}} \right]}_{\substack{\text{Pre-treatment} \\ \text{in WA}}} + \underbrace{\sum_{s \in \text{Control}} \frac{1}{39} \left(\frac{E_{s,k,Post}}{N_{s,Post}} - \frac{E_{s,k,Pre}}{N_{s,Pre}} \right)}_{\substack{\text{Change in control} \\ \text{states}}}$$

where $\frac{E_{skt}}{N_{s,t}}$ is per-capita employment for each dollar wage bin k in state s at time t , and N_{st} is the size of the population (age 16 or over) in state s at time t . To calculate the third part of this expression, the change in control states, we use hourly wage data from the Outgoing Rotation Group of the Current Population Survey (CPS). We will discuss the data in more detail in Section II.C.. For the second part of the expression, the pre-treatment Washington wage distribution, we use administrative data on hourly wages. However, in Appendix Figure C.4 we show that when we use the CPS, we get very similar results. Finally, the first part of this expression, the normalization, is to express the counterfactual employment counts in terms of pre-treatment total employment in Washington. It is worth highlighting that our normalization does not force the area below the counterfactual wage distribution to be the same as the area below the actual wage distribution—in other words, the minimum wage can affect aggregate employment.

In Figure C.1, panel (a) we report the actual (blue filled bar) and the counterfactual (red empty bars) frequency distributions of wages, normalized by the pre-treatment total employment in Washington. We define the pre-treatment period as 1996-1998, and the post-treatment period as 2000-2004. The post-treatment actual wage distribution in Washington state (blue filled bars) shows that very few workers earn less than the mandated wage, and there is a large spike at the new minimum wage at \$9. The post-treatment counterfactual distribution differs considerably. That distribution indicates that in the absence of the minimum wage increase, there would have been more jobs in the \$7 and \$8 bins, but fewer jobs at the \$9 bin and above. Compared to the counterfactual wage distribution, the actual distribution is also elevated \$1 and \$2 above the minimum wage, which suggests that minimum wages induce some modest spillover effects. At the same time, the ripple effect of the minimum wage fades out above \$12, and no difference is found between the actual and counterfactual distribution above that point. Such a relationship between the actual and counterfactual distributions closely resembles the illustration of the shown in Figure I.

The difference between the actual, $e_{WA,k,Post}$, and the counterfactual, $e_{WA,k,Post}^{CF}$, frequency distributions of wages represents the causal effect of the minimum wage on the wage distribution. This difference can be expressed as:

$$\begin{aligned}
e_{WA,k,Post} - e_{WA,k,Post}^{CF} = & \underbrace{\frac{1}{\frac{E_{WA,Pre}}{N_{WA,Pre}}}}_{\text{normalization}} \times \left[\underbrace{\frac{E_{WA,k,Post}}{N_{WA,Post}} - \frac{E_{WA,k,Pre}}{N_{WA,Pre}}}_{\text{Change in treatment}} \right. \\
& \left. - \underbrace{\sum_{s \in \text{Control}} \frac{1}{39} \left(\frac{E_{WA,k,Post}}{N_{s,Post}} - \frac{E_{WA,k,Pre}}{N_{s,Pre}} \right)}_{\text{Change in control}} \right]
\end{aligned} \tag{C.5}$$

which is the classic difference-in-differences estimator underlying the core estimates in the paper. Standard errors are calculated using the procedure proposed by Ferman and Pinto (forthcoming). Appropriate for a single treated unit, their procedure extends the cluster residual bootstrap by correcting for sample-size based heteroskedasticity—an important issue given the very different sample sizes across states in the CPS, and because Washington is based on administrative data.

The blue bars in Panel (b) of Figure C.1 report the differences in job counts for each wage bin. The difference-in-differences estimate shows a clear drop in counts for wage bins just below the new minimum wage. In the upper part of the table we report our estimate of missing jobs, Δb , which is the sum of employment changes, $\sum_{k=\$5}^{\$8} e_{WA,k,Post} - e_{WA,k,Post}^{CF}$, between \$5 and \$8—i.e., under the new minimum wage. These missing jobs paying below \$9 represent around 4.6% of the aggregate pre-treatment Washington employment. We also calculate the number of excess jobs paying between \$9 and \$13, Δa , which is equal to $\sum_{k=\$9}^{\$13} e_{WA,k,Post} - e_{WA,k,Post}^{CF}$. The excess jobs represent around 5.4% of the aggregate pre-treatment Washington employment.

As we explained in the previous section, the effect of the minimum wage on low-wage jobs is equal to the sum of the missing jobs below and the excess jobs above the new minimum wage of \$9. We find that the net employment change is positive—the increase amounted to 0.8% of the pre-treatment aggregate employment in Washington. This reflects a 6.1% (s.e. 10.9%) increase in employment for the workers who earned below the new minimum wage in 1998. We also find that average wages of affected workers at the bottom of the wage distribution increased by around 9.0% (s.e. 18.8%) Coming from a single case study, the precision of these estimates is much lower than in the pooled event study estimates presented in the main paper.

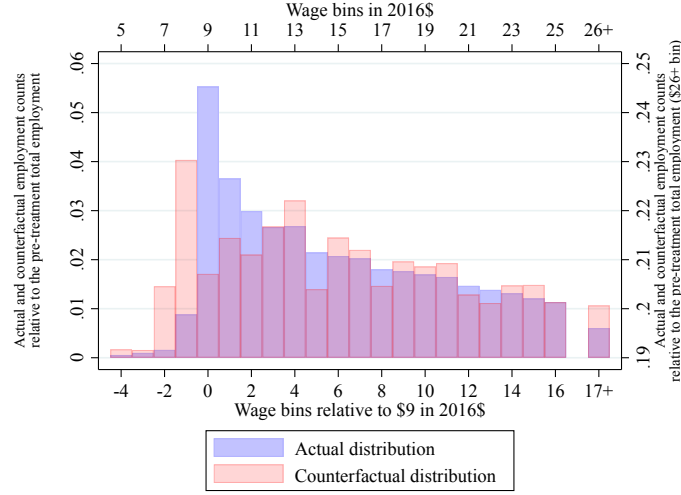
In Panel (b) of Figure 2, the red line shows the running sum of employment changes up to each wage bin. The running sum drops to a sizable, negative value just below the new minimum wage, but returns to around zero once the minimum wage is reached. By around \$2 above the minimum wage, the running sum reaches a small positive value and remains flat thereafter—indicating little change in upper tail employment. This strengthens the case for a causal interpretation of these results.

Finally, we also explore the evolution of missing jobs (red line) and excess jobs (blue line) over time in panel (a) in Figure C.3. The figure shows that excess and missing jobs are close to zero

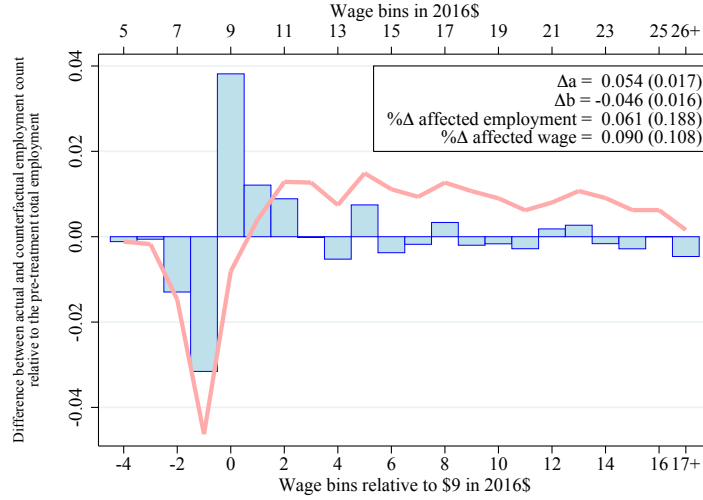
before 1999, and there are no systematic pre-existing trends.⁴² Once the minimum wage is raised in two steps between 1999 and 2000, there is a clear and sustained drop in jobs below the new minimum wage (relative to the counterfactual). Since the minimum wage is indexed to inflation in Washington, the persistence of the drop is not surprising. The evolution of excess jobs after 2000 closely matches the evolution of missing jobs. As a result, the net employment change—which is the sum of missing and excess jobs—is close to zero in all years following the minimum wage increase (see panel (b) in Figure C.3).

⁴²There is a one-time, temporary, drop in excess jobs and an increase in missing jobs in 1996, which likely reflects the fact that the 1996 federal minimum increase from \$4.25 to \$4.75 only affected control states, since Washington’s minimum wage was already at \$4.90 (in current dollars). However, the 1997 federal minimum wage increase to \$5.15 affected both Washington and controls states and hence restored the difference in excess and missing jobs prior to Washington’s state minimum wage increase in 1999 and 2000.

Figure C.1: Employment by Wage Bins in Washington between 2000-2004



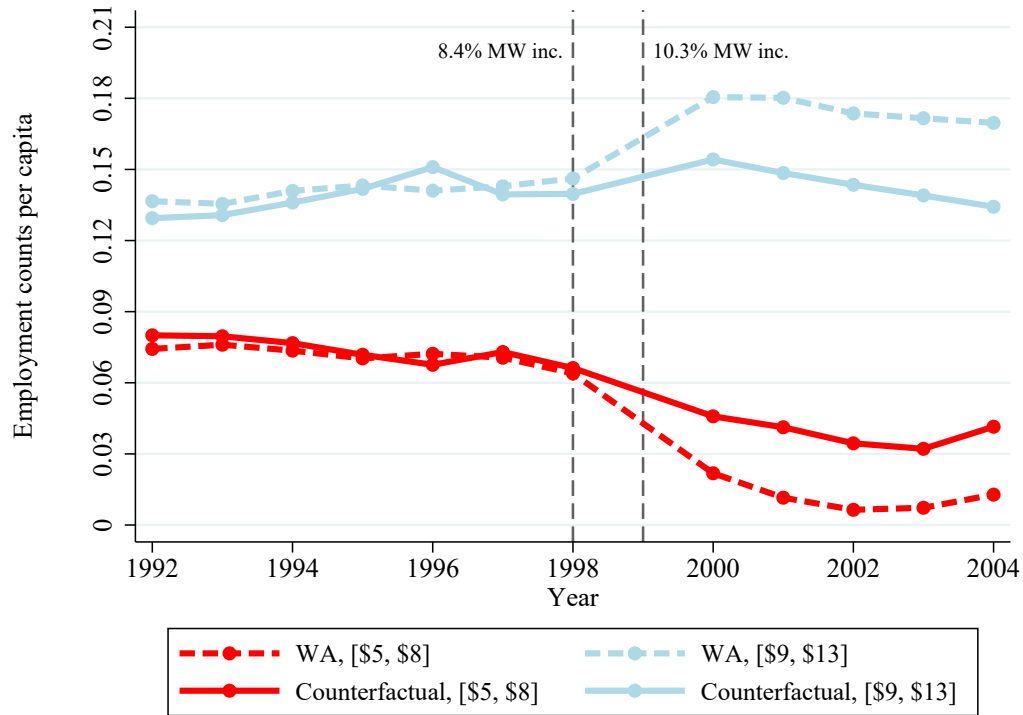
(a) The actual and counterfactual frequency distribution of wages



(b) The difference between the actual and counterfactual frequency distribution of wages

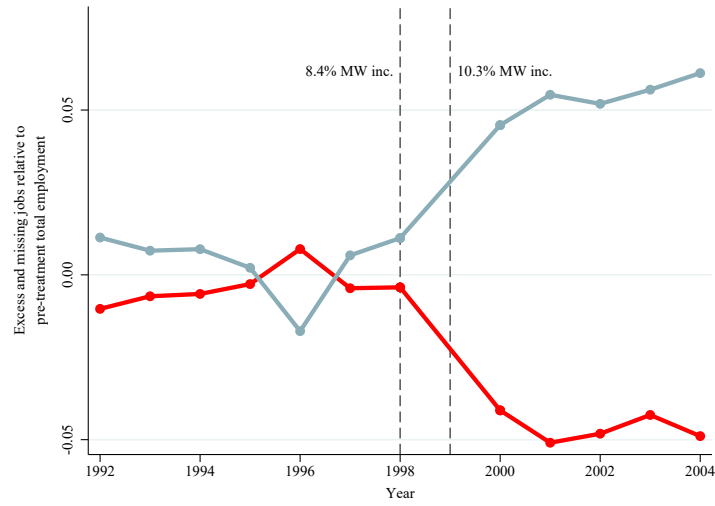
Notes: We examine the effect of the 1999-2000 minimum wage change in Washington state on the frequency distribution of wages (aggregated in \$1 bins), normalized by the 1998 level of employment in Washington. The minimum wage was raised from \$7.51 to \$9.18 (in 2016 values) and it was indexed by inflation afterwards. Panel (a) shows the actual (purple solid bars) and counterfactual (red outlined bars) wage frequency distribution after the minimum wage increases in Washington. The actual distribution (post treatment) plots the average employment between 2000 and 2004 by wage-bin relative to the 1998 total employment in Washington using administrative data on hourly wages between 2000-2004. The counterfactual distribution adds the average change in employment between 2000 and 2004 in states without any minimum wage change to the mean 1996-1998 job counts (see the text for details). The \$26+ bin (the bin that is \$17+ above the new minimum wage) contains all workers earning above \$26, and its values shown on the right y-axis. Panel (b) depicts the difference between the actual and the counterfactual wage distribution. The blue bars show the change in employment at each wage bin (relative to the 1998 total employment in Washington). The red line shows the overall employment changes up to that wage bin. The upper right panel shows the estimates on missing jobs below \$9, Δb ; on the excess jobs between \$9 and \$13, Δa , and on the estimated employment and wage effects. The standard errors are calculated using the method proposed by Ferman and Pinto (forthcoming).

Figure C.2: Comparison of Per-capita Employment Counts of Washington and the Counterfactual

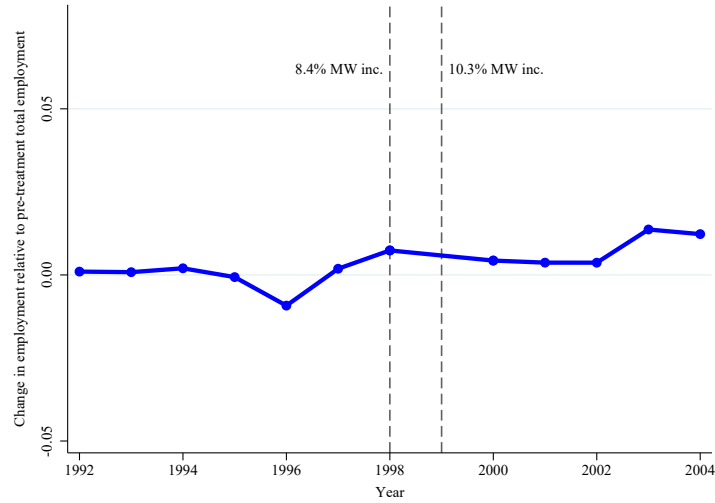


Notes: The figure shows the evolution of the number of jobs per capita with hourly wages between \$5 and \$8, and \$9 and \$13 in Washington and in the counterfactual, with data aggregated in \$1 bins. The counterfactual jobs are calculated using states without any minimum wage change during the 1998-2004 time period. In particular, we add the average change in per capita employment between \$5 and \$8 (and between \$9 and \$13) in the control states to the mean 1996-1998 job counts in Washington state (see the text for details). The two vertical dashed black lines at 1998 and 1999 show the that the minimum wage was raised in 1999 and 2000 in two steps from from \$7.51 to \$9.18 (in 2016 values). The minimum wage was indexed to inflation after 2001. We exclude all observations with imputed wages in the CPS in forming the counterfactual employment counts, except for years 1994 and 1995. Since determining imputed wages is not possible for those years, we use all observations in 1994 and 1995.

Figure C.3: Impact of Minimum Wages on Missing and Excess Jobs, and Employment Change Over time in the Washington Case Study



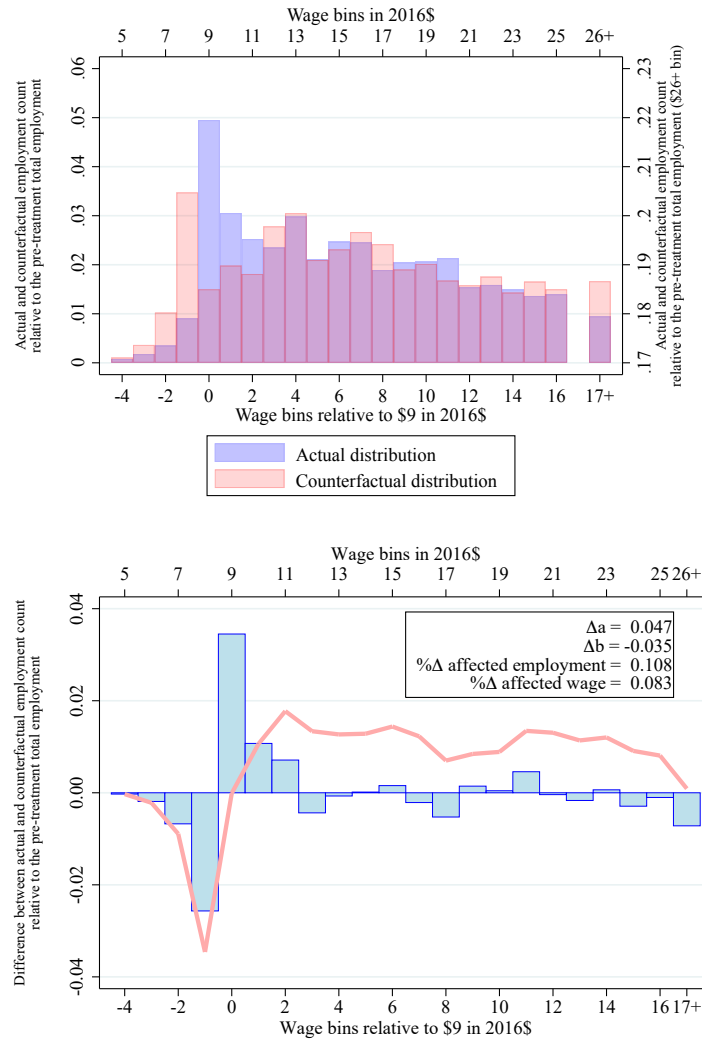
(a) Missing and excess jobs over time



(b) Employment change over time

Notes: The figure shows the evolution of missing jobs, excess jobs, and total employment change over time in Washington state, with data aggregated in \$1 bins. In Panel (a), the red line represents the missing jobs—the difference between the actual and counterfactual wage distribution between \$5 and \$8; while the light blue line shows the excess jobs that is the difference between the actual and counterfactual frequency distributions for wages between \$9 and \$13. In Panel (b), we report the employment change over time (the sum of excess jobs and missing jobs). The counterfactual distribution is calculated by adding the average job change in the control states to the mean 1996-1998 job counts in Washington (see the text for details). The two vertical dashed black lines at 1998 and 1999 show that the minimum wage was raised in 1999 and 2000 in two steps from from \$7.51 to \$9.18 (in 2016 values). The minimum wage was indexed to inflation after 2001. We exclude all observations with imputed wages in the CPS in forming the counterfactual employment counts, except for years 1994 and 1995. Since determining imputed wages is not possible for those years, we use all observation in 1994 and 1995.

Figure C.4: Employment by Wage Bins in Washington between 2010-2004 (Replication of Figure C.1 using CPS data)



Notes: The figure replicates Figure C.1 that examine the effect of the 1999-2000 minimum wage change in Washington on the frequency distribution of wages (aggregated in \$1 bins), normalized by the 1998 level of employment in Washington. The minimum wage was raised from \$7.51 to \$9.18 (in 2016 values) and it was indexed by inflation. Panel (a) shows the actual (purple solid bars) and counterfactual (red outlined bars) frequency wage distribution after the minimum wage increases in Washington. The actual distribution (post treatment) plots the average employment between 2000 and 2004 by wage-bin relative to the 1998 total employment in Washington using CPS data on hourly wages between 2000-2004 (instead of using administrative data as in Figure C.1). The counterfactual distribution adds the average change in employment between 2000 and 2004 in states without any minimum wage change to the mean 1996-1998 job counts (see the text for details). The 26+ bin contains all workers earning above \$26, and its values shown on the right y-axis. Panel (b) depicts the difference between the actual and the counterfactual wage distribution. The blue bars shows the change in employment at each wage bin (relative to the 1998 total employment in Washington). The red line shows the overall employment changes up to that wage bin. The upper left panel shows the estimates on missing number of jobs between \$5 and \$8, Δb ; on the excess number of jobs between \$9 and \$13, Δa , and on the estimated employment and wage effects.

Appendix D

Event-by-event analysis

While the baseline estimates in this paper are an average effect across 138 events estimated by equation (1), our event-by-event analysis estimates separate treatment effects for each of the events. To do so, we first create event-specific annual state panel datasets using the same real wage bin-state-specific employment counts as before. Then we calculate event-specific estimates using separate regressions for each event.

Each event h -specific dataset includes the treated state and all other clean control states for an 8-year panel by event time ($t = -3, \dots, 4$) with the minimum wage increase at $t = 0$. Clean controls are those without any non-trivial minimum wage increase within the 8-year event window. We calculate event-specific per-capita number of jobs in \$1 wage bins relative to the minimum wage for each state-by-year. For each event, we have a similar regression equation as the one used in our baseline estimates:

$$Y_{sjth} = \sum_{\tau=-3}^4 \sum_{k=-4}^4 \alpha_{\tau kh} I_{sjth}^{\tau k} + \mu_{sjh} + \rho_{jth} + \Omega_{sjth} + u_{sjth}, \quad (\text{D.6})$$

where j indicates the j^{th} dollar bin relative to the minimum wage. Y_{sjth} is the per-capita number of jobs in state s , time t , and j^{th} wage bin relative to the minimum wage in dataset h . Ω_{sjth} is an indicator that controls for federal, and small minimum wage events whose 5-year post-treatment periods take place within the data set h . ($\Omega_{sjth} = 1$ for post-treatment periods of these events.) Just like our baseline estimates, we calculate the event-specific change in excess jobs above (Δa_h), change in missing jobs below (Δb_h), and employment change ($\Delta e_h = \Delta a_h + \Delta b_h$) relative to the first year prior to treatment. For instance, the change in the excess number of jobs is given by $\Delta a_h = \frac{1}{5} \sum_{\tau=0}^4 \Delta a_{\tau h} = \frac{1}{5} \sum_{\tau=0}^4 \frac{\sum_{k=0}^4 \alpha_{\tau kh} - \sum_{k=0}^4 \alpha_{-1 kh}}{EPOP_{-1}}$.

Figure D.1 shows the resulting estimated employment changes for each event, along with 95% confidence intervals obtained according to the procedure proposed by Ferman and Pinto (forthcoming). Appropriate for a single treated unit, their procedure extends the cluster residual bootstrap by correcting for heteroskedasticity—an important issue given the very different sample sizes across states in the CPS. Given the very small sample sizes for Washington D.C. in the CPS, we exclude these minimum wage increases from the event-by-event analysis, for a total of 130 events. The figure shows estimates for missing, excess, and total employment changes, where filled in markers represent statistically significant employment changes at the 5 percent level. There is clear evidence of sizable but heterogeneous bites across events: 83% (108) of the missing jobs estimates are negative, and 25% (32) of the events are statistically significant at the 5 percent level. At the same time 21% (27) of the excess jobs estimates are statistically significant, while 78% (100) are positive in sign. Therefore, while there is considerable heterogeneity in the bite of the policy, the distribution of employment estimates is consistent with the sharp null of zero effect everywhere: only 7 (or 5.3%) of events yield statistically significant overall employment changes: 1 is negative

and 6 are positive, and the median estimate is very close to zero.

We can also use the event-by-event estimates to assess whether the lack of leading effects and upper tail employment changes hold event-by-event, and not just on average. Figure D.2 shows leading and upper tail employment changes for 129 events; here one event from Connecticut in 1981 is dropped because it lacks a third leading term. Only 5.4% (7) of the events experience a statistically significant upper tail effects at the 5 percent level, while 7.7% (10) the events experience statistically significant leading effects. Overall, these results are reassuring as they show that the lack of upper tail or leading effects in aggregate is not driven by a mix of unusual positive and negative individual effects. Rather, our findings are consistent with the sharp null of zero upper tail and zero leading effects everywhere.

We also stack all of the event-specific data to calculate an average effect across all 138 events using the a single set of treatment effects $\alpha_{\tau k}$:

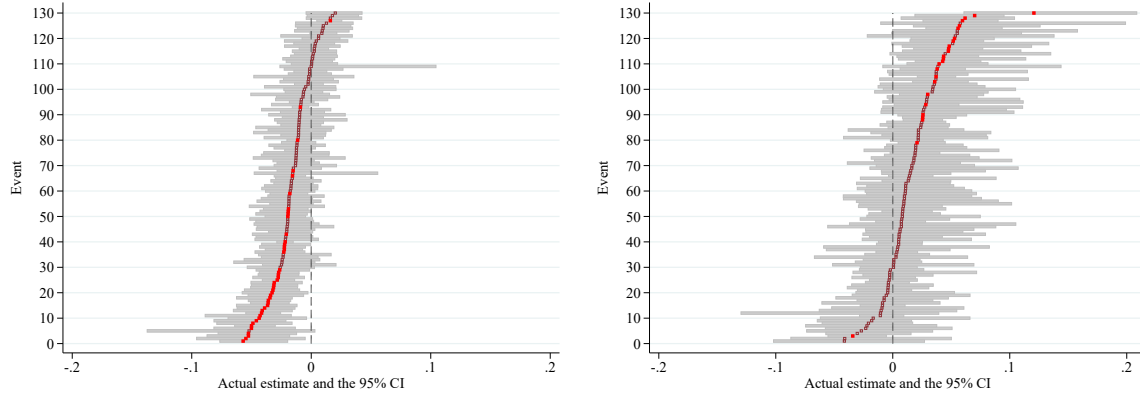
$$Y_{sjth} = \sum_{\tau=-3}^4 \sum_{k=-4}^4 \alpha_{\tau k} I_{sjth}^{\tau k} + \mu_{sjh} + \rho_{jth} + \Omega_{sjth} + u_{sjth}. \quad (\text{D.7})$$

This provides an alternative to our baseline panel specification using a more stringent criteria for admissible control groups, and is more robust to possible problems with a staggered treatment design in the presence of heterogeneous treatment effects. In particular, by aligning events by event-time (and not calendar time), it is equivalent to a setting where the events happen all at once and are not staggered; this prevents negative weighting of some events that may occur with a staggered design (Abraham and Sun, 2018). Moreover, by dropping all control states with any state-level minimum wage increases within the 8 year event window, we further guard against bias due to heterogeneous treatment effects. Moving to the stacked-by-event approach (column 2 in Table D.1) continues to produce a sizable and statistically significant positive wage effect, but an employment effect that is statistically indistinguishable from zero. The minimum wage employment elasticity using the stacked-by-event approach (column 2) is 0.001 (s.e. 0.022), which is fairly similar to the estimate of 0.024 (0.025) in the baseline panel specification (column 1). The own wage elasticity is 0.018 (s.e. 0.546) in column 2 as opposed to 0.411 (s.e. 0.430) in the baseline column 1; here the more stringent stacked-by-event approach is somewhat less precise, though it still rules out an own wage elasticity more negative than -0.88 at the 90% confidence level. The time paths for missing jobs, excess jobs and employment are reported in Figure D.3; again these are quite similar to the baseline Figures III and A.5 and show a sharp and persistent change in missing and excess jobs on the event date, and a flat employment time path before and after the event.

In column (3) we consider the case were we manually average the 138 estimates where each event is weighted by population. The point estimates are very similar to column (2), providing further assurance against the problematic (e.g., negative) implicit weights in the panel estimate. Finally, in column (4) we further refine the sample by only considering events that have a full five year post-treatment sample (i.e., events that occurred on 2012q1 or earlier). The point estimates are quite close to column (2), even though, as expected, the standard errors are now somewhat

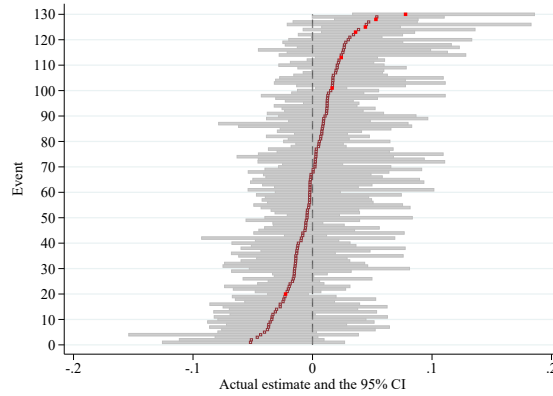
larger. This shows that the small size of our estimates in columns (1) - (3) is not driven by a lack of a sufficiently long post-treatment period in some of the events.

Figure D.1: Event-specific Estimates for Excess Jobs Above, Missing Jobs Below, and Employment Change



(a) Missing jobs below (Δb)

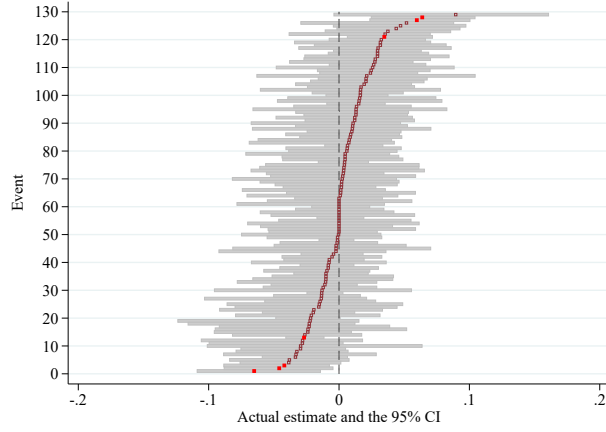
(b) Excess jobs above (Δa)



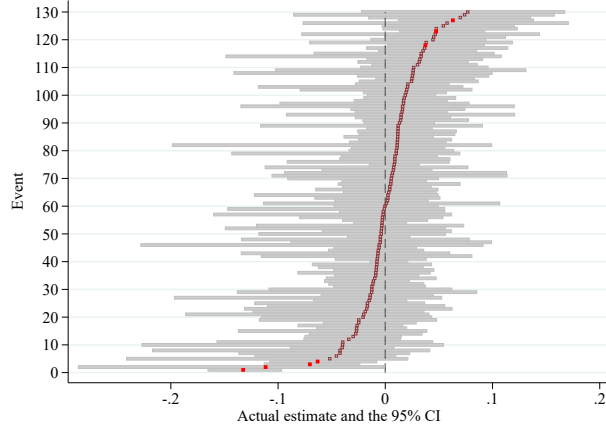
(c) Employment change ($\Delta a + \Delta b$)

Notes: The figure shows the event-specific point (square markers) and confidence interval (gray horizontal bars) estimates for missing jobs below (Δb), excess jobs above (Δa), and employment change ($\Delta a + \Delta b$). The point estimates are calculated using equation D.6, and the confidence intervals are obtained according to the procedure proposed by Ferman and Pinto (forthcoming). The vertical gray dash line indicates the null hypothesis of no effect, and it is rejected with 95% confidence if the confidence intervals do not contain 0. There are 130 events (D.C. events are dropped due to the measurement error concerns). 44/130, 25/130, and 7/130 events yield statistically significant estimates (filled markers) for missing jobs below, excess jobs above, and employment change.

Figure D.2: Leading Estimates and Upper Tail Falsification Tests for Event-specific Estimates



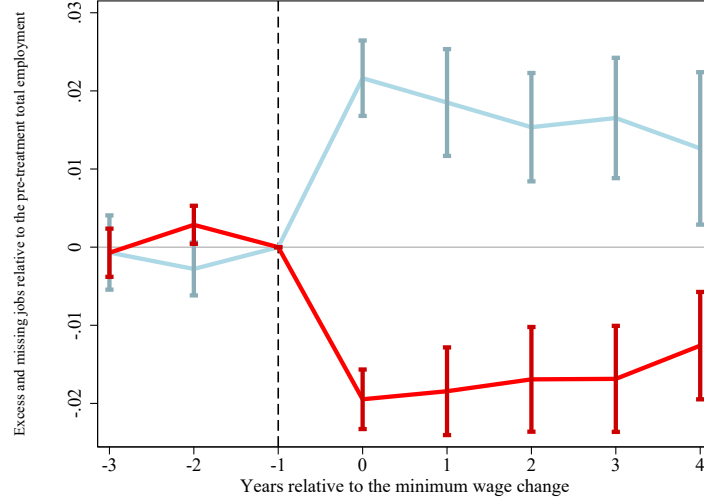
(a) Leading employment change ($\Delta a_{-3} + \Delta b_{-3}$)



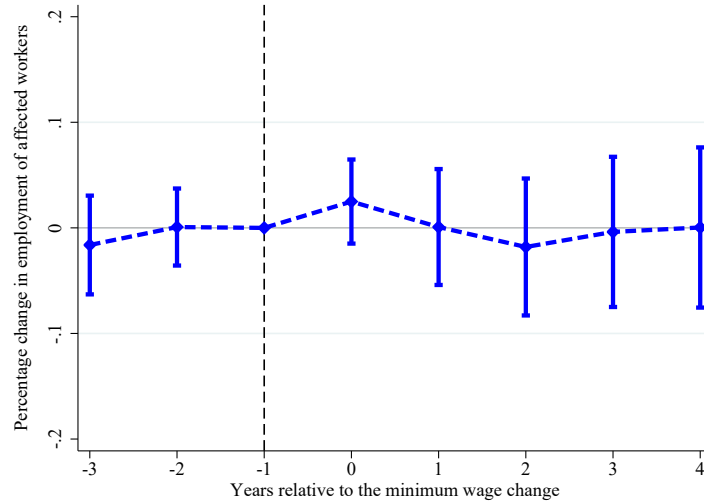
(b) Upper tail ($\frac{\sum_{\tau=0}^{\tau=4} \sum_{k \geq 5} \alpha_{\tau k} - \alpha_{-1k}}{EPOP_{-1}}$)

Notes: The figure shows the event-specific point (square markers) and confidence interval (gray horizontal bars) estimates for leading ($\Delta a_{-3} + \Delta b_{-3}$), and upper tail ($\frac{\sum_{\tau=0}^{\tau=4} \sum_{k \geq 5} \alpha_{\tau k} - \alpha_{-1k}}{EPOP_{-1}}$) employment change. The point estimates are calculated using equation D.6, and the confidence intervals are obtained according to the procedure proposed by Ferman and Pinto (forthcoming). The vertical gray dash line at 0 indicates the null hypothesis of no effect, and it is rejected with 95% confidence if confidence intervals do not contain 0. There are 129 events (D.C. events are dropped due to the measurement error concerns and the minimum wage event that takes place in Connecticut in 1981 does not have the third leading term.). 7/129, and 7/129 events yield statistically significant estimates (filled markers) for leading, and upper tail employment change.

Figure D.3: Impact of Minimum Wages on the Missing and Excess Jobs, and Employment Over Time - Stacked Event-by-Event Estimates



(a) Evolution of the missing and excess jobs



(b) Evolution of the employment of the affected workers

Notes: The figure shows the main results from our stacked analysis (see equation D.7) exploiting 138 state-level minimum wage changes between 1979-2016. Panel (a) shows the effect of a minimum wage increase on the missing jobs below the new minimum wage (blue line) and on the excess jobs at and slightly above it (red line) over time. The blue line shows the evolution of the number of jobs (relative to the total employment 1 year before the treatment) between \$4 below the new minimum wage and the new minimum wage (Δb); and the red lines show the number of jobs between the new minimum wage and \$5 above it (Δa). Panel (b) shows the evolution of employment between \$4 below the new minimum wage and \$5 above it (relative to the total employment 1 year before the treatment), which is equal to the sum of missing jobs below and excess jobs at and slightly above the minimum wage, $\Delta b + \Delta a$. We also show the 95% confidence intervals based on standard errors that are clustered at the state level.

Table D.1: Impact of Minimum Wage Increase on the Average Wage and Employment of Affected workers - Stacked Event-by-Event Estimates

	(1)	(2)	(3)	(4)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.017*** (0.002)	-0.017*** (0.002)	-0.015*** (0.003)
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.017*** (0.003)	0.019*** (0.003)	0.015*** (0.003)
% Δ affected wages	0.068*** (0.010)	0.048*** (0.012)	0.060*** (0.014)	0.042*** (0.013)
% Δ affected employment	0.028 (0.029)	0.001 (0.026)	0.022 (0.041)	-0.001 (0.030)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.001 (0.022)	0.019 (0.035)	-0.001 (0.024)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.018 (0.546)	0.367 (0.613)	-0.017 (0.713)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.086	0.086	0.086
% Δ MW	0.101	0.101	0.101	0.108
Number of events	138	138	138	98
Number of observations	847,314	983,934	983,934	838,584
Set of events	Primary	Primary	Primary	Primary, until 2012q1
Data	All workers, state-by- wage-bin	All workers, stacked	All workers, stacked	All workers, stacked
Specification	Baseline	Pooled stacked	Manual averaging	Pooled stacked

Notes. The table reports the effects of a minimum wage increase based on the event study analysis (see equation 1) and alternative variants of stacked analysis (see equations D.6 and D.7) exploiting 138 state-level minimum wage changes between 1979 and 2016. The table reports five year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment and wages. The first column reproduces column (1) of Table 1 for comparison purposes. Column (2) uses equation 7. Column (3) uses equation 6 and manually averages each event-by-event estimates. Column (4) uses the same regression equation as column (2), but uses only events that have occurred on or before 2012q1 to ensure a full five year post-treatment sample. Robust standard errors in parentheses are clustered by event-by-state in columns (1), (2), and (4). In column (3), we employ the procedure proposed by Ferman and Pinto (forthcoming) to obtain the standard errors. Significance levels are * 0.10, ** 0.05, *** 0.01.

Appendix E

Data Appendix

The primary data set we use in the event study analysis is the individual-level NBER Merged Outgoing Rotation Group of the Current Population Survey for 1979-2016 (CPS). We use variables EARNHRE (hourly wage), EARNWKE (weekly earnings), and UHOURSE (usual hours) to construct our hourly wage variable. For the period after 1995q4, we exclude observations with imputed hourly wages (I25a>0) among those with positive EARNHRE values, and exclude observations for which usual weekly earnings or hours information is imputed (I25a>0 or I25d>0) among those with positive EARNWKE values. There is no information on the imputation between 1994q1 and 1995q3 so we exclude these observations entirely. For the years 1989-1993, we follow the methodology of Hirsch and Schumacher (2004) to determine imputed observations.

The CPS is a survey, where only a subset of workers is interviewed each month; therefore, there is sampling error in the dataset. In addition, as we do not use observations with imputed hourly wages in most of our analysis, the employment counts of the raw CPS data are biased downwards. To reduce the sampling error and also address the undercounting due to dropping imputed observations, our primary sample combines the CPS wage densities with the true state-level employment counts from the QCEW (E). Specifically, in the QCEW benchmarked CPS, the employment counts for a wage bin w is calculated as $\widehat{\frac{E_w}{N}}^{QCEW} = \widehat{f}_w^{CPS} \times \frac{E}{N}$, where $\widehat{f}_w^{CPS}$ is the (discretized) wage density estimated using the CPS: $\widehat{f}_w^{CPS} = Prob(w \leq wage < w + 0.25)$. We also do a similar benchmarking of NAICS-based industry-and-state-specific QCEW employment (between 1990 and 2016) when we conduct sectoral analysis.

In addition, we use micro-aggregated administrative data on hourly wages from Washington state for the case study in Section C.5. This data was provided to us as counts of workers in (nominal) \$0.05 bins between 1992 and 2016 by the state’s Employment Security Department. We convert this data into \$0.25 (real 2016 USD) hourly wage bins for our analysis using the CPI-U-RS. We also use similar micro-aggregated administrative data from Minnesota and Oregon for conducting comparison of data quality and measurement error in Appendix F.

Matched CPS

The CPS outgoing rotation groups are structured so that an individual reports her wage twice, one year apart, in 4th and 8th sample months. We employ the longitudinal aspect of the CPS when separately estimating the impacts of the minimum wage on new entrant and incumbent workers. This requires matching two CPS files. We exactly follow the procedure proposed by Madrian and Lefgren (2000), and use household ID (HHID), household number (HHNUM), person line number in household (LINENO), month in sample (MINSAMP), and month and state variables to match observations in two consecutive CPS files. We confirm the validity of matches by evaluating reported sex, race, and age in the two surveys. If sex or race do not match, or if individual’s age decreases by more than 1 or increases by more than 2, we declare them as “bad matches” and exclude from

the matched sample. Additionally, since matching is not possible from July to December in 1984 and 1985, from January to September in 1985 and 1986, from June to December in 1994 and 1995, or from January to August in 1995 and 1996, we exclude these periods. On average, 72% of the observations in the CPS are matched: around 25% of the individuals are absent in the 8th sample month, while an additional 3% are dropped because they are bad matches. We determine the incumbency of individual from employment status information in the 4th sample month. Similar to our primary CPS sample, we drop observations with imputed wages in the 8th sample month. Overall, the number of worker-level observations is smaller in the matched sample because we only use the 8th sample month in the matched sample, as opposed to both 4th and 8th sample months in the baseline sample.

Industry classifications

Following Mian and Sufi (2014), we use an industry classification with four categories (tradable, non-tradable, construction, and other) based on retail and world trade. According to the classification, an industry is “tradable” if the per worker import plus export value exceeds \$10,000, or if the sum of import and export values of the NAICS 4-digit industry is greater than \$500 million. The retail sector and restaurants compose “non-tradable” industries, whereas the “construction” industries are industries related to construction, land development and real estate. Industries that do not fit in either of these three categories are pooled and labeled as “other”. We merge the CPS with Mian and Sufi (2014) industry classification using the IND80 and IND02 variables in the CPS.

Appendix F

Comparison of Administrative Data to CPS

In our event study analysis, we use the Current Population Survey (CPS), which provides information on wages for a large sample of individuals, after benchmarking to aggregate state-level employment counts in the QCEW. Therefore, there is sampling error in our estimated job counts in each wage bin. In this section we assess the accuracy of CPS based jobs counts by comparing administrative data on job counts from three states with reliable information on hourly wages (Minnesota, Oregon, and Washington).

In Section F.1, we compare the performance of the raw CPS and the QCEW-benchmarked CPS in predicting the counts of workers earning less than \$15 in the administrative data from Minnesota, Oregon and Washington. We show that counts from the QCEW-benchmarked CPS are much closer to the counts from the administrative data than those from the raw CPS: the mean squared prediction error is substantially smaller when we use QCEW-benchmarked CPS data. In Section F.2, we show that the wage distribution from the QCEW-benchmarked CPS closely matches the distribution from the administrative data from the three states. In particular, we show that the number of workers reporting earnings under the state minimum wage is similarly small in both the administrative data and the CPS, which is an important indication of the degree of misreporting in the CPS. In section F.3 we implement structural estimation to further assess the importance of wage misreporting in the administrative data and in the QCEW-benchmarked CPS along the lines of Autor, Manning and Smith (2016). Our estimates show that the implied misreporting is of a similar magnitude in the two data sources. In section F.4 we deconvolve the QCEW-benchmarked CPS using the estimated measurement error model of Autor, Manning and Smith (2016), and provide estimates using this measurement-error-corrected frequency distribution.

F.1 Assessing the Accuracy of the Raw versus the QCEW- benchmarked CPS

We compare the administrative data with the raw CPS, and the QCEW-benchmarked CPS. Because the CPS is a survey, it has substantially greater sampling error than the QCEW which is a near-census of all workers in a state. Also, since we are not using observations with imputed hourly wages in our data sets, state-level employment counts of the raw CPS data are biased downwards. To address both of these problems, our primary sample combines the CPS wage distribution with state-level employment counts in the QCEW. We label the data with the QCEW adjustment as the “QCEW-benchmarked CPS”, and the raw CPS as “CPS-Raw.”⁴³

⁴³We note that the QCEW and CPS have slightly different employment concepts. The CPS measures employment in a reference week while the QCEW measures employment at any time in a quarter. So CPS employment may be slightly lower than QCEW since some people work only parts of a quarter. Therefore, the QCEW-benchmarked CPS is closer to the QCEW employment concept. At the same time, any such gap is likely picked up by the state and time fixed effects. To confirm this, we implement an event study regression where the outcome variable is the gap between CPS and QCEW employment. Events are the same 138 state-level minimum wage changes between 1979-2016 that we use in our benchmark specification. Similar to our benchmark specification we include state and time fixed effects in the regression. The blue line shows the evolution of the gap in the employment rate (relative to the year before the

First, we establish here that QCEW benchmarking of aggregate employment is likely to improve the accuracy of our counts by wage bin. The employment count for wage bin w , E_w , can be rewritten as the product of the (discretized) wage density, $f_w = \text{Prob}(w \leq \text{wage} < w + 0.25)$, and the employment, E , so $E_w = f_w \times E$. The raw CPS-based estimate for per-capita count is $E_w^{CPS} = f_w^{CPS} \times E^{CPS}$. The QCEW benchmarked CPS uses the state-level employment counts from the QCEW which has no measurement error given that includes the near universe of workers; so formally, $E_w^{QCEW} = f_w^{CPS} \times E$. It follows that the mean squared prediction error ($MSPE$) is lower for the QCEW benchmarked CPS than for the raw CPS, if the measurement errors for f_w^{CPS} are uncorrelated with E^{CPS} . The latter condition holds if the source of the error is sampling.

Since our approach mainly focuses on job changes at the bottom of the wage distribution, we assess whether the raw CPS or the QCEW-benchmarked CPS does a better job in predicting the number of workers earning less than \$15. For each quarter t , we calculate the average per-capita numbers of workers earning less than \$15 in the 20 subsequent quarters (i.e., between t and $t + 20$); we also calculate the average for the 4 preceding quarters (i.e., between t and $t - 4$). Then, we subtract the latter from the former and we refer to this as the transformed counts. The employment changes in Table I show the average employment changes in the 20 subsequent quarter after the minimum wage relative to the 4 preceding quarters. Therefore, the transformed counts are closely related to the employment estimates shown in Table I.

In Figure F.1 panels (a) and (b), we show the scatterplot of the transformed counts (per capita) from the administrative data against those from QCEW-benchmarked CPS and the raw CPS, respectively. In addition to a visual depiction, we also regress the transformed administrative counts on the transformed CPS-Raw, and QCEW-benchmarked CPS counts. To assess the accuracy of the data, we use two measures: R^2 and the slope ($\hat{\beta}$). A perfect match between the CPS and the administrative data would yield $R^2 = \hat{\beta} = 1$, or a zero mean-squared prediction error (MSPE). If the CPS correctly predicts the administrative counts on average, but each prediction possesses some error, then $R^2 < 1$ and $\hat{\beta} = 1$. On the other hand, if there is a bias in the CPS counts, then $\hat{\beta} \neq 1$. The QCEW-benchmarked counts are better predictors of the administrative counts than are the raw CPS counts: for the former, the estimated slope is 0.778 and the R^2 is 0.643. In contrast, the raw CPS has a larger bias ($\hat{\beta} = 0.564$) and variance ($R^2 = 0.322$).

In Table F.1, we report the ratio of the MSPE using the raw CPS counts to the MSPE using the QCEW-benchmarked CPS. Besides reporting the MSPE for the transformed count (the 20 subsequent quarter average minus the 4 preceding quarter average) of workers under \$15, we also report the MSPEs for underlying components. Namely, we calculate the MSPEs using counts of workers earnings less than \$15/hour as well as counts of workers in each \$0.25 bins—each averaged over either 4 or 20 quarters. A MSPE ratio above one indicates that the QCEW-benchmarked CPS performs better in predicting the administrative data than the raw CPS. The table shows that this is indeed the case: QCEW-benchmarked CPS performs better in all cases, especially for the

treatment) between the CPS and QCEW. As Figure F.5 shows, there is no systematic change in the gap between CPS and QCEW employment following treatment.

aggregated employment counts under \$15/hour.

F.2 Comparison of the Wage Distribution in the CPS and in the Administrative Data

We assess the sampling and misreporting errors in the CPS by comparing the frequency distribution of hourly wages in the QCEW benchmarked CPS and in the administrative data. In Figure F.2 we plot 5-year averaged per-capita employment counts in \$3 bins relative to the minimum wage. We compare the distributions at this aggregation level, since our main estimates on excess and missing jobs in Table I show 5 year employment changes in \$3 to \$5 bins relative to the minimum wage. The red squares show the distribution in the administrative data while the blue dots show the distribution calculated using QCEW-adjusted CPS. We report the wage distributions in each states separately, as well as in the three states together.

The distributions from the CPS closely match the distributions in the administrative data in all states and in all three five-years periods (2000-2004, 2005-2009, and 2010-2014). A similar number of jobs are present just below the minimum wage in the two data sources, albeit in some cases there are slightly more in the CPS (e.g. in WA 2005-2009). When we pool all three states, the CPS and the administrative data exhibit virtually the same distribution below the minimum wage. Note that in all three of these states, there is no separate tipped minimum wage, and nearly all workers are covered by the state minimum wage laws. Therefore, the presence of jobs paying below the minimum wage may reflect misreporting. If this is the case, then Figure F.2 suggests that the extent of misreporting is quite similar in the CPS and in the administrative data. We formally test this in the next section. At the same time, we should point out that some of the sub-minimum wage jobs may reflect true under-payment. Either way, it is encouraging that the extent of sub-minimum wage jobs in the CPS is very similar to what is found in high quality administrative wage data.

The figures also highlight that the $[0,3)$ bin—which includes workers at and up to \$3 above the minimum wage—contains a somewhat larger number of workers in the administrative data than in the CPS for Washington state; however, for Oregon and Minnesota, the CPS closely matches the number of workers in that bin. As a result, when we pool all three states together, we find that the CPS tends to underestimate the number of jobs at and slightly above the minimum wage. However, this difference is quite stable over time, as further shown below in Figure F.3; as a result, our difference-in-difference estimates are unlikely to be affected by this gap between the two counts. Finally, the CPS tends to place slightly more workers in the middle-income bin ($[MW + \$6, MW + \$21)$), and fewer workers at the high-income bin ($[MW + \$21, \infty)$).

Figure F.3 plots the time paths of the number of jobs below the minimum wage $[MW - \$5, MW)$, and jobs at and above the minimum wage $[MW, MW + \$5)$ relative to the state-level population from both the administrative data and the CPS. Consistent with the previous findings, the job counts below and above in both of the data sets follow very similar paths. When we pool the data across all three states, the evolution of the jobs below the minimum wage lines up perfectly across the two series. The level of jobs at and slightly above the minimum wage is slightly higher in the

CPS, but again, the differences are quite stable over time. As a result, the difference-in-difference estimator implemented in this paper is unlikely to be affected by the small discrepancy between the administrative and the CPS data.

F.3 Assessment of Misreporting of Wages Using Structural Estimation

To compare the potential measurement error in the CPS and in the administrative data for these states, we also implement a structural estimation approach developed by Autor, Manning and Smith (2016). Following Autor, Manning and Smith (2016), we assume that in the absence of the minimum wage, both the observed and the true latent wage distributions are log-normal.⁴⁴ A portion (γ) of the workers report their wages correctly, while others report it with some error. In the absence of a minimum wage, the observed (log) wage can be written as

$$v^* = w^* + D\epsilon$$

where v^* is the observed and w^* is the true latent (log) wage of the worker that would prevail in the absence of a minimum wage. D is a binary variable that is equal to 1 when the wage is misreported, and 0 otherwise. Therefore, $P(D = 0) = \gamma$ measures the probability of reporting wages accurately. When the wage is misreported, the distribution of the (logged) error is again normal, with $\epsilon \sim N(0, \frac{1-\rho^2}{\rho^2})$, where $\rho^2 = \frac{\text{cov}(v^*, w^*)}{\text{var}(v^*)}$, reflects the correlation between the observed and true latent distributions. Both parameters ρ and γ determine how misreporting distorts the observed wage distribution. Here $1 - \gamma$ measures the rate of misreporting, while $\frac{1-\rho^2}{\rho^2}$ measures the variance of the error conditional on misreporting.

We can summarize the overall importance of misreporting by comparing the standard deviation of the true latent distribution (σ_w) and the observed latent distribution (σ). When $\frac{\sigma_w}{\sigma} = 1$, misreporting does not affect the dispersion in observed wages. But when $\frac{\sigma_w}{\sigma} = 0.5$, say, misreporting causes the observed wage distribution's standard deviation to be twice as large that it would if wages were always accurately reported. Autor, Manning and Smith (2016) notes that the ratio can be approximated by ρ and γ as follows:

$$\frac{\sigma_w}{\sigma} = \gamma + \rho(1 - \gamma)$$

We estimate the model parameters γ and ρ for both the administrative data and the CPS. One additional complication in the administrative data is that sometimes small rounding errors in hours can shift a portion of workers to the wage bin below the MW; this will tend to over-state the measurement error in the administrative data (at least in terms of estimating $1 - \gamma$). For this reason, we present two sets of estimates. First we keep the data as is by using wage bins relative to the minimum wage, $[MW, MW + \$0.25)$. Second, we additionally show estimates using

⁴⁴The latent wage distribution refers to the distribution that would prevail in the absence of a minimum wage. The wage is called “observed” when it reflects both the true value as well as the reporting error. Note, however, that the “latent observed” wage distribution is only observed in practice in the absence of a minimum wage.

re-centered \$0.25 wage bins around the minimum wage. The re-centered \$0.25 bin that includes the minimum wage is now defined as $[MW - \$0.10, MW + \$0.15)$. The subsequent re-centered bins are defined as $[MW + \$0.15, MW + \$0.40)$, etc., while the preceding bins are defined as $[MW - \$0.35, MW - \$0.10)$, etc.

Our analysis covers the 1990-2015 period for Washington, and the 1998-2015 period for Minnesota and Oregon: the start dates reflect the earliest years the administrative data are available for each state. Since none of these three states allow tip credits, we do not drop tipped workers from our sample, and use all workers in our analysis.

Table F.2 reports the misreporting rate $(1 - \gamma)$, the variance of the error term, and the ratio of the true and observed standard deviations. In panel A, where we re-center the wage bins, and find that the misreporting rate $1 - \gamma$ is slightly smaller in the CPS (.23) than in the administrative data (0.28).⁴⁵ However, conditional on misreporting, the variance of the errors $\left(\frac{1-\rho^2}{\rho^2}\right)$ is somewhat larger in the CPS (1.46) than in the administrative data (1.25). Putting these two parts together, we find that the ratios of the true to observed standard deviations $\frac{\sigma_w}{\sigma}$ are quite similar in the two datasets: 0.92 in the CPS and 0.91 in the administrative data. In panel B, where we use un-centered wage bins, the CPS estimates are virtually unchanged. However, due to the rounding errors in hours in the administrative data, the estimated misreporting rate $(1-\gamma)$ increases while the variance of the error conditional on misreporting $\left(\frac{1-\rho^2}{\rho^2}\right)$ falls. Overall, the ratio of the true and observed standard deviations for administrative data in panel B (0.90) remains very similar to those reported in panel A (0.91) and to the CPS estimates (0.92).

Overall, the structural estimation results suggest that the extent to which there is misreporting of wages, they are of similar magnitude in the CPS and in high quality administrative wage data. This provides additional support for the validity of our estimates using CPS data.

F.4 Estimates using deconvolved, measurement-error corrected CPS-ORG

In the previous section, we obtained the functional form of the distribution of misreporting error $(D\epsilon)$ in the CPS-ORG. Given an empirical distribution of the observed noisy wage $v = w + D\epsilon$, and an empirical distribution of the error $D\epsilon$, we can obtain an estimated distribution of the error-free wage, w , using the non-parametric deconvolution procedure proposed by Comte and Lacour (2011). Given an empirical sample of errors $D\epsilon$ drawn from an arbitrary distribution (estimated in the previous section), and the sample of noisy observed wages v , the procedure recovers a measurement error corrected distribution. The deconvolution is based on the insight that the inverse-Fourier transform of the unknown distribution of w is a function of the estimable characteristic functions of v and $D\epsilon$. Estimation is performed using penalized deconvolution contrasts and data-driven adaptive model-selection, and implemented using the R package `deamer`.⁴⁶

⁴⁵The CPS estimate is largely in line with Autor, Manning and Smith (2016) who estimate the misreporting rate around 20% between 1979 and 2012 using 50 states.

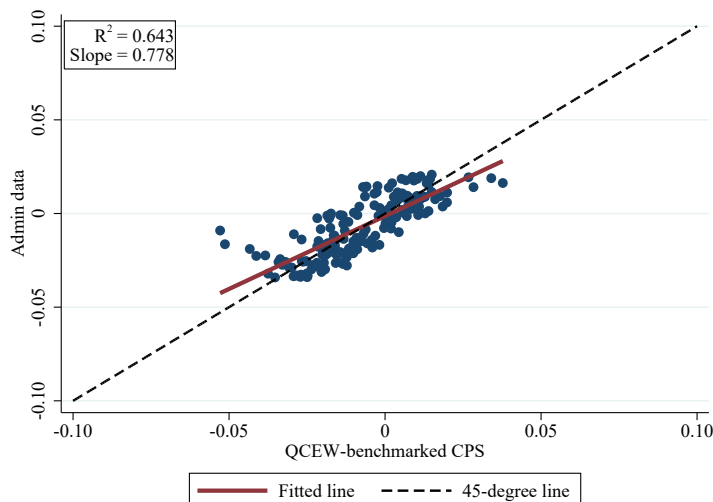
⁴⁶We separately estimate the distribution of true wages for each state-by-quarter using the same distribution function for the measurement error. Estimating annual distribution functions for the error following Autor, Manning and Smith (2016) produces virtually the same results.

Figure F.4 plots the wage distributions of the CPS-ORG and the measurement error corrected (deconvolved) CPS-ORG (MEC-CPS) in \$1 bins relative to the minimum wage averaged over time and states. We make three observations. First, the share of jobs paying below the current minimum wage is smaller in CPS-MEC. This is expected, since the Autor, Manning and Smith (2016) approach uses the share below the minimum wage to estimate the extent of measurement error; so a successful reduction in measurement error should reduce the share earning below the minimum. Numerically, while 2.67% of the workers report working below the minimum wage in the CPS, after the measurement error correction it decreases to 1.57%. Second, the share of workers in the dollar bin of the current minimum wage are similar in both samples, suggesting that the raw CPS performs relatively well in reporting the share of workers at or up to \$0.99 above the minimum. Third, individuals in the raw CPS are more likely to report their wages as \$17 higher than the current minimum wage. The CPS-MEC, on the other hand, find that there are more individuals with hourly wages between \$1 and \$16.99 above the minimum after taking the misreporting error into account.

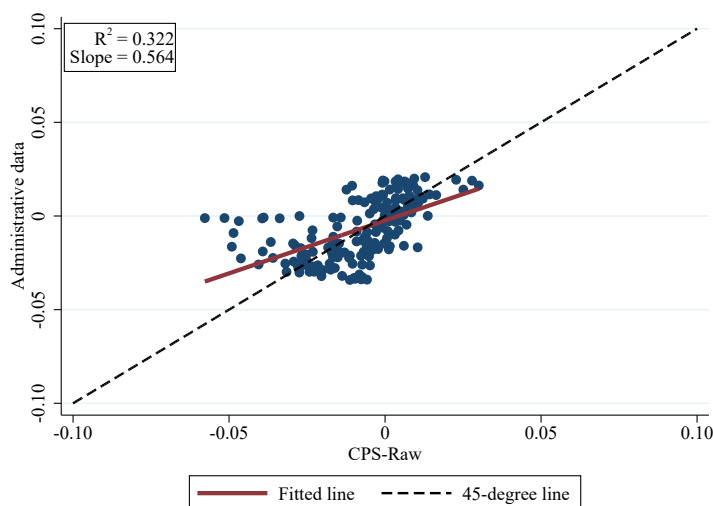
In Table F.3, we compare the baseline estimates with those obtained using the deconvolved data. Column (1) reproduces the baseline estimates reported in Table I column (1). Column (2) reports the results using deconvolved data⁴⁷ The missing and excess jobs estimates are quite similar across columns 1 and 2. The baseline missing jobs estimate of -0.018 (s.e. 0.004) in column 1 is very similar to the measurement error corrected estimate of -0.017 (s.e. 0.004) in column 2. The baseline excess jobs estimates for both columns 1 and 2 are 0.021 (s.e. 0.003). This corroborates our argument that the employment estimates are not substantially affected by measurement error in reported wages. While the baseline employment elasticity with respect to the minimum wage is 0.028 (s.e. 0.029) in column 1, it is 0.037 (s.e. 0.031) after measurement error correction in column 2. The wage effect estimates are also quite similar when we use the deconvolved data. The baseline percentage change in affected wage is 0.068 (s.e. 0.010) in column 1, whereas it is 0.075 (s.e. 0.012) in column 2 using deconvolved data. Overall, these findings underscore that our results are quite robust to the presence of misreporting error in wages. While more precise wage data may uncover more accurate information on the exact size of the wage or spillover effects, the combination of the deconvolution-based estimates and comparisons of the CPS and administrative data suggests any bias due to measurement error is likely to be small.

⁴⁷The deconvolved data uses a slightly different sample that excludes the quarters of events due to the existence of two spikes in those periods. By assumption, the latent wage distribution is log-normal and observed wage distribution can only have one mass point due to the minimum wage. However, if there is a minimum wage event in the quarter, then it is likely that observed wage distribution will have two mass points. In those cases, the deconvolution procedure does not perform well. However, in practice the estimates including the quarter of events are very similar (results not reported).

Figure F.1: Comparison of Administrative with QCEW-benchmarked CPS, and CPS-Raw Counts of Workers Earning less than \$15



(a) Administrative data against QCEW-benchmarked CPS



(b) Administrative data against CPS-Raw

Notes: This figure plots per-capita counts of workers earning less than \$15 in administrative data against QCEW-benchmarked CPS in panel A, and CPS-Raw in panel B. To construct a measure that is comparable to the baseline employment estimate, we transform the counts, and subtract the average number of workers earning less than \$15 (per capita) in the 4 preceding quarters from that in the 20 subsequent quarters. The blue circles indicate each observation, the red straight line is the linear fit, and the black dashed line is the 45-degree line. We report the estimated R^2 and slope from a simple linear regression in the box.

Figure F.2: Frequency Distributions in the Administrative and CPS data

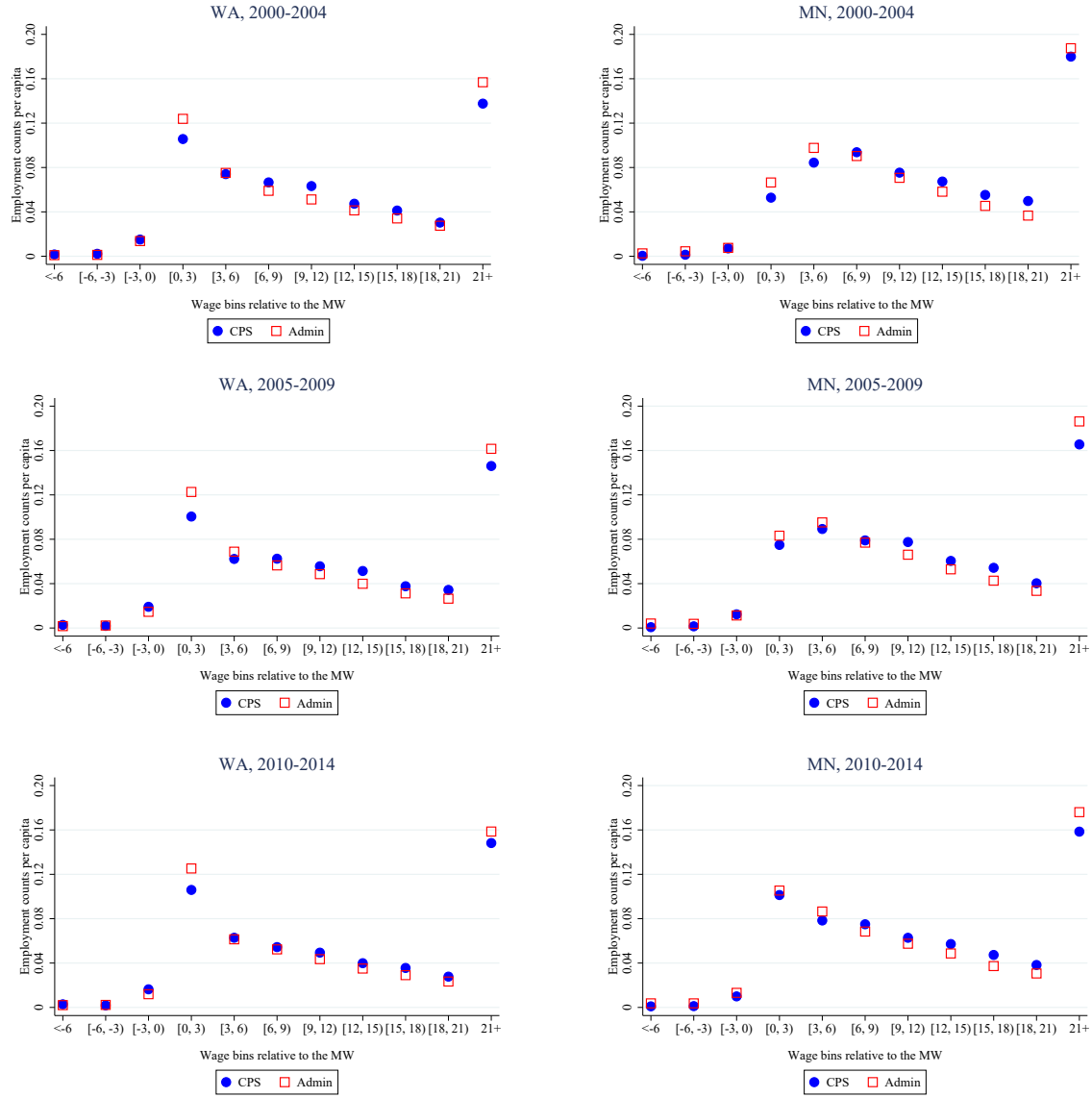
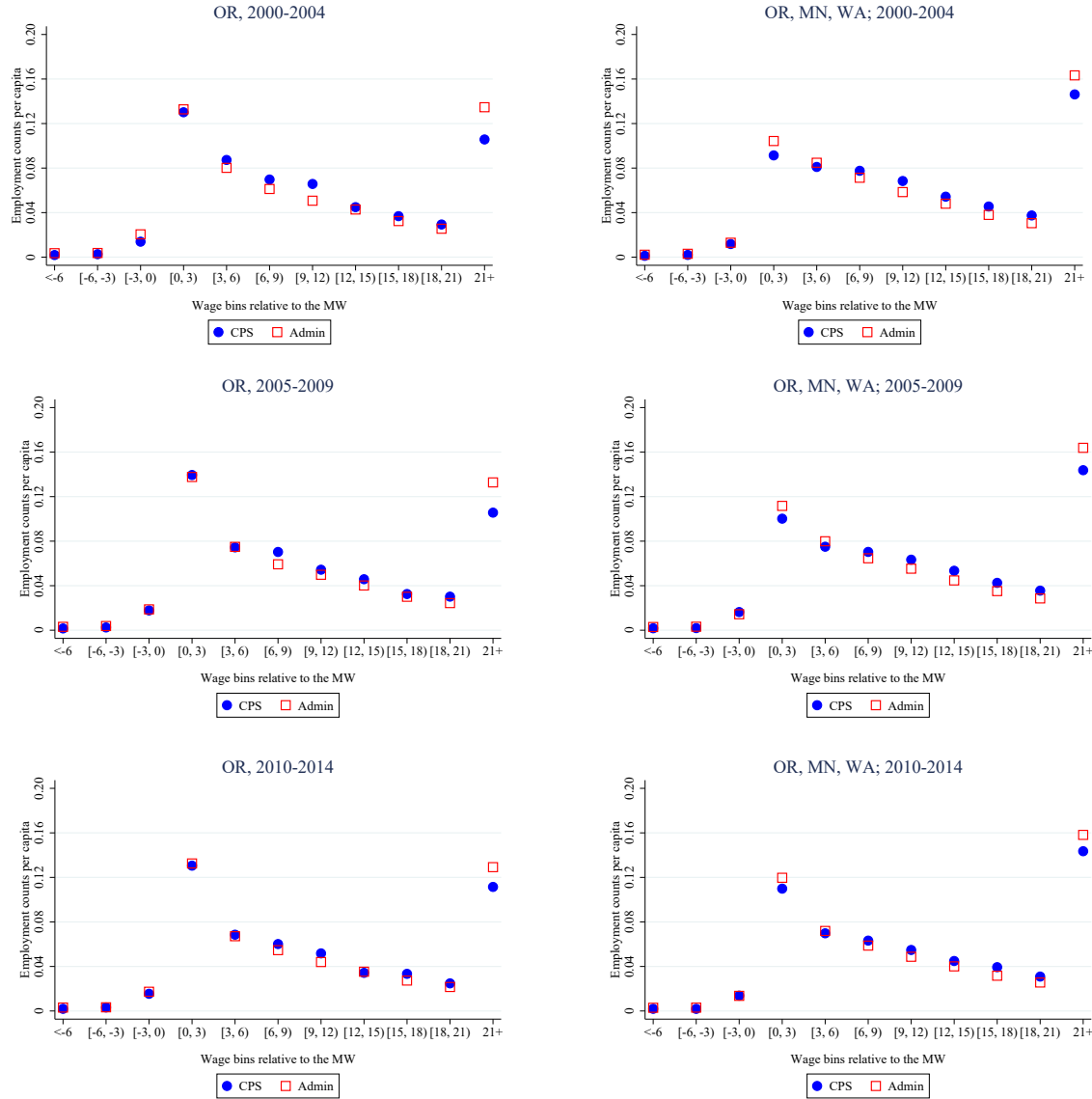
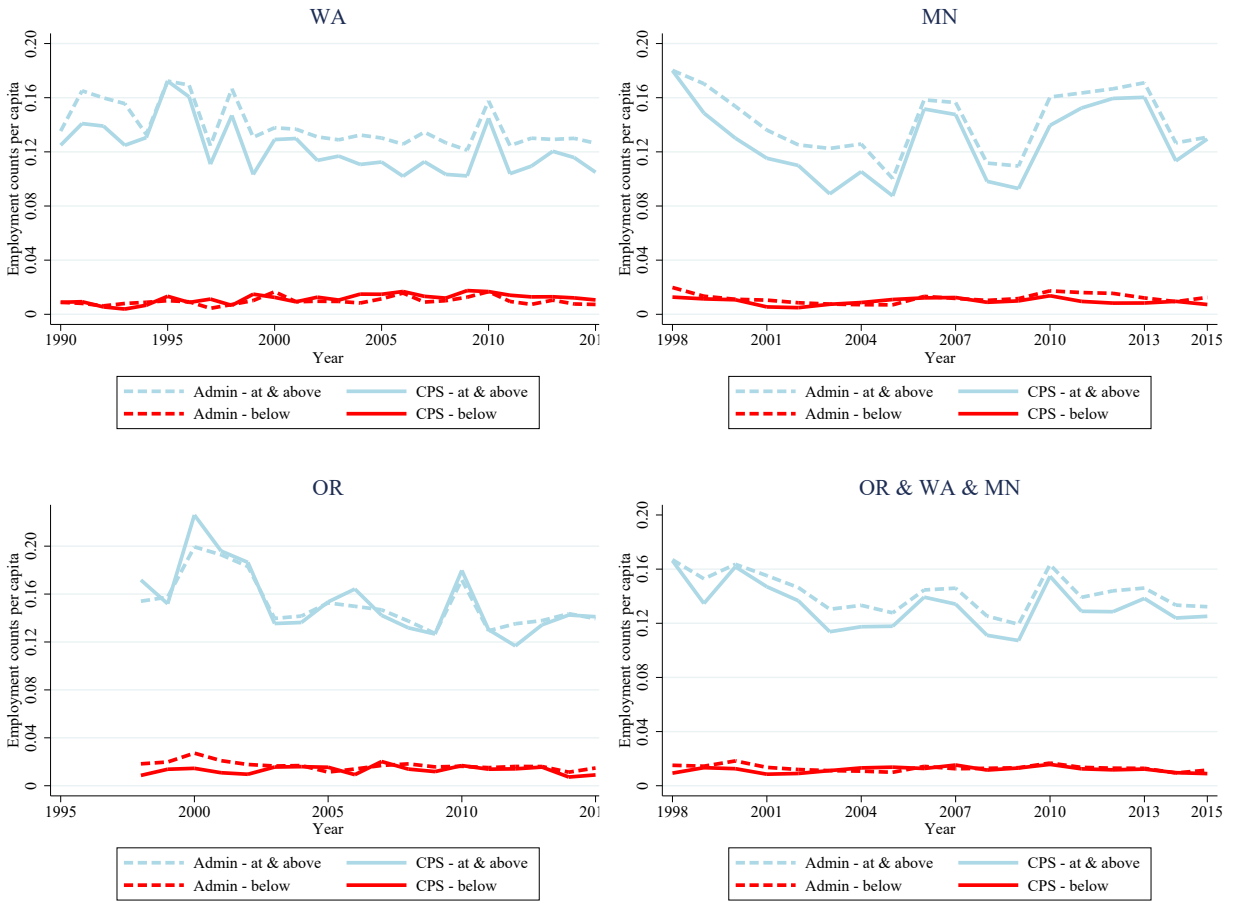


Figure cont'd: Frequency Distributions in the Administrative and CPS data



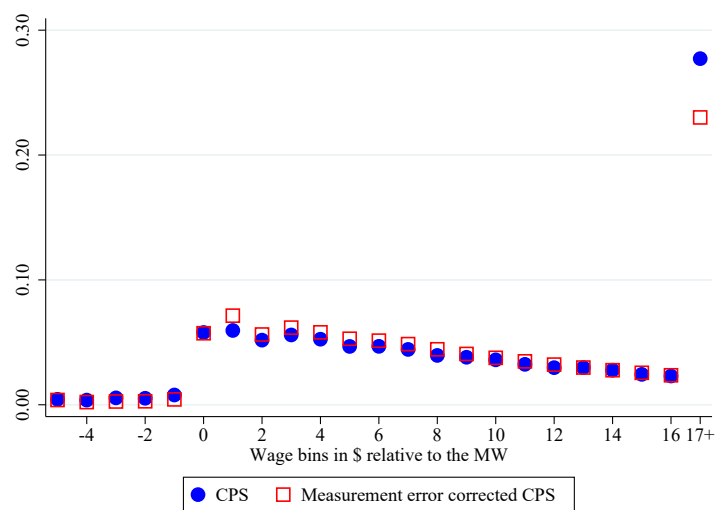
Notes: This figure plots 5-year averaged per-capita administrative and QCEW-benchmarked CPS employment counts of Washington, Minnesota, Oregon, and the three states combined from 2000 to 2014 in \$3 bins relative to the minimum wage. The red squares indicate the administrative data, and the blue circles the QCEW-benchmarked CPS counts.

Figure F.3: Comparing Administrative and CPS data; Time path



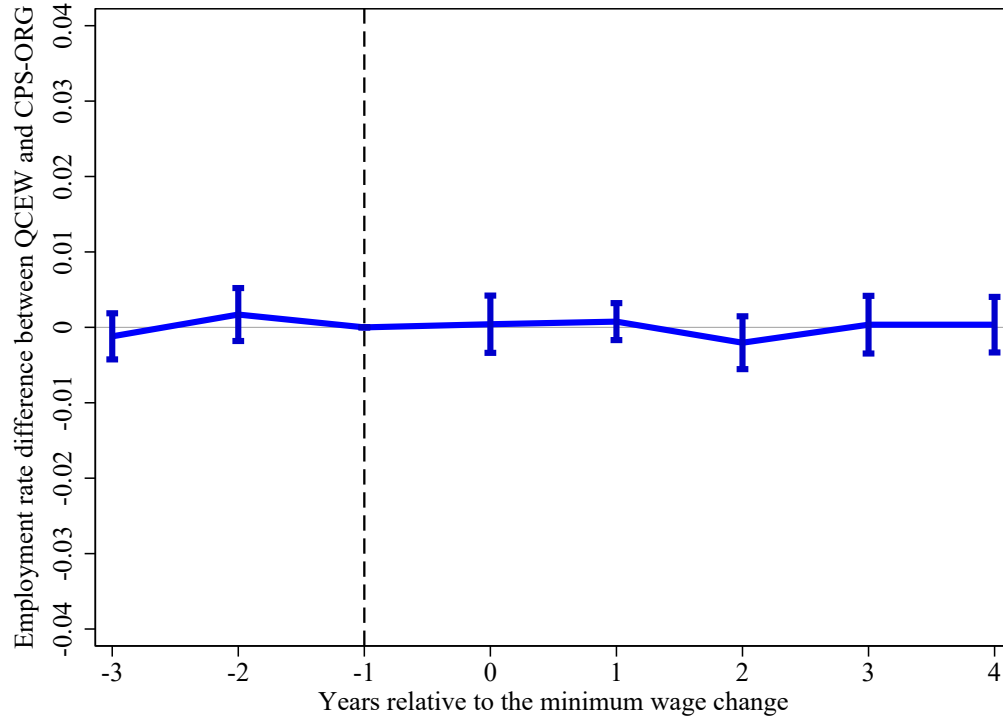
Notes: This figure plots the time paths of the number of jobs below the minimum wage ($MW - \$5, MW$), and jobs at and above the minimum wage ($[MW, MW + \$5]$) relative to the state-level population from both the administrative data and the CPS in three states (MN, OR, WA) separately, and all together.

Figure F.4: Wage Distributions in the CPS and the Measurement Error Corrected CPS



Notes: This figure plots the national wage distributions of the CPS and measurement error corrected CPS combined from 1979 to 2016 in \$1 bins relative to the minimum wage. The measurement error correction process uses the estimates in Table F.2, and the procedure described in Comte and Lacour (2011). The red squares indicate the share of workforce in the particular wage bin in the measurement error corrected CPS data, and the blue circles in the raw CPS.

Figure F.5: Impact of Minimum Wages on the Gap in Employment Between QCEW and CPS



Notes: The figure shows the effect of the minimum wage on the gap in employment rate between QCEW and CPS. In our event study analysis we use QCEW-benchmarked employment. The CPS and the QCEW have somewhat different employment concepts: the CPS asks about employment in a reference week, while the QCEW measures any employment during the quarter. To alleviate the concern that the differences in concepts has an effect on our estimates, we implement an event study regression where the outcome variable is the gap between CPS and QCEW employment. Events are the same 138 state-level minimum wage changes between 1979-2016 that we use in our benchmark specification. Similar to our benchmark specification we include state and time fixed effects in the regression. The blue line shows the evolution of the gap in the employment rate (relative to the year before the treatment) between the CPS and QCEW. We also show the 95% confidence interval based on standard errors that are clustered at the state level.

Table F.1: MSPE Ratios of CPS-Raw to QCEW-Adjusted CPS

Data structure	MSPE ratio: Raw/Benchmarked
Employment count by \$0.25 bins, averaged across 4 quarters	1.637
Employment count by \$0.25 bins, averaged across 20 quarters	3.875
Employment count under \$15, averaged across 4 quarters	7.212
Employment count under \$15, averaged across 20 quarters	7.394
Transformed employment count under \$15: average of 20 subsequent quarters minus the average of 4 preceding quarters	2.141

Notes. This table reports estimated mean squared prediction error (MSPE) ratios of the raw CPS to the QCEW-benchmarked CPS. For each dataset (raw and QCEW-benchmarked), the MSPE comes from predicting the (per-capita) administrative counts with the CPS based ones. The first two lines report the results from state-by-quarter-by-25-cent-wage-bin aggregated, and the last three lines state-by-quarter aggregated data. The transformed count is designed to be comparable to our baseline employment estimates, which compares employment in the 20 quarter following an event to the 4 quarter prior to the event. In all cases, we only consider wage bins under \$15/hour in real, 2016\$.

Table F.2: Structural Estimation of the Autor, Manning and Smith (2016) Model of Measurement Error in Wages: Evidence from CPS and Administrative Data

	Misreporting rate	Conditional error variance	Ratio of std. deviations of true to observed latent distribution
Dataset	$1-\gamma$	$\frac{1-\rho^2}{\rho^2}$	$\frac{\sigma_w}{\sigma}$
A. Re-centered \$0.25 wage bins			
CPS	0.232	1.462	0.916
Administrative data	0.277	1.251	0.908
B. \$0.25 wage bins			
CPS	0.218	1.484	0.920
Administrative data	0.343	1.076	0.895

Notes. We assess the misreporting in the CPS and in the administrative data by implementing the procedure in Autor et al. (2016). To alleviate the effect of rounding of reported hours worked in the administrative data, we re-center the \$0.25 wage bins around the minimum wage in Panel A, while in Panel B we report estimates using wage bins that are not re-centered around the minimum wage. This latter is what we use in our main analysis. We report $1-\gamma$, the misreporting rate, in Column 1; $(1-\rho^2)/\rho^2$, the variance of the error conditional on misreporting in Column 2; and the ratio of the standard deviation of the true latent distribution (w) and the observed latent distribution in Column 3.

Table F.3: Impact of Minimum Wages on Employment and Wages Using Deconvolved Data

	(1)	(2)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.017*** (0.004)
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.021*** (0.003)
% Δ affected wages	0.068*** (0.010)	0.075*** (0.012)
% Δ affected employment	0.028 (0.029)	0.046 (0.038)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.037 (0.031)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.613 (0.502)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.082
% Δ MW	0.101	0.101
Number of events	138	138
Number of observations	847,314	831,285
<i>Sample</i>		
Measurement error corrected		Y

Notes. The table reports the effects of a minimum wage increase based on the event study analysis (see equation 1) exploiting 138 state-level minimum wage changes between 1979 and 2016. The table reports five year averaged post-treatment estimates on missing jobs up to \$4 below the new minimum wage, excess jobs at and up to \$5 above it, employment and wages. Column (1) reproduces the baseline estimates in Table 1 column (1). Column (2) estimates the same parameters, but uses the data deconvolved according to the procedure proposed by Comte and Lacour (2011). In column (2), we also exclude the quarters of events due to the existence of two spikes in those periods, as explained in footnote 47. To implement the procedure, we rely on the estimates in Table F.2. All specifications include wage-bin-by-state and wage-bin-by period fixed effects. Regressions are weighted by state-quarter aggregated population. Standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Line-by-line description. The first two rows report the change in number of missing jobs below the new minimum wage (Δb), and excess jobs above the new minimum wage (Δa) relative to the pre-treatment total employment. The third row, the percentage change in average wages in the affected bins, (% ΔW), is calculated using equation 2 in Section 2.2. The fourth row, percentage change in employment in the affected bins is calculated by dividing change in employment by jobs below the new minimum wage ($\frac{\Delta a + \Delta b}{\bar{b}_{-1}}$). The fifth row, employment elasticity with respect to the minimum wage is calculated as $\frac{\Delta a + \Delta b}{\% \Delta MW}$ whereas the sixth row, employment elasticity with respect to the wage, reports $\frac{1}{\% \Delta W} \frac{\Delta a + \Delta b}{\bar{b}_{-1}}$. The line on the number of observations shows the number of quarter-bin cells used for estimation, while the number of workers refers to the underlying CPS sample used to calculate job counts in these cells.

Appendix G

Reconciling the Results from the Two-Way Fixed Effects Panel Regression with Log Minimum Wage and Event Based Regressions

This Appendix provides an explanation for the difference between the classic two-way fixed effects panel regression on log minimum wage (shown in Figure VI) and the event study used in our benchmark specification (shown in Figure II). We first establish that the difference between the two estimates is not driven by using discrete versus continuous treatment measures, or any artifact of binning the wages. Rather, they differ in how they use variation in the outcome outside of the event windows. Then, we evaluate the credibility of the two specifications by examining whether the parallel trends assumption holds before the reform; whether the results are sensitive to including different trends and control variables; whether the results are sensitive to the period chosen; and whether the employment changes are concentrated at the bottom of the wage distribution. This analysis highlights that the credibility of the two-way fixed effects estimator with log minimum wage (TWFE-logMW) can be questioned in our context: we show that the TWFE-logMW results are driven by pre-existing trends; are sensitive to the inclusion of additional controls and to the use of different sample periods; and are driven by employment changes at the upper part of the wage distribution. At the same time, the event based specification (EB) provides plausible estimates in all cases. As a result, we conclude that the EB approach is preferred to the TWFE-logMW in our context. We then provide an explanation for why the TWFE-logMW and EB estimates differ. The TWFE-logMW specification is sensitive to shocks to upper tail employment in the 1980s and early 1990s in Democratic-leaning states, which contaminate the TWFE-logMW estimates using the full 1979-2016 sample. Even though most minimum wage variation comes from after 1992, these shocks affect the estimation of the fixed effects. In contrast, the TWFE-logMW specification in the 1993-2016 sample is not affected by these shocks. The EB specification is not affected by these shocks either because it uses variation locally within the event window around the minimum wage events. We finish this Appendix by providing additional insights into how the shocks from the 1980s and early 1990s, along with the use of the TWFE-logMW specification, help explain some of the key controversies in the minimum wage literature—including estimates for teen employment.

G.1 Bridging the TWFE-logMW and the benchmark specification

We begin our analysis by assessing the contribution of various factors that drive the differences in estimates between our benchmark event-based bunching (EB-bunching) and the TWFE-logMW specifications. The benchmark specification in the paper estimates bin-by-bin employment changes relative to the minimum wage. The EB-bunching specification is calculated using the following regression (see the details in Section II.B.):

$$\frac{E_{sjt}}{N_{st}} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} I_{sjt}^{\tau k} + \mu_{sj} + \rho_{jt} + \Omega_{sjt} + u_{sjt} \quad (\text{G.1})$$

where E_{sjt} is the employment in \$0.25 wage bin j in state s and at quarter t , while N_{st} is the size of the population in state s and quarter t . The treatment dummy $I_{sjt}^{\tau k}$ equals to 1 if the minimum wage was raised τ years from date t and for the \$0.25 wage bins j that fall between k and $k + 1$ dollars of the new minimum wage. In this regression we control for state-by-wage bin and period-by-wage bin effects, μ_{sj} and ρ_{jt} , and for small or federal increases Ω_{sjt} . We use the estimates based on this specification to produce our key results (e.g. Figure II).

In Section IV we discuss the employment changes along the wage distribution using the TWFE-logMW specification (see Figure VI). For brevity, in this appendix we focus on the estimates for overall employment, E_{st} . We estimate the following specification:

$$\frac{E_{st}}{N_{st}} = \sum_{\tau=-2}^4 \alpha_{\tau} \log MW_{s,t-\tau} + \mu_s + \rho_t + u_{st} \quad (\text{G.2})$$

where E_{st} is the employment in state s at time t ; μ_s is state fixed effects and ρ_t are time effects.

There are two key differences between the EB-bunching specification shown in equation G.1 and the TWFE-logMW shown in equation G.2. First, the benchmark EB-bunching specification identifies the employment responses based on bin-by-bin employment changes around the minimum wage. To examine whether this makes a difference we estimate an event study where the outcome variable is the state level aggregate employment change (similarly to the TWFE-logMW). In particular, we estimate the following regression:

$$\frac{E_{st}}{N_{st}} = \sum_{\tau=-3}^4 \alpha_{\tau} I_{st}^{\tau} + \mu_s + \rho_t + \Omega_{st} + u_{st} \quad (\text{G.3})$$

Note that this is estimated using state-year data, the treatment variable, I_{st}^{τ} , is now defined at the state level and not at the wage-bin level, and we include state fixed effects, μ_s , and time effects, ρ_t , instead of bin-specific fixed effects. We refer to this specification as EB-state-discrete; in the main text, we also refer to this specification as EB-aggregate when the outcome is aggregate employment.

The estimated employment elasticities with respect to the minimum wage are shown in Panel A of Table G.1.⁴⁸ In Column (1) we report the benchmark EB-bunching estimates (shown in equation G.1). Panel A in Column (2) show the event study estimates on state-level employment (the EB-state-discrete specification shown in equation G.3). The employment effects are virtually the same: 0.024 in the benchmark (shown in Column 1) and 0.016 in the state-level EB (shown in Column 2), which highlight that relying on bin-by-bin estimates and controlling for state-by-wage bin, μ_{sj} , and period-by-wage bin effects ρ_{jt} is not what drives the discrepancy between our benchmark EB-bunching estimate and the TWFE-logMW estimate.

The second key difference between the benchmark EB-bunching estimate and TWFE-logMW is that the former defines each treatment using a dummy variable, while the latter uses a continuous treatment definition. To bridge the two specifications, we report employment estimates based on two

⁴⁸In this Section we focus on the the average employment changes 5 years after the minimum wage hike.

intermediate steps. First we examine whether using a continuous treatment measure, but keeping to a event-based (EB) specification makes a difference. In particular, we run the following regression:

$$\frac{E_{sjt}}{N_{st}} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} I_{sjt}^{\tau k} \Delta \log MW_{s,t-\tau} + \mu_{sj} + \rho_{jt} + \Omega_{sjt} + u_{sjt} \quad (\text{G.4})$$

where we define the treatment as $I_{sjt}^{\tau k} \Delta \log MW_{s,t-\tau}$ instead of $I_{sjt}^{\tau k}$. In other words, instead of a dummy for treatment, now the treatment switches from 0 to $\Delta \log MW_{s,t-\tau}$ at event date $\tau = 0$. Column (3) in Table G.1 shows that the estimate using continuous treatment definition (0.024) is virtually the same as our benchmark estimate (0.024). We refer to this as the EB-bunching-continuous specification.

Second, we also explore whether a similar modification of the event based estimate on aggregate employment makes a difference, by estimating the following EB-state-continuous specification:

$$\frac{E_{st}}{N_{st}} = \sum_{\tau=-3}^4 \beta_{\tau} I_{st}^{\tau} \Delta \log MW_{s,t-\tau} + \mu_s + \rho_t + u_{st} \quad (\text{G.5})$$

Column (4) in Panel A in Table G.1 show that redefining the treatment in that regression makes only a minor difference: the estimate of 0.024 in the benchmark EB-bunching specification (equation G.1) changes to 0.008 in the EB specification with aggregate employment and continuous treatment (equation G.5).

In Column (6) in Table G.1 we report the TWFE-logMW estimates. This produces a large disemployment estimate (-0.089) in line with the analysis in Section IV in the main paper. As is clear from Table G.1, the discrepancy between our estimates is not driven by use of data by wage bins, or by the continuous versus discrete treatment definition. In Panel B of Table G.1 we also report estimates on state-level employment below \$15 and state-level employment above \$15. The results highlight that the below \$15 employment change is always close to zero even in the TWFE-logMW specification. At the same time the employment changes above \$15 are small and insignificant in the EB specifications, but large negative in the TWFE-logMW specification. So what drives the difference between the TWFE-logMW specification G.2 on the one hand, and specifications G.1, G.3, G.4, and G.5 on the other? Even though equation G.2 has 4 lags and 2 leads like the other specifications, it uses variation across observations throughout the sample period, including distant observations far away from event dates. This is because the first lead and the last lag are “binned up.” In contrast, the other four observations specifically use variation within the event window.

We also estimate the distributed lag model estimated in first differences (FD):

$$\Delta \left(\frac{E_{st}}{N_{st}} \right) = \sum_{\tau=-2}^4 \alpha_{\tau} \Delta \log MW_{s,t-\tau} + \rho_t + u_{st} \quad (\text{G.6})$$

Unlike the TWFE-logMW estimates, FD estimator does not compare employment levels across observations that are decades apart, and compare within the lead/lag window. As shown in column (5), that the FD specification produces employment estimate of 0.031 (s.e. 0.031), which is similar to

the EB specification. These results highlight that the key factor driving the difference in estimates across the empirical designs is the role of employment comparisons with distant observations outside of the event window.

G.2 Credibility of the TWFE-logMW and the event study designs

Since we have two empirical designs that provide very different estimates, it is important to assess the credibility of the two estimates. To simplify the discussion we will compare the TWFE-logMW specification (shown in equation G.2) to the event based estimates on aggregate employment (shown in equation G.3). As we documented above, the focus on the low-wage bins is not driving the difference between the two designs, since the aggregate employment estimates from the event-based design are similarly small as the benchmark event-based bunching estimates.

The crucial assumption made in all difference-in-differences style estimation is that the treated and untreated states would follow parallel trends in the absence of the policy change. While testing this directly is not possible, a standard way to assess the credibility of this assumption is to examine pre-existing trends. Figure G.1 plots the time path of employment elasticities with respect to the minimum wage for the TWFE-logMW (panel a) and for the EB-aggregate (panel b). Note that interpretation of the last lag and the first lead is different in the two empirical design. Since increases in nominal minimum wages, $\log(MW)$, are always permanent, the last lag in the distributed lag model (such as TWFE-logMW reflects the “long term effect” - the weighted average of effect at or after 4 years following a minimum wage increase. Moreover, since we normalize the estimates relative to the one year before the minimum wage, $-\alpha_{-1}$ measures the average employment occurring at 3 (or more) years prior to the minimum wage increase. At the same time the event study estimates only focus on employment changes around the event window and so the last lag and first lead specifically reflect employment changes in that period.

The time path of the estimates shows that the TWFE-logMW estimator produces a spurious, positive leading effect three (or more) years prior to the minimum wage increase. This shows that there were large employment reductions substantially prior to minimum wage increases, which can impart a bias on the treatment effect estimated using the TWFE-logMW model; moreover, because we are “binning up” the leads and lags at -3 and +4, respectively, biases associated with these binned estimates can impart a bias on the estimated leads and lags, producing a spurious dynamic pattern even within the event window. These sizable and statistically significant pre-treatment and post-treatment effects are not present in our event based estimates (see panel b). Additionally, as shown in Figure G.3, the leading effect obtains only for high wage employment (above \$15) in the TWFE-logMW model.

Another standard way to test the credibility of an estimate is to assess its robustness to alternative specifications. In Table G.3 we report estimates with additional controls such as state-specific linear trends (Column 2) or with average major industry and broad occupation shares from 1979-1980 interacted with time periods (Column 3). We also explore the effect of restricting the sample to the post 1992 periods when most minimum wage changes occurred in our sample (Column 4). In

all these specifications, we find that both the TWFE-logMW and EB estimates suggest close to zero disemployment effect, which highlights that the large negative employment estimates in the TWFE-logMW are not robust to small modification of the empirical design. In other words, the large negative TWFE-logMW estimates arise only from inclusion of the 1979-1992 period, even though most of the minimum wage variation occurs after 1992. In contrast, for the 1993-2016 period, the TWFE-logMW specification passes the credibility tests, including showing no spurious leads (see and no large upper tail effects (see Figure G.5); and precisely in the sample where it passes these credibility tests, it suggests little impact on aggregate employment, including in the long run.

Finally, we examine the source of disemployment effect to assess the credibility of the two empirical designs. In the main text, we already discussed that the employment changes in the EB design occur where we expect the minimum wage should play a role. At the same time, the large negative estimate in the TWFE-logMW is driven by employment changes at the upper tail of the wage distribution. We extend this analysis here by providing direct evidence on the Card and Krueger (CK) probability groups. In Figure G.2 we show the effect of the minimum wage for the low versus the medium/high probability groups. We estimate the effect of TWFE-logMW (panel a and b) and the EB model (panel c and d) for each wage bin. The results highlight that the overall effect in the TWFE-logMW model is not only concentrated in high wage jobs but also for the “wrong” workers: namely, the large employment change occurs for only the workers least demographically likely to be earning near the minimum wage. At the same time, the EB estimates show that the minimum wage mainly affects workers most demographically likely to be earning near the minimum wage.

These results highlight that the TWFE-logMW specification, when it is implemented using the entire-sample between 1979-2016, produces a spurious negative estimate on employment. At the same time the EB design passes the standard credibility tests.

G.3 What drives the TWFE-logMW estimates

We see that TWFE-logMW when it is applied to the entire sample, produces a large negative employment effect, while when we restrict the sample to the 1993-2016 period, the TWFE-logMW estimates become small and statistically indistinguishable from zero. This is noteworthy because there were few state minimum wage changes prior to 1993.

Building on this, we next perform an exercise to demonstrate the bias in the TWFE-logMW estimates using the 1979-2016 sample. As the first step, column 3 shows results using the full 1979-2016 sample of data, but excluding the ten states that experienced any minimum wage increases prior to 1993. The pattern of results remains the same in column 3, with the TWFE-logMW specification estimating large employment declines due to the minimum wage. In columns 4 and 5, we decompose the estimate in column 3 by decomposing the outcome variable and running two separate regressions. (Note that the estimates in columns 4 and 5 are designed to add up to the estimates in column 3.) In column 4, we use actual employment data until 1992 for the forty states that did not have any minimum wage event 1979-1992. For 1993-2016, we set employment outcomes

to exactly 0 in all states. Because there were no minimum wage events prior to 1993 in this sample, and because the employment outcomes are exactly constant after 1993, the causal employment effect should be zero. Yet, column 4 shows that the TWFE-logMW specification still estimates a sizable negative employment effect, in contrast to the first-differenced and event-based specifications. Put differently, minimum wage events in 1993 and onwards appears to affect employment changes in 1979-1992. Column 5 does the opposite, and replaces all employment outcomes before 1993 with 0, and uses the actual employment rate in 1993-2016. In this case, variations in the variable of interest and the dependent variable take place in the same time period, and both the TWFE-logMW and the EB specifications indicate no disemployment effect. Finally, in column 6 we show that the spurious negative results in column 4 are not due to anticipation effects: here we consider states without minimum wage events prior to 1996 (instead of 1993 as in column 4); this reduces the sample to 39 instead of 40 states but the results are similar. Finally, in contrast to the TWFE-logMW case (panel A), our EB estimates (panel C) easily pass this test.

To summarize, this exercise shows that estimated disemployment in the TWFE-logMW specification is entirely due to employment shocks in the 1980s that were correlated with future minimum wage increases decades later, thereby affecting the estimation of the state fixed effects. This is why the restriction to an explicit event window as in the EB specification guards against the bias afflicting the TWFE-logMW specification. This is also why the inclusion of state trends or controls for historical industry/occupation shares interacted with periods substantially reduces the likely bias in that specification.

G.4 The Partisan Tilt of the 1990-1991 Recession and the Confounder

Why are state-level employment rates in 1979-1992 correlated with minimum wage events in the post-1996 period? To understand what drives this correlation we plot the time paths of the minimum wage (Panel (a)) and employment rates (Panel (b)) of the low and high minimum wage states in Figure G.4. The 15 states where the federal minimum wage laws applied during 1996-2016 are classified as low minimum wage states, and the remaining 36 states as high minimum wage states. Figure G.4 shows that the employment rate of the latter states are elevated relative to the former between the mid-1980s and the early 1990s, even as the level of minimum wages were almost the same across the two set of states in this period. The elevated employment level in the mid-1980s affects the TWFE-logMW model covering 1979-2016. However, the divergence between low minimum wage and high minimum wage states ended quickly during the 1990-1991 recession. Since then the employment rates follow parallel trends, even though there is a clear divergence in the level of the minimum wage between low and high minimum wage states in the 2000's. The timing of divergence between high and low minimum wage states highlights that the bias in the TWFE-logMW estimates is related to the differential impact of the 1990-1991 recession on (future) low and high minimum wage states.

Why is the drop in employment in the 1990-1991 recession related to future minimum wage changes in the 2000s? It is possible that the 1990-1991 recession was so severe in some states

that it changed the political landscape and opened up the door for parties supporting minimum wages. Another explanation is that the 1990-1991 recession just happened to be more pronounced in Democratic-leaning states—states that would also be more inclined to raise the minimum wage in the early 2000s following a long period of federal inaction. Table G.6 aims to test the empirical relevance of these explanations by examining the determinants of having a state-level minimum wage higher than the federal level in the post-1996s using a linear probability model. Column (1) shows that states that are harder hit by the 1990-1991 recession are more likely to have a state-level minimum wage after 1996, confirming our previous observation about Figure G.4. The model reports that for each percentage point decline in employment rate in 1990-1991, the probability of a state to be a high minimum wage state increases by 4.2% (s.e. 1.3%). However, including political leanings variables in columns (2) and (3) substantially decreases the estimate and renders it statistically indistinguishable from zero. In column (2), the unionization rate in the 1980s variable substantially decreases the size of the 1990s shock estimate and renders it statistically insignificant. In column (3), we include the average of the Partisan Voting Index (PVI) in 2000s. The PVI shows the difference between Republican Party and Democratic Party candidates' vote shares in the state. To address potential concerns related to long-run effects of the recession on political leanings, we instrument the average PVI in the 2000s with that of the 1988. In this case, the coefficient of the severity of the recession has changed its sign, become negative and statistically insignificant. This suggests that the severity of the 1990-1991 recession did not have a causal impact on future state-level minimum wage changes.

Overall, these findings clarify that the large, negative TWFE-logMW estimate from the full 1979-2016 sample is driven by upper tail shocks in the 1980s—substantially prior to most minimum wage increases we study. Moreover, these shocks are predicted by a state's historical industrial/occupational structure. Importantly, these shocks died out substantially prior to most minimum wage changes we study: indeed, as we have shown, these shocks do not produce any pre-existing trends or upper tail employment changes within the 8-year window used in our event-based analysis. However, they do substantially bias the TWFE-logMW estimator that is sensitive to persistent shocks occurring many years before the actual treatment events.

G.5 Relation to other findings in the minimum wage literature

The argument that the TWFE-logMW specification can sometimes produce spurious findings is not new to this paper. However, we provide some new insights about the nature of the problem by highlighting how shocks from the 1980s and early 1990s—long before most minimum wage variation occurred—tend to drive estimates from this specification. Here we relate this point to some key findings in the minimum wage literature.

Teen employment

The existing literature finds teen employment estimates to be sensitive to specifications. Estimates using a TWFE-logMW specification has produced more negative estimates (Neumark et al., 2014)

while inclusion of controls for state-specific trends or other controls for heterogeneity tend to suggest estimates close to zero (Allegretto et al., 2017). Since our estimates for teens (whether focused on low wage jobs or not) do not suggest disemployment effects—with or without any trend controls—in Table G.7 we provide a reconciliation with the existing teen literature. These estimates are based on the same distributed lag structure as before for TWFE-logMW, EB and FD specifications, with 2 annual leads, 4 annual lags, and the contemporaneous treatment measure.

The 1979-2016 estimates are large and negative in the TWFE-logMW specification, with an elasticity of -0.238 (s.e. 0.088). In contrast, the FD elasticity of 0.092 (s.e. 0.122) is positive in sign and not statistically significant. Both of these are consistent with findings reported in Allegretto et al. (2017). In addition, the EB estimate of 0.163 (s.e. 0.115) is also similarly positive and not statistically different from zero. Additionally, the TWFE-logMW specification in the full sample is highly sensitive to the inclusion of state-specific linear trends: inclusion of these trends produces an estimated elasticity of 0.065 (s.e. 0.128). This, too, is consistent with findings in Allegretto et al. (2017) and reflects disagreements about the right way to control for heterogeneity.

However, consistent with our findings on aggregate employment in this paper, when we consider the 1993-2016 period, none of the estimates are statistically different from zero across various specifications. The TWFE-logMW estimate from this sample of -0.024 (s.e. 0.153) is close to zero and not statistically significant. The FD estimate of 0.059 (s.e. 0.137) and EB estimate of 0.162 (s.e. 0.135) continue to be positively signed and not significant in this subsample. Therefore, if we consider the time period where most of the minimum wage increases have occurred, the estimates across all standard specifications suggest little teen dis-employment from minimum wage increases. This highlights how the same shock during the 1980s and early 1990s discussed above has also driven the sensitivity of the teen estimates. Moreover, the necessity to properly control for violations of the parallel trends assumption seems to arise from inclusion of a period with relatively little minimum wage variation (i.e., 1980s and early 1990s). And this primarily affects specifications like the TWFE-logMW which make distant comparisons. As far as we know, this point has not been recognized in the literature.

Estimates using border county design

Dube et al. (2010) (hereafter DLR) also argue that the estimates from a TWFE-logMW specification in the 1990-2006 produces spurious negative employment effects for restaurant employment, based on the presence of large negative leading effects similar to what we find here. They propose using a border-discontinuity design that compares outcomes across contiguous border county pairs to reduce the bias in the TWFE-logMW specification.

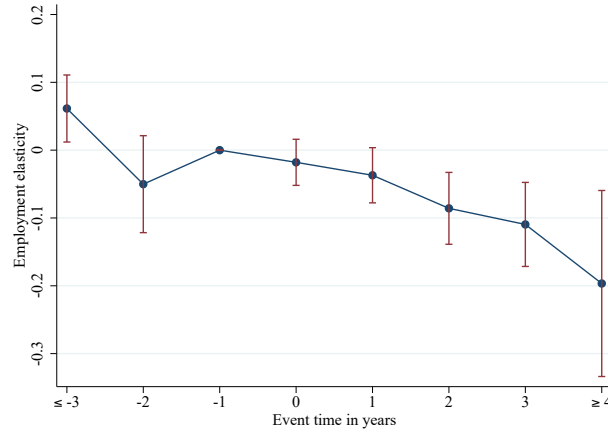
In Table G.8 we use their county-level data to establish several findings on aggregate employment, in addition to restaurant employment reported in DLR. First, similar to this paper, Figure 1 in DLR demonstrates that aggregate employment growth was systematically lower in high minimum wage states during the 1990-1991 recession, but quite similar afterwards. This highlights the same likely source of bias in their sample due to the early years that we have documented above.

Using county-level QCEW dataset from the DLR replication package, we estimate distributed lag models 16 quarters of lags and 8 quarters of leads. We calculate the estimate for the long run (16th quarter) effect net of the 4-quarter average just prior to treatment, similar to those reported in this paper. We find the TWFE-logMW estimate for aggregate employment is large and negative at -0.139 (s.e. 0.091). However, when we estimate the model in first differences, the FD estimate is close to zero at 0.023 (s.e. 0.094), consistent with what we have found in this paper using state-level data. Similarly, when we use an event-based approach like in this paper, we also find a small EB estimate of -0.024 (s.e. 0.052). Second, we find that when we estimate the model using contiguous border county pairs, the border county pair (BCP) specification similarly produces estimates that are close to zero 0.001(s.e. 0.073). If comparing either highly similar areas (i.e, border counties) or looking locally around the time of the policy change avoids biases, combining both approach would have a “double robust” property. We find that the BCP-EB specification (column 5) produces an aggregate employment elasticity of 0.030 (s.e. 0.054), which is quite close to the baseline estimate in this paper, though less precise.

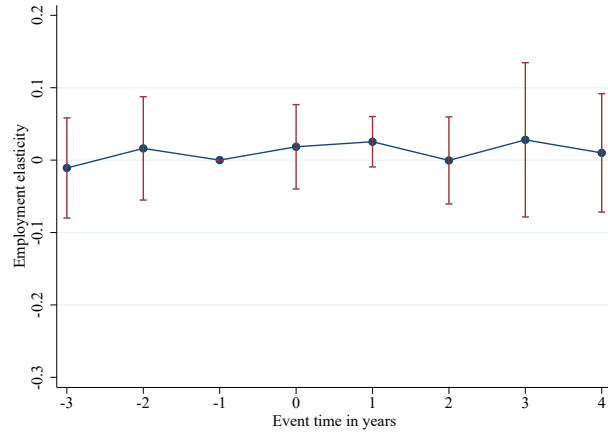
The estimates for restaurant employment follow a similar pattern. While the TWFE-logMW estimate is large and negative, -0.289 (s.e. 0.113), the EB, FD, BCP, and BCP-EB estimates are 0.002 (s.e. 0.050), -0.055 (s.e. 0.094), -0.016 (s.e. 0.082) and 0.043 (s.e., 0.055) respectively. An event-based approach looking within a window around the minimum wage increases and that allows for up to a 16 quarters post-treatment period finds no evidence of losses in restaurant jobs, even when using a panel of all-counties used in DLR.

Overall, these results confirm that using specifications like EB or FD that avoid making distant comparisons or specifications that compare across highly similar areas (like just across the state border as in BCP) appear to avoid a bias from the shocks far outside the event window. And the shocks in the 1980s and early 1990s recession seem to drive the key violation of parallel trends in this literature. This is true for aggregate employment as well as for highly affected groups like restaurants or teens.

Figure G.1: Estimated Impacts of Minimum Wages on Aggregate Employment Over Time Using Alternative Specifications



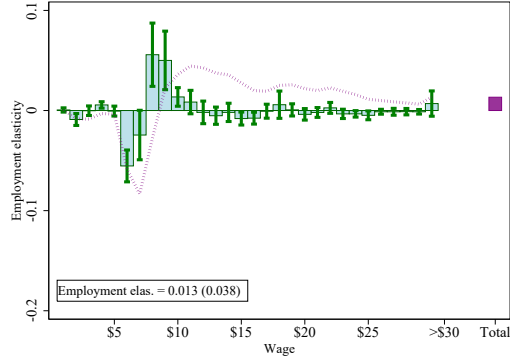
(a) TWFE-logMW



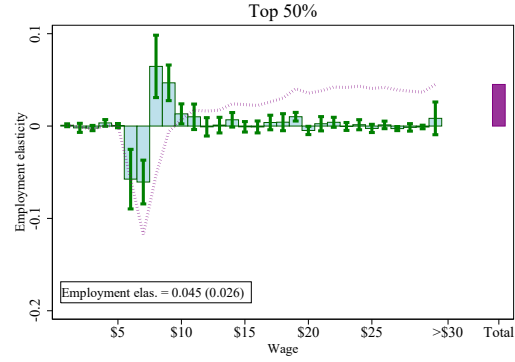
(b) Event Based Estimate

Notes: The figure shows the effect of the minimum wage on aggregate employment over time. Panel (a) uses the TWFE-logMW (equation G.2) regression of state-level aggregate employment rate on the state-level contemporaneous log minimum wage, as well as on 4 annual lags and 2 annual leads. Panel (b) uses the EB specification (equation G.3), and regresses 4 annual lags and 3 annual leads in the event dummies. The blue markers show cumulative employment elasticities by event date. These cumulative effects are calculated by successively summing the coefficients on leads and lags of log minimum wage (panel a) or event dummies (panel b), and then dividing them by the sample average employment-to-population rate. Furthermore, the cumulative elasticity at event date -1 is normalized to 0, which is why the panel (a) shows a 3rd year or earlier (≤ -3) estimate. The red error bars indicate the 95% confidence intervals around the point estimates, calculated using clustered standard errors at the state level. All regressions are weighted by sample average state population.

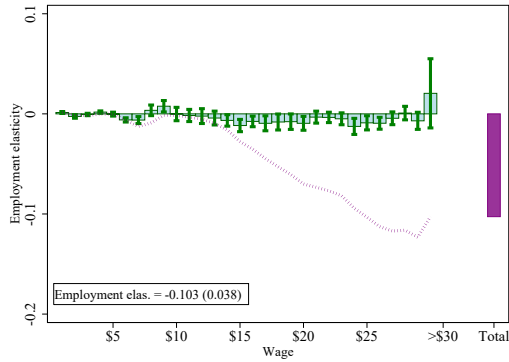
Figure G.2: Impact of Minimum Wages on the Wage Distribution by Predicted Probability Groups



(a) High and middle probability groups for the TWFE-logMW



(b) High and middle probability groups for the EB



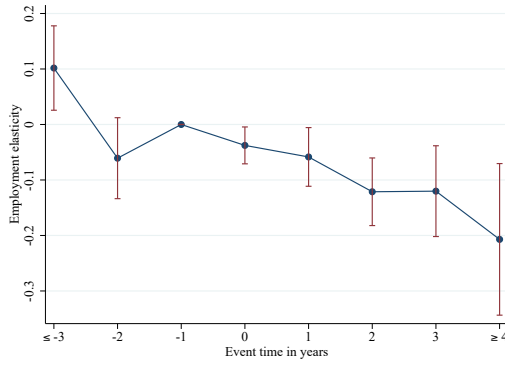
(c) Low probability for the TWFE-logMW



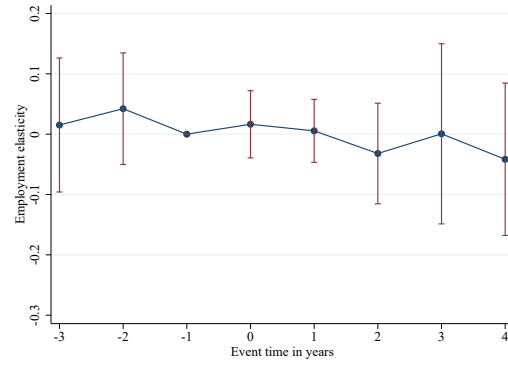
(d) Low probability for the EB

Notes: The figure shows the effect of the minimum wage on the wage distribution of the Card and Kruger probability groups in fixed effects (TWFE-logMW) and event-based specifications (EB). Panels (a) and (c) estimate the regression on the contemporaneous log minimum wage, as well as on 4 annual lags and 2 annual leads. Panels (b) and (d) use the EB specification (equation G.3), and regresses 4 annual lags and 3 annual leads in the event dummies. For each wage bin we run a separate regression, where the outcome is the number of jobs per capita in that state-wage bin. In panels (a) and (c) the cumulative response for each event date 0, 1, ..., 4 is formed by successively adding the coefficients for the contemporaneous and lagged log minimum wages. In panels (b) and (d), the responses for each event date 0, 1, ..., 4 are captured by the corresponding α_τ . The green histogram bars show the mean of these cumulative responses for event dates 0, 1, ..., 4 relative to the event date -1, divided by the sample average employment-to-population rate —and represents the average elasticity of employment in each wage bin with respect to the minimum wage in the post-treatment period. The 95% confidence intervals around the point estimates are calculated using clustered standard errors at the state level. The dashed purple line plots the running sum of the employment effects of the minimum wage up until the particular wage bin. The rightmost purple bar in each of the graphs decomposes the post-averaged elasticity of the overall state employment-to-population with respect to minimum wage by the groups, where the latter is obtained from the regressions where outcome variable is the state level employment-to-population rate. All regressions are weighted by the sample average state population.

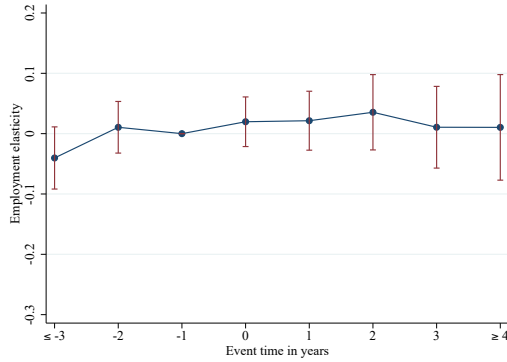
Figure G.3: Impact of Minimum Wages on Lower- and Upper-tail Employment Over Time for Fixed Effects Specification



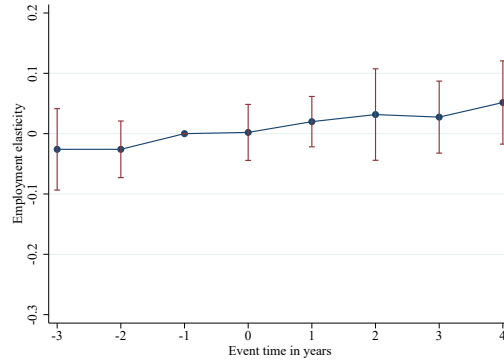
(a) TWFE-logMW, Employment Above \$15



(b) EB, Employment Above \$15



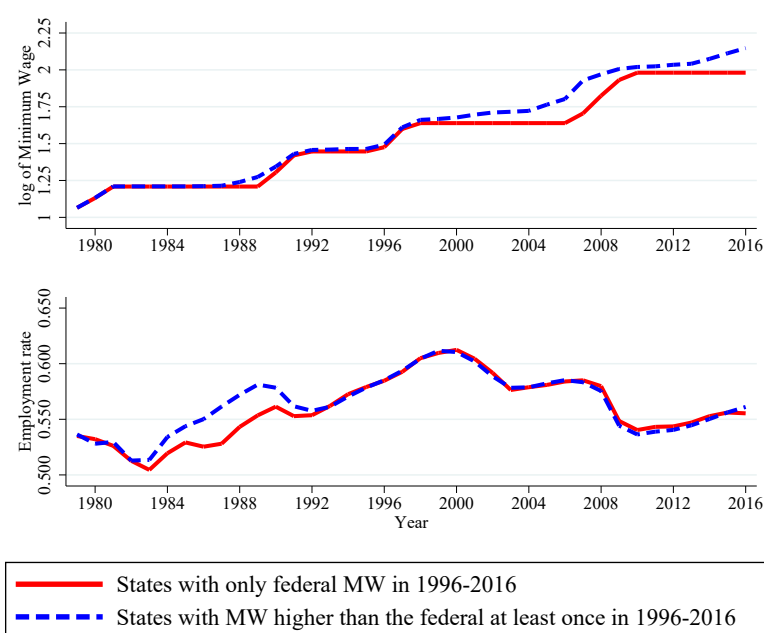
(c) TWFE-logMW, Employment Below \$15



(d) EB, Employment Below \$15

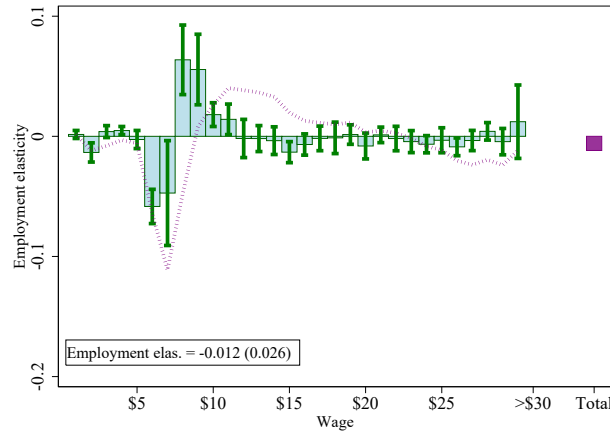
Notes: The figure shows the effect of the minimum wage on the number of jobs at or above (panels (a) and (b)), and below \$15 (panel (c) and (d)) over time in the fixed effects (TWFE-logMW) and event-based specifications. Panels (a) and (c) estimate regressions of state-level total number of jobs below, and at or above \$15 over state population on the state-level contemporaneous log minimum wage, as well as on 4 annual lags and 2 annual leads. Panels (b) and (d) use the EB specification (equation G.3), and regresses 4 annual lags and 3 annual leads in the event dummies. The blue markers show cumulative employment elasticities by event date. In panels (a) and (c), the cumulative effects are calculated by successively summing the coefficients on leads and lags of log minimum wage, and then dividing them by the sample average employment-to-population rate. Furthermore, the cumulative elasticity at event date -1 is normalized to 0; this is why the figure shows a 3rd year or earlier (" ≤ -3 ") estimate. In panels (b) and (d), the responses for each event date are captured by the corresponding α_τ . The red error bars indicate the 95% confidence intervals around the point estimates, calculated using clustered standard errors at the state level. All regressions are weighted by sample average state population.

Figure G.4: Time Paths of the Statutory Minimum Wage and Employment Rate in High and Low Minimum Wage States



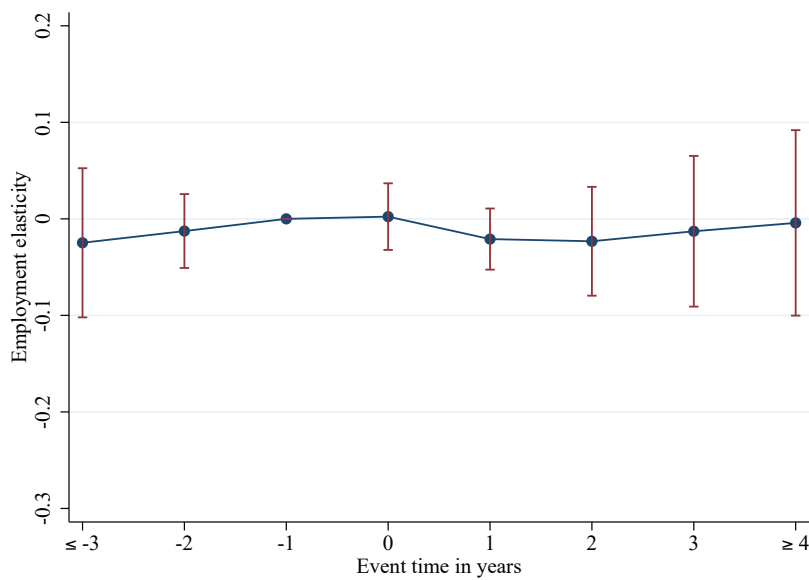
Notes: The figure shows the time paths of the average statutory log minimum wage (Panel (a)) and employment rate (Panel (b)) in 15 states where the federal minimum wage law applies in 1996 (low minimum wage states) and onward, and in 36 remaining states that had state-level minimum wages higher than the federal level at least once in 1996-2016 (high minimum wage states). In both graphs, the straight red lines correspond to the low minimum wage states, and the dash blue lines to the high minimum wage states.

Figure G.5: Impact of Minimum Wages on the Wage Distribution for TWFE-logMW Specification - 1993-2016 Sample



Notes: This figure is based on the same specification as Figure VI, but restricted to the 1993-2016 period. The figure shows the effect of the minimum wage on the wage distribution in fixed effects (TWFE-logMW) specification. We estimate two-way (state and year) fixed effects regressions on the contemporaneous log minimum wage, as well as on 4 annual lags and 2 annual leads. For each wage bin we run a separate regression, where the outcome is the number of jobs per capita in that state-wage bin. The cumulative response for each event date 0, 1,...,4 is formed by successively adding the coefficients for the contemporaneous and lagged log minimum wages. The green histogram bars show the mean of these cumulative responses for event dates 0, 1,...,4, divided by the sample average employment-to-population rate—and represents the average elasticity of employment in each wage bin with respect to the minimum wage in the post-treatment period. The 95% confidence intervals around the point estimates are calculated using clustered standard errors at the state level. The dashed purple line plots the running sum of the employment effects of the minimum wage up until the particular wage bin. The rightmost purple bar is the elasticity of the overall state employment-to-population with respect to minimum wage, obtained from regressions where the outcome variable is the state level employment-to-population rate. In the bottom left corner we also report the point estimate on this elasticity with standard errors that are clustered at the state level. Regressions are weighted by state population. The figure highlights that large aggregate disemployment effects are often driven by shifts in employment at the upper tail of the wage distribution.

Figure G.6: Impact of Minimum Wages on Employment Over Time for TWFE-logMW Specification
- 1993-2016 Sample



Notes: This figure is comparable to panel (a) of Figure G.1, except that it restricts the sample to the 1993-2016 period. The figure shows the effect of the minimum wage on aggregate employment over time using the TWFE-logMW (equation G.2) regression of state-level aggregate employment rate on the state-level contemporaneous log minimum wage, as well as on 4 annual lags and 2 annual leads. The blue markers show cumulative employment elasticities by event date. These cumulative effects are calculated by successively summing the coefficients on leads and lags of log minimum wage, and then dividing them by the sample average employment-to-population rate. Furthermore, the cumulative elasticity at event date -1 is normalized to 0, which is why the panel (a) shows a 3rd year or earlier (≤ -3) estimate. The red error bars indicate the 95% confidence intervals around the point estimates, calculated using clustered standard errors at the state level. All regressions are weighted by sample average state population.

Table G.1: Employment Elasticities with Respect to the Minimum Wage, Event-based and Continuous Variation

	EB bunching (1)	EB state discrete (2)	EB bunching continuous (3)	EB state continuous (4)	FD (5)	TWFE-log(MW) (6)
[$MW\$4$, $MW + \$5$)	0.024 (0.025)		0.024 (0.020)			
Aggregate		0.016 (0.029)		0.008 (0.025)	0.031 (0.031)	-0.089*** (0.025)
<i>By wage bin</i>						
Below \$15		0.027 (0.022)		0.020 (0.016)	-0.005 (0.020)	0.020 (0.028)
Over \$15		-0.010 (0.042)		-0.012 (0.033)	0.035 (0.023)	-0.109*** (0.030)
Number of observations	847,314	7,242	847,314	7,242	1,479	1,530
Period estimated	1979-2016	1979-2016	1979-2016	1979-2016	1979-2016	1979-2016
Equation	G.1	G.3	G.4	G.5	G.6	G.2

Notes. The table reports estimated employment elasticities of minimum wage from alternative approaches. Column 1 reports our baseline estimates (Column 1 in Table 1) that is derived by using local employment changes within a \$9 window around the new minimum wage. Column 2 use the same event study design as in Column 1 (see equation 2), but estimate the effect on below \$15 employment counts, on above \$15 employment counts, and on aggregate employment counts. In Column 3 we use the 8-year event window around the minimum wage like in Column 1, but use a continuous treatment measure, where we multiply the wage-bin-state-specific treatment indicators by the change in log minimum wage. Column 4 reports the results using continuous treatment measure for the below \$15 employment counts, above \$15 employment counts, and overall employment counts (see equation G.1). Column 5 uses a first-differences estimator with the change in the log minimum wage (equation G.6). For comparison, Column 6 reports the results using two way fixed effects estimator with log minimum wage (equation G.2) shown in Figure 9. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.2: Employment Elasticities with Respect to the Minimum Wage, Event-based and Continuous Variation - by Probability Groups

	EB-State Discrete (1)	EB-State Continuous (2)	FD (3)	TWFE-log(MW) (4)
<i>By demographically predicted wage</i>				
Predicted low-wage workers	0.019** (0.009)	0.022* (0.013)	0.015 (0.010)	-0.014 (0.009)
Predicted middle-wage workers	0.026 (0.020)	0.001 (0.014)	0.019 (0.024)	0.027 (0.035)
Predicted high-wage workers	-0.029 (0.023)	-0.015 (0.022)	-0.004 (0.013)	-0.103*** (0.038)
Number of observations	7,242	7,242	1,479	1,530
Period estimated	1979-2016	1979-2016	1979-2016	1979-2016
Equation	G.3	G.5	G.6	G.2

Notes. Column 1 uses the same event study design our baseline approach (see equation 2) but estimates the effect on the aggregate employment of three Card and Krueger probability groups. Column 2 reports the results using continuous treatment measure. For comparison, Column 3 and 4 report the results using a first-differenced (equation **G.6**) or two way fixed effects estimator with log minimum wage (equation **G.2**) shown in Figure 9. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.3: Aggregate Employment Elasticities with Respect to the Minimum Wage: Robustness of Alternative Model Specifications to Controls

	Baseline	State-specific linear trends	Base period occ. & ind. shares	Post-1992 sample
	(1)	(2)	(3)	(4)
Panel A: TWFE-log(MW)				
Emp. elas. wrt MW	-0.089*** (0.025)	0.010 (0.036)	-0.025 (0.029)	-0.012 (0.027)
Number of observations	1,530	1,530	1,530	1,020
Panel B: EB				
Emp. elas. wrt MW	0.016 (0.029)	0.022 (0.023)	0.022 (0.028)	-0.009 (0.018)
Number of observations	7,242	7,242	7,242	4,538
Panel C: FD				
Emp. elas. wrt MW	0.027 (0.031)	0.037 (0.035)	0.034 (0.023)	0.017 (0.030)
Number of observations	1,479	1,479	1,479	1,020
Number of states	51	51	51	51
Period estimated	1979-2016	1979-2016	1979-2016	1993-2016

Notes. The table reports estimated aggregate employment elasticities of minimum wage from alternative approaches. Each column and panel is a separately estimated model specification. Panel A shows the results using TWFE-logMW specification (see equation G.2), Panel B the event-based specification (equation G.3), while Panel C shows the first-differenced specification (equation G.6). Column 1 reports the results obtained from the baseline specifications, while column 2 augments the model with state-specific linear trends. Column 3 additionally controls for 1979-1980 major industry and occupation shares interacted with time fixed effects. Column 4 is similar to the first column, but limits the time span of the sample to 1993-2016. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.4: Aggregate Employment Elasticities with Respect to the Minimum Wage: Robustness of Alternative Model Specifications to Presidential Voting Index

	Baseline (1)	With continuous PVI control (2)
Panel A: TWFE-log(MW)		
Emp. elas. wrt MW	-0.089*** (0.025)	-0.027 (0.022)
Number of observations	1,530	1,530
Panel B: EB		
Emp. elas. wrt MW	0.016 (0.029)	0.009 (0.028)
Number of observations	7,242	7,242
Panel C: FD		
Emp. elas. wrt MW	0.027 (0.031)	0.029 (0.033)
Number of states	51	51
Period estimated	1979-2016	1979-2016
Number of observations	1,479	1,479

Notes. The table shows the robustness of the estimated minimum wage elasticities for aggregate employment using TWFE-logMW (Panel A), EB (Panel B), and FD (Panel C) specifications to the presidential voting index control. The first column reproduces the first column of Table [G.3](#), while column 2 augments it with presidential voting index variable. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.5: Estimated Impacts of Minimum Wages on Actual, and Simulated Employment Using Alternative Specifications

	Actual	Post 1992 sample	Excl. states with pre-1993 events		Excl. states with pre-1996 events
	(1)	(2)	(3)	(4)	(5)
Panel A: TWFE-logMW					
Emp. elas. wrt MW	-0.089*** (0.025)	-0.012 (0.027)	-0.106*** (0.037)	-0.091** (0.041)	-0.015 (0.043)
Number of observations	1,530	1,020	1,200	1,200	1,200
Panel B: EB					
Emp. elas. wrt MW	0.016 (0.029)	-0.009 (0.018)	-0.027 (0.039)	-0.001 (0.023)	-0.026 (0.032)
Number of observations	7,242	4,538	5,680	5,680	5,680
Panel C: FD					
Emp. elas. wrt MW	0.031 (0.031)	0.021 (0.030)	-0.034 (0.021)	0.006 (0.018)	-0.040 (0.024)
Number of observations	1,479	1,020	1,160	1,160	1,160
Number of states	51	51	40	40	40
Period estimated	1979-2016	1993-2016	1979-2016	1979-2016	1979-2016
Outcome variable	Actual epop	Actual epop	Actual epop	Simulated epop (1993 onwards 0)	Simulated epop (0 until 1992) (1993 onwards 0)

Notes. The table reports the effect of a minimum wage increase on the actual and simulated employment using the fixed effects (Panel A), the event-based (Panel B), and first-differences specifications (Panel C). The first column reports the estimates using the actual employment data using the full sample from 1979 to 2016. The second column excludes all the pre-1993 years. The third column employs the entire time span of the data, yet leaves out the states that have a minimum wage event before 1993. The fourth column replaces the actual outcome variable in the previous column with the simulated employment data, where we use the actual data until 1992, and replace it with 0 from 1993 onwards. The fifth column uses an alternative simulated data, we use the actual data from 1992, but replace it with 0 before 1993. The sixth column replicates the analysis based on the same simulated data as in Column 4, but it excludes all the states that experience a minimum wage increase before 1996 to account for potential anticipation effects. Regressions are weighted by state-quarter aggregated population. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.6: Determinants of Having State-level Minimum Wage in Post-1996

	(1)	(2)	(3)
Severity of the 1990-1991 shock	4.204*** (1.319)	1.473 (1.864)	-2.770 (2.306)
Unionization rate in the 1980s		3.007** (1.124)	
Average wage in the 1980s		0.001 (0.001)	
HSL share in the 1980s		-0.838 (1.496)	
PVI: Rep-Dem Vote Share in the 2000s			-2.035*** (0.576)
IV specification			Y
First stage F-statistic			74.737
Number of observations	51	51	51

Notes. The table reports the probability of having a state minimum wage higher than the federal level in any year after 1996. The predictor "severity of the 1990-1991 shock" is the percentage point decline in state-level employment due to the 1990-1991 recession. "Unionization rate in the 1980s", "Average wage in the 1980s", and "HSL share in the 1980s" are the average unionization rate, wage, and the share of individuals with high school or less education in the 1980s, respectively. "PVI: Rep - Dem Vote Share in the 2000s" is a partisan voting index that shows the difference between Republican Party and Democratic Party candidates' vote shares in the state. The first two columns use least squares estimators, and column (3) a two-stage least squares (2SLS) regression. In the 2SLS regression we use the partisan voting index from 1988 as an instrumental variable. Regressions are weighted by state averaged population. Robust standard errors are in parentheses; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.7: Robustness of Alternative Model Specifications to Controls; Teen Sample

	Baseline	State-specific linear trends	Post-1992 sample	Post-1992 sample & State-specific linear trends
	(1)	(2)	(3)	(4)
Panel A: TWFE-log(MW)				
Emp. elas. wrt MW	-0.238*** (0.088)	0.065 (0.128)	-0.024 (0.153)	0.124 (0.128)
Number of observations	1,530	1,530	1,020	1,020
Panel B: EB				
Emp. elas. wrt MW	0.163 (0.115)	0.162 (0.100)	0.162 (0.135)	0.152 (0.121)
Number of observations	7,242	7,242	4,538	4,538
Panel C: FD				
Emp. elas. wrt MW	0.094 (0.122)	0.143 (0.129)	0.059 (0.137)	0.081 (0.139)
Number of observations	1,479	1,479	1,020	1,020
Number of states	51	51	51	51
Period estimated	1979-2016	1979-2016	1993-2016	1993-2016

Notes. The table reports estimated teen employment elasticities of minimum wage from alternative approaches. Each column and panel is a separately estimated model specification. Panel A shows the results using TWFE-logMW specification (see equation G.2), Panel B the event-based specification (equation G.3), while Panel C shows the first-differenced specification (equation G.6). Column 1 reports the results obtained from the baseline specifications, while column 2 augments the model with state-specific linear trends. Column 3 is similar to the first column, but limits the time span of the sample to 1993-2016. Column 4 also limits the sample to 1993-2016, and includes state specific linear trends. Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

Table G.8: Reconciliation with DLR (2010) Findings

Specification:	TWFE-logMW	Event-based	First-differences	BCP	BCP & EB
	(1)	(2)	(3)	(4)	(5)
Restaurant emp. elasticity	-0.289** (0.113)	0.002 (0.056)	-0.055 (0.094)	-0.016 (0.082)	0.043 (0.055)
Number of observations	88,320	91,080	86,940	40,448	41,316
Overall emp. elasticity	-0.131 (0.091)	-0.024 (0.052)	0.023 (0.094)	0.001 (0.073)	0.030 (0.054)
Number of observations	197,631	203,807	194,542	144,768	148,896

Notes. This table shows estimated restaurant and aggregate employment elasticities of minimum wage using fixed effects (TWFE-logMW), event-based (EB), first differenced (FD), and border county-pairs (BCP) specifications. The BCP specifications exactly follows the proposed specification of Dube, Lester, and Reich (2010). Robust standard errors in parentheses are clustered by state; significance levels are * 0.10, ** 0.05, *** 0.01.

References

- Aaronson, Daniel, and Eric French. 2007. "Product market evidence on the employment effects of the minimum wage." *Journal of Labor Economics*, 25(1): 167–200.
- Abraham, Sarah, and Liyang Sun. 2018. "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects."
- Addison, John T., McKinley L. Blackburn, and Chad D. Cotti. 2012. "The Effect of Minimum Wages on Labour Market Outcomes: County-Level Estimates from the Restaurant-and-Bar Sector." *British Journal of Industrial Relations* 50(3): 412–435.
- Allegretto, Sylvia, Arindrajit Dube, and Michael Reich. 2011. "Do minimum wages really reduce teen employment? Accounting for heterogeneity and selectivity in state panel data." *Industrial Relations: A Journal of Economy and Society* 50(2): 205–240.
- Allegretto, Sylvia, Arindrajit Dube, Michael Reich, and Ben Zipperer. 2017. "Credible research designs for minimum wage studies: A response to Neumark, Salas, and Wascher." *ILR Review*, 70(3): 559–592.
- Autor, David H., Alan Manning, and Christopher L. Smith. 2016. "The Contribution of the minimum wage to U.S. wage inequality over three decades: a reassessment." *American Economic Journal: Applied Economics*, 8(1): 58–99.
- Bell, Linda A. "The impact of minimum wages in Mexico and Colombia." *Journal of Labor Economics* 15(3): 102–135.
- Burkhauser, Richard V., Kenneth A. Couch, and David C. Wittenburg. 2000. "A reassessment of the new economics of the minimum wage literature with monthly data from the Current Population Survey." *Journal of Labor Economics*, 18(4): 653–680.
- Campolieti, Michele, Morley Gunderson, and Chris Riddell. 2006. "Minimum wage impacts from a prespecified research design: Canada 1981-1997." *Industrial Relations: A Journal of Economy and Society* 45(2): 195–216.
- Card, David. 1992a. "Do minimum wages reduce employment? A case study of California, 1987–89." *ILR Review* 46(1): 38–54.
- Card, David. 1992b. "Using regional variation in wages to measure the effects of the federal minimum wage." *ILR Review* 46(1): 22–37.
- Card, David, Lawrence F. Katz, and Alan B. Krueger. 1994. "Comment on David Neumark and William Wascher, 'Employment effects of minimum and subminimum wages: Panel data on state minimum wage laws.'" *ILR Review* 47(3): 487–497.
- Comte, Fabienne, and Claire Lacour. 2011. "Data-driven density estimation in the presence of additive noise with unknown distribution." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(4): 601–627.
- Currie, Janet, and Bruce C. Fallick. 1996. "The Minimum Wage and the Employment of Youth." *Journal of Human Resources* 31(2): 404–428.

- Dickens, Richard, Stephen Machin, and Alan Manning. 1998. "Estimating the effect of minimum wages on employment from the distribution of wages: A critical view." *Labour Economics*, 5(2): 109–134.
- Dube, Arindrajit, T. William Lester, and Michael Reich. 2010. "Minimum wage effects across state borders: estimates using contiguous counties." *The Review of Economics and Statistics*, 92(4): 945–964.
- Dube, Arindrajit, Suresh Naidu, and Michael Reich. 2007. "The economic effects of a citywide minimum wage." *ILR Review* 60(4): 522–543.
- Eriksson, Tor, and Mariola Pytlikova. 2004. "Firm-level Consequences of Large Minimum-wage Increases in the Czech and Slovak Republics." *Labour* 18(1): 75–103.
- Fang, Tony, and Carl Lin. 2015. "Minimum wages and employment in China." *IZA Journal of Labor Policy* 4(1): 22.
- Ferman, Bruno, and Cristine Pinto. Forthcoming. "Inference in differences-in-differences with few treated groups and heteroskedasticity." *The Review of Economics and Statistics*.
- Giuliano, Laura. 2013. "Minimum wage effects on employment, substitution, and the teenage labor supply: Evidence from personnel data." *Journal of Labor Economics* 31(1): 155–194.
- Harasztsi, Péter, and Attila Lindner. Forthcoming. "Who pays for the minimum wage?" *American Economic Review*.
- Hirsch, Barry T., Bruce E. Kaufman, and Tetyana Zelenska. 2015. "Minimum wage channels of adjustment." *Industrial Relations: A Journal of Economy and Society* 54(2): 199–239.
- Hirsch, Barry T, and Edward J Schumacher. 2004. "Match bias in wage gap estimates due to earnings imputation." *Journal of Labor Economics*, 22(3): 689–722.
- Katz, Lawrence F, and Kevin M Murphy. 1992. "Changes in relative wages, 1963–1987: supply and demand factors." *The Quarterly Journal of Economics*, 107(1): 35–78.
- Kim, Taeil, and Lowell J. Taylor. 1995. "The employment effect in retail trade of California's 1988 minimum wage increase." *Journal of Business & Economic Statistics*, 13(2): 175–182.
- Madrian, Brigitte C, and Lars John Lefgren. 2000. "An approach to longitudinally matching Current Population Survey (CPS) respondents." *Journal of Economic and Social Measurement*, 26(1): 31–62.
- Meyer, Robert H, and David A Wise. 1983. "The Effects of the Minimum Wage on the Employment and Earnings of Youth." *Journal of Labor Economics*, 1(1): 66–100.
- Mian, Atif, and Amir Sufi. 2014. "What explains the 2007–2009 drop in employment?" *Econometrica*, 82(6): 2197–2223.
- Neumark, David, and Olena Nizalova. 2007. "Minimum wage effects in the longer run." *Journal of Human Resources*, 42(2): 435–452.
- Neumark, David, JM Ian Salas, and William Wascher. 2014. "Revisiting the Minimum Wage-Employment Debate: Throwing Out the Baby with the Bathwater?" *ILR Review*, 67(3_suppl): 608–648.
- Pereira, Sonia C. 2003. "The impact of minimum wages on youth employment in Portugal."

European Economic Review, 47(2), pp.229–244.

Sabia, Joseph J. 2009. “The effects of minimum wage increases on retail employment and hours: New evidence from monthly CPS data.” *Journal of Labor Research*, 30(1), pp.75–97.

Solon, Gary, Steven J Haider, and Jeffrey M Wooldridge. 2015. “What are we weighting for?” *Journal of Human Resources*, 50(2): 301–316.

Vaghul, Kavya, and Ben Zipperer. 2016. “Historical state and sub-state minimum wage data.” Washington Center for Equitable Growth Working Paper.